CITY OF CADILLAC

MAHL/LOCAL LIMITS REPORT

PREPARED FOR:

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Project No. 854440

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1.0 BACKGROUND & PURPOSE

The City of Cadillac wastewater system currently provides wastewater collection and treatment service within the City limits and to portions of the townships of Haring, Selma, Cherry Grove, and Clam Lake. The City of Cadillac has an active Industrial Pretreatment Program (IPP) due to the robust commercial and industrial activity in the area. The program procedures and local limits were adopted in 1994. The Michigan Department of Environment, Great Lakes, and Energy (EGLE) has requested the City update the program.

The first step in updating the IPP for EGLE approval was to evaluate the Maximum Allowable Headworks Loadings (MAHLs) for Pollutants of Concern. MAHLs were evaluated for selected compatible and incompatible (e.g. metals) pollutants in order to determine appropriate Local Limits to include in an updated Sewer Use Ordinance and updated industrial discharge permits. The City currently has 15 permitted industrial facilities.

- AAR Cadillac Manufacturing (Haynes St.)
- AAR Cadillac Manufacturing (6th Ave.)
- ARVCO Container Corp.
- AKWEL
- Avon Protection
- Borg Warner of Cadillac
- Cadillac Castings, Inc. (CCI)
- Cadillac Renewable Energy (CRE)
- Fiamm Technologies, Inc.
- Hutchinson Antivibration Systems
- Michigan Rubber Products, Inc.
- Munson Healthcare – Cadillac
- Piranha Hose, Inc.
- Rec Boat Holdings – Trailer Division
- Spencer Plastics

The City currently has Local Limits for the following metals: Arsenic, Cadmium, Chromium, Copper, Total Cyanide, Mercury, Molybdenum, Nickel, Silver, Lead, Zinc; and the following organics: 1,2-Dichlorobenzene, Chloroform, Ethylbenzene, Lindane, PCBs, Phenol, Styrene, Toluene, and Xylene. It is recommended that local limits also be evaluated for Bis(2-ethylhexylphthalate), BOD_5, Total Suspended Solids, Ammonia, and Phosphorus due to NPDES permit limitations for these parameters, and for selenium based on biosolids land application requirements.

This report discusses the approaches used, data, and calculations used for the MAHL Evaluation and Local Limits recommendations for the City of Cadillac.

The MAHL evaluation was done in accordance with EGLE and US EPA guidelines. Detailed descriptions and rationale are provided where procedures or methods varied from those standard procedures outlined in the guidance.

2.0 DATA COLLECTION

Historic WWTP and industrial user data was used in conjunction with data collected specifically for this MAHL evaluation in accordance with the EGLE-approved sampling plan. WWTP influent, primary effluent, final effluent and representative domestic/background wastewater samples were collected in October 2022 to provide a basis for the MAHL evaluation.

The average non-compatible pollutant concentrations (mg/L) for WWTP influent, primary effluent, final effluent, and domestic/background wastewater samples are shown in Table 1. It should be noted that secondary sludge is co-settled with primary sludge, resulting in higher primary effluent concentrations than facilities that don't co-settle.

Table 1 – Non-Compatibles - Data Collection Average Results

PARAMETER	BACKGROUND SAMPLES (mg/L)	WWTP INFLUENT (mg/L)	PRIMARY EFFLUENT (mg/L)	FINAL EFFLUENT (mg/L)
Arsenic	0.0017	0.0023	0.0049	0.0002
Cadmium	0.0004	0.0002	0.0004	0.0001
Chromium	0.0046	0.0056	0.0198	0.0001
Copper	0.167	0.066	0.1485	0.0058
Cyanide	0.0027	0.0025	0.0025	0.0173
Lead	0.0143	0.0025	0.0041	0.0005
Mercury	0.00022	0.0001	0.00031	0.0001
Molybdenum	0.0017	0.0064	0.0079	0.0053
Nickel	0.0057	0.003	0.007	0.0036
Selenium	0.0009	0.0009	0.0013	0.0007
Silver	0.001	0.0003	0.0005	0.0003
Zinc	0.352	0.1036	0.203	0.015

*Note that analytical results below the method detection limit were taken as half of the detection limit when calculating the average concentration for these parameters.

Samples for specific organic pollutants in the background and at the WWTP were also collected in October 2022, and summarized in Table 2.

Table 2 – Organic Pollutants - Data Collection Average Results

PARAMETER	BACKGROUND SAMPLES (mg/L)	WWTP INFLUENT (mg/L)	PRIMARY EFFLUENT (mg/L)	FINAL EFFLUENT (mg/L)
PCBs	<0.00011	<0.00012	<0.00011	<0.0001
Chloroform	<0.0048	<0.0048	<0.0048	<0.0048
1,2-Dichlorobenzene	<0.005	<0.005	<0.005	<0.005
Ethylbenzene	<0.005	<0.005	<0.005	<0.005
Styrene	<0.005	<0.005	<0.005	<0.005
Toluene	0.0049	<0.005	0.071	<0.005
Xylene	<0.010	<0.010	<0.010	<0.010
Total Phenols	0.071	0.055	0.157	<0.20
Lindane	<0.000053	<0.000062	<0.000053	<0.00051

*Note that analytical results below the method detection limit were taken as half of the detection limit when calculating the average concentration for these parameters. If all data was non-detect for a particular parameter, it is shown as < RL in the table.

Samples were collected from each of the currently permitted industrial user discharges and analyzed for the organic pollutants to help determine whether these parameters need to continue to have local limits. In general, phenols were the only parameter with detections in the IU discharge; therefore, phenols will be the only organic parameter included in the local limits evaluation. The full analytical results are included in Appendix A for reference.

WWTP operating data for 2021-2012 were also included in the MAHL evaluation for compatibles pollutants. The average value for the WWTP influent, primary effluent and final effluent for 2021-2022 is included in Table 3. Table 3 also presents the average concentration for the domestic/background samples based on the October 2022 MAHL sampling.

Table 3 – Compatibles - Data Collection Average Results

PARAMETER	BACKGROUND SAMPLES (mg/L)	WWTP INFLUENT (mg/L)	PRIMARY EFFLUENT (mg/L)	FINAL EFFLUENT (mg/L)
BOD_5	342	270	313	7
TSS	899	252	293	7.5
Phosphorus	10	4.2	4.6	0.22
Ammonia	34	24	24	0.6

The full analytical results are presented in Appendix A for reference.

3.0 NON-COMPATIBLE POLLUTANTS – METHODS

The non-compatible pollutant MAHL results were determined using the EGLE-developed spreadsheet model, customized by Fleis & VandenBrink Engineering, Inc. (F&V) for use at the City of Cadillac.

The following parameters are key user inputs in the non-compatibles MAHL spreadsheet:

- Influent flows, including: domestic/background, Significant Industrial Users and other non-domestic flows
- Domestic/background, industrial, and WWTP influent wastewater characteristics
- Biosolids flows and percent solids
- Treatment plant % removals

Table 4 summarizes the key flow and miscellaneous variables used in the Cadillac MAHL.

Table 4 – Variables and Values Used for Non-Compatible Pollutant MAHL

VARIABLE	VALUE	UNITS
Q_{dom}, Domestic/Background Flow (est.)	0.857	MGD
Q_{ind}, Total Industrial (SIU/CIU) Flow	0.863	MGD
Q_{POTW}, Total Avg WWTP Influent Flow	1.72	MGD
Q_{Max}, Maximum Design WWTP Influent Flow	4.5	MGD
Q_{dig}, Sludge Flow to Digester	0.07	MGD
Q_{Sludge}, Sludge Flow to Disposal	0.0025	MGD
Sludge (to disposal) % Solids	3.0	%

Table 5 summarizes the percent removals used and the source reference for the value selected to be used in the calculations. Site specific percent removals were used where applicable and appropriate. If insufficient data (i.e. influent and effluent were both less than detectable) was available to determine a site-specific percent removal, an appropriate literature value (e.g. US EPA Local Limits Development Guidance appendices) was used.

Table 5 – Treatment Plant % Removals

PARAMETER	PRIMARY % REMOVAL*	OVERALL % REMOVAL	SOURCE (primary, overall)
Arsenic	-112%	93%	Site Specific, Site Specific
Cadmium	-101%	52%	Site Specific, Site Specific
Chromium, Total	-256%	98%	Site Specific, Site Specific
Copper	-125%	91%	Site Specific, Site Specific
Cyanide	0%	69%	Site Specific, Literature
Lead	-67%	80%	Site Specific, Site Specific
Mercury	-217%	60%	Site Specific, Literature
Molybdenum	-24%	17%	Site Specific, Site Specific
Nickel	-100%	42%	Site Specific, Literature
Selenium	-44%	24%	Site Specific, Site Specific
Silver	-87%	75%	Site Specific, Literature
Zinc	-96%	86%	Site Specific, Site Specific
Phenols	-189%	8%	Site Specific, Site Specific

*Site specific data indicate negative removal efficiencies for the primary effluent because secondary sludge is co-settled

The allowable headworks loading for each parameter was calculated for the following criteria:

- NPDES Effluent Limits (copper, mercury, selenium)
- Secondary treatment inhibition
- Nitrification treatment inhibition
- Anaerobic Digester Inhibition
- Water quality based on chronic toxicity pass-through
- Water quality based on acute toxicity pass-through
- Biosolids (sludge) quality for land application (Class A)

3.1 Non-compatible Pollutants – Loading Calculations

The formula and a table summarizing the Allowable Headworks Loading (AHL) results for each criterion are described below.

3.2 Allowable Loading Based on NPDES Permit Limits

$$L_{NPDES} = \frac{Q_{POTW} * 8.34 * C_{NPDES}}{1 - R_{Avg}}$$

Where:

L_{NPDES} = Loading based on NPDES permit limits (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

C_{NPDES} = NPDES permit limit (mg/L)

R_{Avg} = Removal % for the WWTP (expressed as a decimal)

Table 6 – MAHLs based on NPDES Permit Limits

PARAMETER	C_{NPDES} (mg/L)	MAHL (lb/day)
Copper	0.011	1.79
Mercury	0.000002	0.0001
Selenium	0.0057	0.107
Bis(2-ethyhexylphthalate)*	0.031	0.4445

*No analytical data was collected for Bis(2-ethylhexylphthalate), therefore, the removal efficiency was assumed to be 0% to be conservative.

3.3 ALLOWABLE LOADING BASED ON SECONDARY BIOLOGICAL TREATMENT INHIBITION

$$L_{INHIB,Sec} = \frac{Q_{POTW} * 8.34 * C_{INHIB,Sec}}{1 - R_{PRIM}}$$

Where:

$L_{INHIB,\ Sec}$ = Loading based on biological secondary inhibition (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

$C_{INHIB,\ Sec}$ = Concentration that inhibits biological treatment of the secondary process (mg/L), literature values from *EPA Local Limits Development Guidance, Appendix G* and *EPA Guidance Manual for Preventing Interference at POTW (Table 2-1)*

R_{PRIM} = Removal % across the primary process (expressed as a decimal)

Table 7 – MAHLs based on Secondary Treatment Inhibition

PARAMETER	C_{INHIB} (mg/L)	MAHL (lb/day)
Arsenic	0.1	1.42
Cadmium	1.0	7.14
Chromium, Total	1.0	4.02
Copper	1.0	6.37
Cyanide	0.1	1.43
Lead	1.0	8.60
Mercury	0.1	0.45
Molybdenum	NA	NA
Nickel	1.0	7.17
Selenium	NA	NA
Silver	0.25	1.92
Zinc	5.0	36.6
Phenols	50.0	248

3.4 Allowable Loading Based on Nitrification Inhibition

$$L_{INHIB,Nit} = \frac{Q_{POTW} * 8.34 * C_{INHIB,Nit}}{1 - R_{Prim}}$$

Where:

$L_{INHIB,\ Nit}$ = Loading based on nitrification inhibition (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

$C_{INHIB,\ Nit}$ = Concentration that inhibits nitrification (mg/L), literature values from *EPA Local Limits Development Guidance, Appendix G* and *EPA Guidance Manual for Preventing Interference at POTW (Table 2-1)*

R_{PRIM} = Removal % across the primary process (expressed as a decimal)

Table 8 – MAHLs based on Nitrification Treatment Inhibition

PARAMETER	C_{INHIB} (mg/L)	MAHL (lb/day)
Arsenic	1.5	299
Cadmium	5.2	156
Chromium, Total	1.0	796
Copper	0.48	78.5
Cyanide	0.5	23.1
Lead	0.5	35.1
Mercury	NA	NA
Molybdenum	NA	NA
Nickel	0.5	12.3
Selenium	NA	NA
Silver	NA	NA
Zinc	0.5	49.7
Phenols	4.0	62.6

3.5 ALLOWABLE LOADING BASED ON ANAEROBIC DIGESTER INHIBITION

$$L_{INHIB,Dig} = \frac{Q_{Dig} * 8.34 * C_{INHIB,Dig}}{R_{Avg} * F}$$

Where:

$L_{INHIB,\ Dig}$ = Loading based on digester inhibition (lbs/day)

Q_{Dig} = Total sludge flow to the digester (MGD)

$C_{INHIB,\ Dig}$ = Concentration that inhibits anaerobic digestion (mg/L), literature values from *EPA Local Limits Development Guidance, Appendix G* and *EPA Guidance Manual for Preventing Interference at POTW (Table 2-1)*

R_{Avg} = Removal % across the WWTP (expressed as a decimal)

F = Fraction of overall removal due to sorption, set equal to 1.0 to arrive at the most conservative AHL

Table 9 – MAHLs based on Anaerobic Digester Inhibition

PARAMETER	C_{INHIB} (mg/L)	MAHL (lb/day)
Arsenic	1.0	0.63
Cadmium	1.0	1.11
Chromium, Total	5.0	2.97
Copper	40	25.6
Cyanide	30	25.4
Lead	50	36.7
Mercury	NA	NA
Molybdenum	NA	NA
Nickel	2.0	2.78
Selenium	NA	NA
Silver	13	10.1
Zinc	40	27.3
Phenols	NA	NA

3.6 Allowable Loading Based on Water Quality (Chronic Toxicity Pass-Through)

Chronic Loading Formula:

$$L_{Chron} = Q_{POTW} * 8.34 * \frac{WQB\quad C/1000}{(1-R_{Avg})}$$

Where:

$L_{Chronic}$ = Loading based on chronic toxicity pass through (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

R_{avg} = average overall removal fraction for the treatment plant (fraction)

$WQBEL_C$ = Water quality based effluent limit for chronic toxicity protection (µg/L), provided by EGLE for Cadillac, refer to Interoffice Memo dated 4/19/2022, included in Appendix B for reference.

Table 10 – MAHLs based on Chronic Toxicity Pass-through

PARAMETER	$WQBEL_C$ (mg/L)	MAHL (lb/day)
Arsenic	0.016	3.19
Cadmium	0.004	0.12
Chromium, Total	0.094	74.8
Copper	0.011	1.8
Cyanide	0.006	0.27
Lead	0.094	6.6
Mercury	0.0000013	0.00004
Molybdenum	3.6	129
Nickel	0.047	1.16
Selenium	0.006	0.11
Silver	0.001	0.08
Zinc	0.20	20.9
Phenols	0.51	7.9

3.7 Allowable Loading Based on Water Quality (Acute Toxicity Pass-Through)

Acute Toxicity Loading Formula:

$$L_{Acute} = Q_{POTW} * 8.34 * \frac{WQBEL_A/1000}{(1-R_{Avg})}$$

Where:

L_{Acute} = Loading based on acute toxicity pass through (lbs/day)

Q_{POTW} = Total flow to the WWTP (MGD)

R_{avg} = average overall removal fraction for the treatment plant (expressed as a fraction)

$WQBEL_A$ = Water quality-based effluent limit for acute toxicity protection (µg/L), provided by EGLE for Mt. Cadillac, refer to Interoffice Memo dated 4/19/2022, included in Appendix B for reference.

Table 11 – MAHLs based on Acute Toxicity Pass-through

PARAMETER	$WQBEL_A$ (mg/L)	MAHL (lb/day)
Arsenic	0.68	135
Cadmium	0.012	0.36
Chromium, Total	1.3	1,035
Copper	0.029	4.74
Cyanide	0.044	2.03
Lead	0.8	58
Mercury	NA	NA
Molybdenum	58	2,079
Nickel	0.8	19
Selenium	0.12	2.27
Silver	0.022	1.26
Zinc	0.36	36
Phenols	6.8	106

3.8 Allowable Loading Based on Biosolids Quality

$$L_{Sludge} = Q_{Sludge} * \frac{TSS_{Sludge}}{100} * 8.34 * \frac{C_{Sludge}}{R_{avg} * F_{Sorp}}$$

Where:

L_{Sludge} = Loading based on biosolids quality (lbs/day)

Q_{Sludge} = Average daily sludge generation rate (MGD)

TSS_{Sludge} = Average sludge % solids

C_{Sludge} = Target standard for biosolids quality (mg/kg) – use Part 24 “Clean” or “Ceiling” limit as applicable. Part 24 Table 3 values for biosolids were used for this evaluation because Cadillac prefers to land apply its biosolids as a Class A product

R_{avg} = average overall removal fraction for the treatment plant (expressed as a fraction)

F_{Sorp} = fraction of removal that occurs by sorption to biosolids (this was assumed to be 1.0 for all parameters)

Table 12 – MAHLs based on Biosolids Quality for Land Application of Biosolids

PARAMETER	C_{Sludge} (mg/kg)	MAHL (lb/day)
Arsenic	41	0.027
Cadmium	39	0.046
Chromium, Total	NA	NA
Copper	1500	1.014
Cyanide	NA	NA
Lead	300	0.232
Mercury	17	0.017
Molybdenum	75	0.268
Nickel	420	0.616
Selenium	100	0.256
Silver	NA	NA
Zinc	2800	2.018
Phenols	NA	NA

The most restrictive loading was chosen as the controlling MAHL and safety factors were then applied to determine the MAIL as follows.

Table 13 – Controlling MAHL, Criteria, MAIL, and Safety Factor

PARAMETER	Controlling Criteria	MAHL (lb/day)	Domestic Loading (lb/day)	Safety Factor (S.F.)	MAIL after SF (lb/day)
Arsenic	Biosolids	0.027	0.007	10%	**0.017**
Cadmium	Biosolids	0.046	0.001	10%	**0.041**
Chromium, Total	Digestion Inhibition	2.97	0.007	10%	**2.67**
Copper	Biosolids	1.01	0.501	10%	**0.412**
Cyanide	Chronic Toxicity	0.273	0.018	10%	**0.228**
Lead	Biosolids	0.232	0.005	10%	**0.204**
Mercury	Chronic Toxicity	0.00004	0.0007	10%	**-**
Molybdenum	Biosolids	0.268	0.006	10%	**0.236**
Nickel	Biosolids	0.617	0.015	10%	**0.540**
Selenium	Chronic Toxicity/NPDES	0.108	0.004	10%	**0.093**
Silver	Chronic Toxicity	0.080	0.0018	10%	**0.070**
Zinc	Biosolids	2.02	0.701	10%	**1.115**
Phenols	Chronic Toxicity	7.98	0.717	10%	**6.46**
Bis(2-ethylhexylphthalate)	NPDES Limits	0.444	-	20%	**-**

*Safety Factor as defined in Section 6.2.3 of US EPA's *Local Limits Guidance*

4.0 Non-compatible Pollutants – Allocation & Local Limit Development

After the MAHL was determined for each of the non-compatible parameters, the Maximum Allowable Industrial Loading (MAIL) was determined by subtracting out the domestic background portion of the total loading.

The City has chosen to utilize a uniform allocation of the available MAIL. The available MAIL was converted to into a calculated local limit by dividing by the current total industrial flow of 0.8506 MGD and the conversion factor 8.34. The calculated Local Limits for the non-compatible pollutants are summarized and compared with the City's current local limit in Table 14 below.

Table 14 – Draft Standard Local Limits – Non-compatible Pollutants

POLLUTANT	Current Local Limit (Daily Max) (µg/L)	Calculated Local Limit (Daily Max) (µg/L)	Proposed Local Limit (Daily Max) (µg/L)
Arsenic	26	2	2
Cadmium	2	6	2
Chromium	369	373	370
Copper	99	58	58
Cyanide, Total	4	32	4
Lead	100	29	29
Mercury	ND*	-	ND*
Molybdenum	118	33	33
Nickel	185	76	76
Selenium	-	13	13
Silver	3	10	3
Zinc	835	156	156
Phenols	1236	912	Report
Bis(2-ethylhexylphthalate)	-	25	Report

* Monitoring for mercury shall be in accordance with the following test methods: The discharge of mercury at or above the quantification level of 0.2 ug/l shall represent an exceedance of the local limit. Mercury sampling procedures, preservation and handling, and analytical protocol for compliance monitoring shall be in accordance with U.S. EPA Method 245.1, unless screening using Method 1631 is required by the City. The quantification level shall be 0.2 ug/l for Method 245.1 or 0.5 nanograms/liter (ng/l) for Method 1631, unless higher levels are appropriate due to sampling matrix interference.

It should be noted that a typical method for determining the Local Limit was not feasible for mercury because the observed domestic loading exceeds the calculated MAHL. Mercury will be assigned a non-detectable limit in accordance with the State of Michigan and City of Cadillac's pollutant minimization program for mercury.

The Bis(2-ethylhexylphthalate) local limit was calculated by dividing the MAHL by the total POTW flow since the contribution from background compared to industrial sources is not currently known.

It is proposed that the City monitor phenols and Bis(2-ethylhexylphthalate) in the WWTP effluent and permitted IU discharges annually to determine whether the calculated local limit needs to be implemented.

5.0 Compatible Pollutants – Methods

MAHL evaluations for 5-day biochemical oxygen demand, total suspended solids, phosphorus, and ammonia were also completed.

Table 3 in the Data Collection section above summarizes the compatible pollutant sampling results for WWTP influent, primary effluent, final effluent, and domestic/background sources.

The MAHLs for the compatible pollutants were determined based on the treatment plant capacity or NPDES limitations, whichever loading was lowest. The allowable headworks loading based on the NPDES permit limit was calculated in accordance with the EPA Local Limits Development Guidance using the following equation:

NPDES Limit Formula: $L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW}}{(1-R)}$

Where:

L_{NPDES} = Loading based on NPDES permit limit (lbs/day)

C_{NPDES} = NPDES permit limit

Q_{POTW} = Total flow to the WWTP (MGD)

R = average or minimum overall removal fraction for the treatment plant (fraction)

R_{avg} is used for monthly average calculations, R_{min} is used for daily maximum calculations

5.1 Biochemical Oxygen Demand MAHL and Local Limit

The City's 1975 O&M Manual Basis of Design indicates that the WWTP is designed to treat an average BOD_5 loading of 3,300 lb BOD_5/day. Fine bubble aeration was added to the plant in 1996, resulting in a calculated BOD capacity of 4,748 lb/day based the size of the aeration tanks and the *10 States Standards* loading of 40 lbs BOD per 1000 cf of aeration.

The estimated 1996 design basis AHL was compared to seasonal NPDES Limit formula calculations based on available performance data for the City. The calculations are included in Appendix A and summarized below.

Table 15 – NPDES Permit Based AHLs for BOD_5

Permit Basis	Season	C_{NPDES}	% Removal	L_{NPDES}
Monthly Average	June-September	7 mg/L	97.7	4,366 lb/d
Monthly Average	October-November	11 mg/L	97.5	6,312 lb/d
Monthly Average	December-March	25 mg/L	97.2	12,808 lb/d
Monthly Average	April-May	17 mg/L	96.8	7,621 lb/d
Daily Max	June-September	10 mg/L	91.6	4,468 lb/d
Daily Max	October-November	17 mg/L	91.1	7,169 lb/d
Daily Max	December-March	34 mg/L	91.5	15,012 lb/d
Daily Max	April-May	25 mg/L	90.6	9,981 lb/d

The Monthly Average, June-September NPDES permit limit is the controlling criteria for the MAHL. The controlling AHL and MAHL is 4,366 lb/day.

The MAIL for BOD_5 was calculated as the MAHL (4,366 lb/day) minus the current average domestic/background loading (1,452 lb/day) and reserve loading (50 lb/day).

The calculated MAIL for BOD_5 is 2,864 lb day. The MAIL was distributed uniformly to all industrial users, resulting in a calculated BOD local of 401 mg/L. The proposed local limit is 400 mg/L.

5.2 Total Suspended Solids MAHL and Local Limit

The MAHL for TSS was determined similarly to the method described for the BOD_5 MAHL. The 1972 WWTP design capacity for TSS was 4,200 lb/day; however, additional primary and secondary clarification has been added since then, and tertiary treatment has also been upgraded.

The MAHL and local limits are based on seasonal NPDES permit limits for TSS.

Table 16 – NPDES Permit Based AHLs for TSS

Permit Basis	Season	C_{NPDES}	% Removal	L_{NPDES}
Monthly Average	June-September	20 mg/L	97.0	9,563 lb/d
Monthly Average	October-May	30 mg/L	96.3	11,631 lb/d
7-Day Max	June-September	30 mg/L	84.5	7,264 lb/d
7-Day Max	October-May	45 mg/L	88.2	14,312 lb/d

The 7-day Max, June-September NPDES permit limit is the controlling criteria for the MAHL. After applying a 10% safety factor, the MAHL is 6,537 lb/day.

The MAIL for TSS was calculated as the MAHL (6,537 lb/day) minus the current average domestic loading (1,430 lb/d) and reserve loading (100 lb/day).

The calculated MAIL for TSS is 5,007 lb day. As with BOD, a standard local limit was developed for TSS for all non-domestic users, by dividing the MAIL by the industrial flow. This yields an allowable concentration of 701 mg/L. The proposed standard local limit for TSS is 700 mg/L.

5.3 Ammonia MAHL and Local Limit

The MAHL for Ammonia nitrogen was determined similarly to the method described for the BOD_5 and TSS MAHL calculations. The 1972 WWTP O&M manual lists the max day ammonia design basis as 751 lb/day.

This design capacity was compared to the seasonal NPDES permit limits for ammonia. The spreadsheet calculations are attached for reference.

Table 17 – NPDES Permit Based AHLs for Ammonia

Permit Basis	Season	C_{NPDES}	% Removal	L_{NPDES}
Monthly Average	June-September	0.9 mg/L	98.5	861 lb/d
Monthly Average	October-November	1.6 mg/L	97.7	998 lb/d
Monthly Average	December-March	3.7 mg/L	96.3	1,434 lb/d
Monthly Average	April-May	4.8 mg/L	97.0	2,295 lb/d
Daily Max	June-September	2.0	96.1	1,925 lb/d
Daily Max	October-November	2.8	78.8	496 lb/d

The daily max seasonal October-November NPDES permit limit is the controlling criteria for the MAHL. The controlling AHL is 496 lb/day. After applying a 5% safety factor, the MAHL is 471 lb ammonia/day.

The MAIL for ammonia was calculated as the MAHL (471 lb/day) minus the current average domestic loading (238 lb/day). The calculated MAIL for ammonia is 233 lb/day.

As with BOD and TSS, a standard local limit was developed for all industrial users. The maximum concentration allowable for this limit was calculated by dividing the MAIL by the industrial flow rate. This yields an allowable concentration of 33 mg/L. The proposed standard local limit for ammonia is 33 mg/L.

5.4 Phosphorus MAHL and Local Limit

The MAHL for phosphorus was determined similarly to the method described for the other compatible pollutant MAHLs. The 1972 design basis report indicates the WWTP has a phosphorus treatment capacity of 667.2 lb/day.

The design capacity was compared to the NPDES permit based AHLs, summarized in Table 18.

Table 18 – NPDES Permit Based AHLs for Phosphorus

Permit Basis	Season	C_{NPDES}	% Removal	L_{NPDES}
Monthly Average	May-March	0.5 mg/L	94.7	135 lb/d
Monthly Average	April	0.65 mg/L	95.2	194 lb/d

The controlling AHL for phosphorus is the May-March monthly average permit limit of 0.5 mg/L. The MAIL for phosphorus was calculated as the MAHL (135 lb/day) minus the current average domestic loading (40 lb/day). The calculated MAIL for phosphorus is 89 lb/day, which was divided uniformly to industrial users. The calculated local limit for phosphorus is 12 mg/L.

5.5 Compatible Pollutant MAHL & Local Limits Summary

The average MAHL, domestic loading, and MAILs for the compatible pollutants are summarized in Table 19, along with the proposed daily maximum local limit based on uniform allocation of the MAIL.

Table 19 – Compatible MAHL, MAIL, and Safety Factor

PARAMETER	Safety Factor	MAHL (lb/day)	Background Loading (lb/day)	Reserve Loading (lb/day)	MAIL (lb/day)	Daily Max (mg/L)
BOD_5	-	4,366	1,452	50	2,864	400
TSS	10%	6,537	1,430	100	5,007	700
Ammonia	5%	471	238	-	233	33
Phosphorus	5%	129	40	-	89	12

Appendix A

City of Cadillac
Maximum Allowable Headworks Loading Evaluation
Table A1 - Flows and Data Summary

Q_{POTW}	1.72	MGD	2021 Avg
Q_{Ind}	0.857	MGD	
$Q_{AAR\ Facility\ +\ Rinse\ Wa}$	0.298	MGD	
$Q_{AAR\ 6th\ St}$	0.006	MGD	
Q_{ARVCO}	0.005	MGD	
Q_{AVON}	0.004	MGD	
$Q_{Akwell\ 5th\ St}$	0.066	MGD	
$Q_{Akwell\ 7th\ St}$	0.178	MGD	
$Q_{Borg\ Warner}$	0.017	MGD	
$Q_{Cadillac\ Castings}$	0.097	MGD, est.	
$Q_{Cadillac\ Renewable\ Eng}$	0.065	MGD	
Q_{Fiamm}	0.002	MGD	
$Q_{Hutchinson}$	0.008	MGD	
$Q_{Michigan\ Rubber}$	0.005	MGD	
Q_{Munson}	0.027	MGD	
$Q_{Piranha}$	0.077	MGD	
$Q_{Spencer\ Rec\ Boats}$	0.0003	MGD	
$Q_{Spencer\ Plastics}$	0.0006	MGD	
$Q_{domestic/commercial}$	0.863	MGD	
$Q_{max,\ design}$	4.50	MGD	
$Q_{max,\ observed\ 2021\text{-}20}$	2.85	MGD	
Q_{dig}	0.070	MGD	
Q_{sludge}	0.0025	MGD	
Sludge % TS	3%		
95% exceedence	2.6	cfs	
Clam River	2	MGD	

Parameter	Background #1 (mg/L)	Background #2 (mg/L)	Influent (mg/L)	Primary Effluent (mg/L)	Final Effluent (mg/L)
Arsenic	0.0024	0.0010	0.0023	0.0049	0.0002
Cadmium	0.0006	0.0001	0.00021	0.0004	0.0001
Chromium	0.0082	0.0010	0.0056	0.0198	0.0001
Copper	0.2648	0.0696	0.0659667	0.1485	0.0058
Cyanide, Total	0.0030	0.0025	0.0025	0.0025	0.0173
Cyanide, Availabl	0.006	0.005	0.003	0.005	0.0016
Lead	0.0278	0.0007	0.0025	0.0041	0.0005
Mercury	0.00034	0.00010	0.0001	0.00031667	0.0001
Molybdenum	0.00263	0.00077	0.0063833	0.0079	0.0053
Nickel	0.0094	0.0021	0.003	0.007	0.0036
Selenium	0.0012	0.0006	0.0009	0.0013	0.0007
Silver	0.0018	0.0003	0.0003	0.0005	0.0003
Zinc	0.6062	0.0974	0.1036	0.2029	0.0150
cBOD5	482	202	270	309	6.8
TSS	1599	199	250	267	5.8
Ammonia	34	33	26	4.3	3.6
Phosphorus	15	6	4.3	0.34	0.22
Phenol	0.043	0.100	0.055	0.16	0.05

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TABLE A2

Local Limits Determination Based on NPDES Monthly Effluent Limits

	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE							MAXIMUM LOADING		INDUSTRIAL LOADING		
						Domestic and Commercial						
Pollutant	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Average Removal Efficiency (%) (Ravg)	Nonvolatile Removal Fraction (%)	NPDES Monthly Limit (mg/l) (Ccrit)	Conc. (mg/l) (Cdom)	Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)	Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8506	1.720	92.8	100	-	0.0010	0.8632	-	0.0074	-	-	10
Cadmium	0.8506	1.720	52.4	100	-	0.0001	0.8632	-	0.0007	-	-	10
Chromium	0.8506	1.720	98.2	100	-	0.0010	0.8632	-	0.0072	-	-	10
Copper	0.8506	1.720	91.2	100	0.011	0.0696	0.8632	1.7992	0.5005	0.8489	0.1197	25
Cyanide	0.8506	1.720	69.0	100		0.0025	0.8632	-	0.0180	-	-	10
Lead	0.8506	1.720	79.6	100	-	0.0007	0.8632	-	0.0052	-	-	10
Mercury	0.8506	1.720	50.0	100	0.000002	0.0001	0.8632	0.0001	0.0007	-0.0007	-0.0001	10
Molybdenum	0.8506	1.720	17.2	100	-	0.0008	0.8632	-	0.0055	-	-	10
Nickel	0.8506	1.720	42.0	100	-	0.0021	0.8632	-	0.0149	-	-	10
Selenium	0.8506	1.720	24.1	100	0.0057	0.0006	0.8632	0.1077	0.0043	0.0926	0.0131	10
Silver	0.8506	1.720	75.0	100	-	0.0003	0.8632	-	0.0018	-	-	10
Zinc	0.8506	1.720	85.6	100	-	0.0974	0.8632	-	0.7011	-	-	10
Phenol	0.8506	1.720	8.4	100	-	0.0997	0.8632	-	0.7173	-	-	10
Bis(2-ethylhexylphthalate)	0.8506	1.720	0.0	100	0.031			0.4445			0.0248	20

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.

(Q_{potw}) POTW's average influent flow in MGD.

(R_{avg}) Average removal efficiency across POTW as percent.

(C_{crit}) NPDES monthly maximum permit limit for a particular pollutant in mg/l.

(Q_{dom}) Domestic/commercial background flow in MGD.

(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.

(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).

(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).

(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.

(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.

$F_{(nonvol)}$ Fraction of overall POTW removal not due to volatilization as a percent

(SF) Safety factor as a percent.

8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{crit} * Q_{potw}}{(1 - R_{potw}) * F_{(nonvol)}}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * \ 8.337}$$

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TABLE A3

Local Limits Determination Based on Secondary Treatment Inhibition Level

	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE						MAXIMUM LOADING				INDUSTRIAL LOADING		
					Domestic and Commercial								
Pollutant	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Removal Efficiency (%) (Rprim)	Sec. Treatment Inhibition Level (mg/l) (Ccrit)	Conc. (mg/l) (Cdom)	Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8506	1.720	-1	0.1	0.0010	0.8632	1.418	0.007			1.269	0.18	10
Cadmium	0.8506	1.720	-101	1	0.0001	0.8632	7.141	0.001			6.427	0.91	10
Chromium	0.8506	1.720	-256	1	0.0010	0.8632	4.023	0.007			3.613	0.51	10
Copper	0.8506	1.720	-125	1	0.070	0.8632	6.369	0.501			4.276	0.60	25
Cyanide	0.8506	1.720	0	0.1	0.003	0.8632	1.434	0.018			1.273	0.18	10
Lead	0.8506	1.720	-67	1	0.0007	0.8632	8.604	0.005			7.738	1.09	10
Mercury	0.8506	1.720	-217	0.1	0.0001	0.8632	0.453	0.001			0.407	0.06	10
Molybdenum	0.8506	1.720	-24		0.0008	0.8632	-	0.006			-	-	10
Nickel	0.8506	1.720	-100	1	0.0021	0.8632	7.170	0.015			6.438	0.91	10
Selenium	0.8506	1.720	-44		0.0006	0.8632	-	0.004			-	-	10
Silver	0.8506	1.720	-87	0.25	0.0003	0.8632	1.920	0.002			1.727	0.24	10
Zinc	0.8506	1.720	-96	5	0.097	0.8632	36.615	0.701			32.3	4.55	10
Phenol	0.8506	1.720	-189	50	0.100	0.8632	248.048	0.717			222.5	31.4	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.

(Q_{potw}) POTW's average influent flow in MGD.

(R_{prim}) Removal efficiency across across primary treatment as percent.

(C_{crit}) Activated sludge threshold inhibition level, mg/l. Source: US EPA Local Limits Development Guidance, Appendix G.

(Q_{dom}) Domestic/commercial background flow in MGD.

(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.

(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).

(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).

(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.

(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.

(SF) Safety factor as a percent.

8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{crit} * Q_{potw}}{1 - R_{prim}}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

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TABLE A4

Local Limits Determination Based on Nitrification Inhibition Level

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE						MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Removal Efficiency (%) (Rsec)	Nitrification Inhibition Level (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8568	1.720	93	1.5	0.0010	0.8632	298.981	0.007			269.08	37.67	10
Cadmium	0.8568	1.720	52	5.2	0.0001	0.8632	156.589	0.001			140.93	19.73	10
Chromium	0.8568	1.720	98	1	0.001	0.8632	795.850	0.007			716.26	100.28	10
Copper	0.8568	1.720	91	0.48	0.0696	0.8632	78.510	0.501			70.16	9.82	10
Cyanide	0.8568	1.720	69	0.5	0.0025	0.8632	23.128	0.018			20.80	2.91	10
Lead	0.8568	1.720	80	0.5	0.0007	0.8632	35.132	0.005			31.61	4.43	10
Mercury	0.8568	1.720	50		0.0001	0.8632	-	0.001			-	-	10
Molybdenum	0.8568	1.720	60		0.0008	0.8632	-	0.006			-	-	10
Nickel	0.8568	1.720	42	0.5	0.0021	0.8632	12.362	0.015			11.11	1.56	10
Selenium	0.8568	1.720	24		0.0006	0.8632	-	0.004			-	-	10
Silver	0.8568	1.720	75		0.0003	0.8632	-	0.002			-	-	10
Zinc	0.8568	1.720	86	0.5	0.097	0.8632	49.685	0.701			44.02	6.16	10
Phenol	0.8568	1.720	8	4.0	0.100	0.8632	62.587	0.717			55.61	7.79	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.

(Q_{potw}) POTW's average influent flow in MGD.

(R_{sec}) Removal efficiency across primary treatment and secondary treatment as percent.

(C_{crit}) Nitrification threshold inhibition level, mg/l.

(Q_{dom}) Domestic/commercial background flow in MGD.

(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.

(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).

(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).

(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.

(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.

(SF) Safety factor as a percent.

8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{crit} * Q_{potw}}{1 - R_{sec}}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

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TABLE A5

Local Limits Determination Based on Anaerobic Digester Inhibition Level

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE								MAXIMUM LOADING						
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Sludge Flow to Digester (MGD) (Qdig)	Sorption Removal Fraction (%)	Removal Efficiency (%) (Rpotw)	Anaerobic Digester Inhibition Level (mg/l) (Ccrit)	Domestic and Conc. (mg/l) (Cdom)	Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8506	1.720	0.07	100	93	1	0.0010	0.8632	0.629	0.0074			0.55851	0.08	10
Cadmium	0.8506	1.720	0.07	100	52	1	0.0001	0.8632	1.114	0.0007			1.00199	0.14	10
Chromium	0.8506	1.720	0.07	100	98	5	0.0010	0.8632	2.97	0.0072			2.66714	0.38	10
Copper	0.8506	1.720	0.07	100	91	40	0.0696	0.8632	25.59	0.5005			22.52759	3.18	10
Cyanide	0.8506	1.720	0.07	100	69	30	0.0025	0.8632	25.373	0.0180			22.81814	3.22	10
Lead	0.8506	1.720	0.07	100	80	50	0.0007	0.8632	36.7	0.0052			32.99012	4.65	10
Mercury	0.8506	1.720	0.07	100	50		0.0001	0.8632	-	0.0007			-	-	10
Molybdenum	0.8506	1.720	0.07	100	60		0.0008	0.8632	-	0.0055			-	-	10
Nickel	0.8506	1.720	0.07	100	42	2	0.0021	0.8632	2.78	0.0149			2.48623	0.35	10
Selenium	0.8506	1.720	0.07	100	24		0.0006	0.8632	-	0.0043			-	-	10
Silver	0.8506	1.720	0.07	100	75	13	0.0003	0.8632	10.12	0.0018			9.10220	1.28	10
Zinc	0.8506	1.720	0.07	100	86	40	0.0974	0.8632	27.3	0.7011			23.85116	3.36	10
Phenol	0.8506	1.720	0.07	100	8		0.100	0.8632	-	0.7173			-	-	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.

(Q_{potw}) POTW's average influent flow in MGD.

(Q_{dig}) Sludge flow to digester in MGD.

(R_{potw}) Removal efficiency across POTW as percent.

(C_{crit}) Anaerobic digester threshold inhibition level in mg/l.

(Q_{dom}) Domestic/commercial background flow in MGD.

(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.

(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).

(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).

(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.

(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.

$F_{(sorp)}$ Fraction of overall removal due to sorption as a percent

(SF) Safety factor as a percent.

8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{crit} * Q_{dig}}{R_{potw} * F_{sorp}}$$

$$L_{ind} = L_{hw} * (1-SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

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TABLE A6

Local Limits Determination Based on Chronic Water Quality Standards

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE									MAXIMIM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	Avg POTW Flow (MGD) (Qpotw)	20yr Avg POTW Flow (MGD) (Qmax)	River 95% Exceedence Flow (MGD)	Nonvolatile Removal Fraction (%)	Average Removal Efficiency (%) (Ravg)	Chronic WQBEL (mg/l) (Ccrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Available Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8506	1.720	4.5	1.7	100	93	0.016	0.0010	0.8632	3.19	0.007			2.863	0.4	10
Cadmium	0.8506	1.720	4.5	1.7	100	52	0.004	0.0001	0.8632	0.12	0.001			0.110	0.02	10
Chromium	0.8506	1.720	4.5	1.7	100	98	0.094	0.0010	0.8632	74.81	0.007			67.322	9	10
Copper	0.8506	1.720	4.5	1.7	100	91	0.011	0.0696	0.8632	1.80	0.501			0.849	0.1	25
Cyanide	0.8506	1.720	4.5	1.7	100	69	0.006	0.0025	0.8632	0.27	0.018			0.228	0.03	10
Lead	0.8506	1.720	4.5	1.7	100	80	0.094	0.0007	0.8632	6.60	0.005			5.939	1	10
Mercury	0.8506	1.720	4.5	1.7	100	50	0.0000013	0.0001	0.8632	0.00004	0.0007			-0.001	-0.0001	10
Molybdenum	0.8506	1.720	4.5	1.7	100	60	3.600	0.0008	0.8632	129.06	0.006			116.146	16	10
Nickel	0.8506	1.720	4.5	1.7	100	42	0.047	0.0021	0.8632	1.16	0.015			1.031	0	10
Selenium	0.8506	1.720	4.5	1.7	100	24	0.006	0.0006	0.8632	0.11	0.004			0.093	0.01	10
Silver	0.8506	1.720	4.5	1.7	100	75	0.001	0.0003	0.8632	0.080	0.002			0.070	0.01	10
Zinc	0.8506	1.720	4.5	1.7	100	86	0.2	0.0974	0.8632	20.87	0.701			18.080	2.5	10
Phenol	0.8506	1.720	4.5	1.7	100	8	0.51	0.100	0.8632	7.98	0.717			6.465	0.9	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.

(Q_{potw}) POTW's average influent flow in MGD.

(Q_{95ex}) MDEQ Designated 95% exceedence flow for receiving stream (MGD)

(Ravg) Average removal efficiency across POTW as percent.

(C_{crit}) State chronic water quality standard for a particular pollutant in mg/l. Reference: EGLE Interoffice Memo 11/25/2019

(Q_{dom}) Domestic/commercial background flow in MGD.

(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.

(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).

(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).

(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.

(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.

$F_{(nonvol)}$ Fraction of overall POTW removal not due to volatilization as a percent

(SF) Safety factor as a percent.

(Q_{max}) Maximim permitted POTW flow

8.337 Unit conversion factor

$$C_{crit} = \frac{C_{std} * (Q_{max} + 25\%*Q_{95ex})}{Q_{max}}$$

$$L_{hw} = \frac{8.337 * Q_{potw} * C_{crit}}{(1 - R_{avg} * F_{nonvol})}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

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TABLE A7

Local Limits Determination Based on Acute Water Quality Standards

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE									MAXIMIM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	Avg POTW Flow (MGD) (Qpotw)	Max POTW Flow (MGD) (Qmax)	River 95% Exceedence Flow (MGD)	Nonvolatile Removal Fraction (%)	Average Removal Efficiency (%) (Ravg)	Acute WQBEL (mg/l) (Ccrit)	Domestic & Commercial Conc. (mg/l) (Cdom)	Domestic & Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8506	1.720	4.5	1.7	100	93	0.680	0.0010	0.8632	135.5	0.0074			121.98	17	10
Cadmium	0.8506	1.720	4.5	1.7	100	52	0.012	0.00010	0.8632	0.361	0.0007			0.32	0.05	10
Chromium	0.8506	1.720	4.5	1.7	100	98	1.3	0.001	0.8632	1035	0.0072			931.14	131	10
Copper	0.8506	1.720	4.5	1.7	100	91	0.029	0.0696	0.8632	4.74	0.5005			3.77	0.53	10
Cyanide	0.8506	1.720	4.5	1.7	100	69	0.044	0.003	0.8632	2.035	0.0180			1.81	0.26	10
Lead	0.8506	1.720	4.5	1.7	100	80	0.8	0.0007	0.8632	58	0.0052			51.85	7.3	10
Mercury	0.8506	1.720	4.5	1.7	100	50	-	0.0001	0.8632	-	0.0007			-	-	10
Molybdenum	0.8506	1.720	4.5	1.7	100	60	58	0.0008	0.8632	2079	0.0055			1871.32	264	10
Nickel	0.8506	1.720	4.5	1.7	100	42	0.8	0.0021	0.8632	19	0.0149			16.90	2.4	10
Selenium	0.8506	1.720	4.5	1.7	100	24	0.120	0.00060	0.8632	2.266	0.0043			2.04	0.29	10
Silver	0.8506	1.720	4.5	1.7	100	75	0.022	0.00025	0.8632	1.262	0.0018			1.13	0.16	10
Zinc	0.8506	1.720	4.5	1.7	100	86	0.360	0.097	0.8632	36	0.7011			31.49	4.4	10
Phenol	0.8506	1.720	4.5	1.7	100	8	6.800	0.100	0.8632	106	0.7173			95.04	13.4	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.
(Q_{potw}) POTW's average influent flow in MGD.
(Q_{95ex}) MDEQ Designated 95% exceedence flow for receiving stream (MGD)
(Ravg) Average removal efficiency across POTW as percent.
(C_{crit}) State acute water quality standard for a particular pollutant in mg/l. Reference: EGLE Interoffice Memo 11/25/2019
(Q_{dom}) Domestic/commercial background flow in MGD.
(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.
(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).
(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).
(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.
(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.
$F_{(nonvol)}$ Fraction of overall POTW removal not due to volatilization as a percent
(SF) Safety factor as a percent.
(Q_{max}) Maximim permitted POTW flow
8.337 Unit conversion factor

$$C_{crit} = C_{std}$$

$$L_{hw} = \frac{8.337 * Q_{potw} * C_{crit}}{(1 - R_{avg} * F_{nonvol})}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

TABLE A8

Local Limits Determination Based on USEPA 503 Sludge Regulations

Pollutant	ENVIRONMENTAL CRITERIA AND PROCESS DATA BASE										MAXIMUM LOADING				INDUSTRIAL LOADING		
	IU Pollut. Flow (MGD) (Qind)	POTW Flow (MGD) (Qpotw)	Sludge Flow (MGD) (Qsldg)	Sorption Removal Fraction (%)	Percent Solids (%) (PS)	Average Digester Removal (%)	Maximum Removal Efficiency (%) (Rmax)	503 Sludge Criteria (mg/kg) (Cslcrit)	Domestic and Commercial Conc. (mg/l) (Cdom)	Domestic and Commercial Flow (MGD) (Qdom)	Allowable Headworks (lbs/day) (Lhw)	Domestic/ Commercial (lbs/day) (Ldom)			Allowable Loading (lbs/day) (Lind)	Local Limit (mg/l) (Cind)	Safety Factor (%) (SF)
Arsenic	0.8506	1.720	0.0025	100	3.0	0	93	41	0.0010	0.8632	0.0272	0.0074			0.017	0.002	10
Cadmium	0.8506	1.720	0.0025	100	3.0	0	52	39	0.0001	0.8632	0.0459	0.0007			0.041	0.006	10
Chromium	0.8506	1.720	0.0025	100	3.0	0	98		0.001	0.8632	-	0.0072			-	-	10
Copper	0.8506	1.720	0.0025	100	3.0	0	91	1500	0.070	0.8632	1.0140	0.5005			0.412	0.06	10
Cyanide	0.8506	1.720	0.0025	100	3.0	0	69		0.003	0.8632	-	0.0180			-	-	10
Lead	0.8506	1.720	0.0025	100	3.0	0	80	300	0.0007	0.8632	0.2325	0.0052			0.204	0.03	10
Mercury	0.8506	1.720	0.0025	100	3.0	0	60	17	0.0001	0.8632	0.0175	0.0007			0.015	0.002	10
Molybdenum	0.8506	1.720	0.0025	100	3.0	0	17	75	0.0008	0.8632	0.2684	0.0055			0.236	0.033	10
Nickel	0.8506	1.720	0.0025	100	3.0	0	42	420	0.0021	0.8632	0.6167	0.0149			0.540	0.076	10
Selenium	0.8506	1.720	0.0025	100	3.0	0	24	100	0.0006	0.8632	0.2562	0.0043			0.226	0.032	10
Silver	0.8506	1.720	0.0025	100	3.0	0	75		0.0003	0.8632	-	0.0018			-	-	10
Zinc	0.8506	1.720	0.0025	100	3.0	0	86	2800	0.0974	0.8632	2.0180	0.7011			1.115	0.157	10
Phenol	0.8506	1.720	0.0025	100	3.0	0	8		0.0997	0.8632	-	0.7173			-	-	10

(Q_{ind}) Industrial User total plant discharge flow in Million Gallons per Day (MGD) that contains a particular pollutant.

(Q_{potw}) POTW's average influent flow in MGD.

(Q_{sldg}) Sludge flow to disposal in MGD.

(PS) Percent solids of sludge to disposal.

(R_{max}) Maximum removal efficiency across POTW as a percent.

(C_{slcrit}) 503 sludge criteria in mg/kg dry sludge.

(Q_{dom}) Domestic/commercial background flow in MGD.

(C_{dom}) Domestic/commercial background concentration for a particular pollutant in mg/l.

(L_{hw}) Maximum allowable headworks pollutant loading to the POTW in pounds per day (lbs/day).

(L_{dom}) Domestic/commercial background loading to the POTW for a particular pollutant in pounds per day (lbs/day).

(L_{ind}) Maximum allowable industrial loading to the POTW in pounds per day.

(C_{ind}) Industrial allowable local limit for a given pollutant in mg/l.

$F_{(sorp)}$ Fraction of overall removal due to sorption as a percent

$R_{(dig)}$ Average removal fraction of digester as a percent

(SF) Safety factor as a percent.

8.337 Unit conversion factor

$$L_{hw} = \frac{8.337 * C_{slcrit} * (PS/100) * Q_{sldg}}{R_{max} * F_{sorp} * (1 - R_{dig})}$$

$$L_{ind} = L_{hw} * (1\text{-}SF/100) - L_{dom}$$

$$C_{ind} = \frac{L_{ind}}{Q_{ind} * 8.337}$$

TABLE A9
Summary of Determined Limits

				Arsenic	Cadmium	Chromium	Copper	Cyanide	Lead	Mercury	Molybdenum	Nickel	Selenium	Silver	Zinc	Phenol
NPDES Monthly Limits	Allowable Headworks Load	(lbs/day)	(Lhw)	-	-	-	1.799	-	-	0.0000574	-	-	0.108	-	-	-
	Domestic/Commercial	(lbs/day)	(Ldom)	0.007	0.001	0.007	0.501	0.018	0.005	0.0007	0.006	0.015	0.004	0.0018	0.701	0.717
	Allowable Industrial Load	(lbs/day)	(Lind)	-	-	-	0.849	-	-	-0.00067	-	-	0.093	-	-	-
	Local Limit (Cind)	(mg/l)	(Cind)	-	-	-	0.12	-	-	-0.0001	-	-	0.013	-	-	-
Secondary Treatment Inhibition	Allowable Headworks Load	(lbs/day)	(Lhw)	1.418	7.141	4.023	6.369	1.434	8.604	0.45283	-	7.170	-	1.920	36.615	248.0
	Domestic/Commercial	(lbs/day)	(Ldom)	0.007	0.001	0.007	0.501	0.018	0.005	0.0007197	0.006	0.015	0.004	0.0018	0.701	0.717
	Allowable Industrial Load	(lbs/day)	(Lind)	1.269	6.427	3.613	4.276	1.273	7.738	0.40683	-	6.438	-	1.727	32.252	222.5
	Local Limit (Cind)	(mg/l)	(Cind)	0.179	0.906	0.510	0.603	0.179	1.091	0.05737	-	0.908	-	0.243	4.548	31.4
Nitrification Inhibition	Allowable Headworks Load	(lbs/day)	(Lhw)	298.981	156.589	795.850	78.510	23.128	35.132	-	-	12.362	-	-	49.685	62.6
	Domestic/Commercial	(lbs/day)	(Ldom)	0.007	0.001	0.007	0.501	0.018	0.005	0.00072	0.006	0.015	0.004	0.002	0.701	0.717
	Allowable Industrial Load	(lbs/day)	(Lind)	269.076	140.929	716.258	70.159	20.798	31.614	-	-	11.111	-	-	44.016	55.6
	Local Limit (Cind)	(mg/l)	(Cind)	37.671	19.730	100.277	9.822	2.912	4.426	-	-	1.556	-	-	-	7.8
Digester Inhibition	Allowable Headworks Load	(lbs/day)	(Lhw)	0.629	1.114	2.971	25.587	25.373	36.661	-	-	2.779	-	10.116	27.280	-
	Domestic/Commercial	(lbs/day)	(Ldom)	0.007	0.001	0.007	0.501	0.018	0.005	0.0007	0.0055	0.015	0.004	0.002	0.701	0.717
	Allowable Industrial Load	(lbs/day)	(Lind)	0.559	1.002	2.667	22.528	22.818	32.990	-	-	2.486	-	9.102	23.851	-
	Local Limit (Cind)	(mg/l)	(Cind)	0.079	0.141	0.376	3.177	3.218	4.652	-	-	0.351	-	1.283	3.363	-
Chronic Water Quality Standards	Allowable Headworks Load	(lbs/day)	(Lhw)	3.189	0.123	74.810	1.799	0.273	6.605	0.00004	129.057	1.162	0.108	0.0803	20.868	7.98
	Domestic/Commercial	(lbs/day)	(Ldom)	0.007	0.001	0.007	0.501	0.018	0.005	0.00072	0.006	0.015	0.004	0.0018	0.701	0.717
	Allowable Industrial Load	(lbs/day)	(Lind)	2.863	0.110	67.322	0.849	0.228	5.939	-0.00069	116.146	1.031	0.093	0.07047	18.080	6.46
	Local Limit (Cind)	(mg/l)	(Cind)	0.404	0.016	9.493	0.120	0.032	0.837	-0.00010	16.3774	0.145	0.013	0.0099	2.549	0.91
Acute Water Quality Standards	Allowable Headworks Load	(lbs/day)	(Lhw)	135.538	0.361	1034.605	4.743	2.035	57.617	-	2079.248	18.790	2.266	1.262	35.773	106.4
	Domestic/Commercial	(lbs/day)	(Ldom)	0.007	0.001	0.007	0.501	0.018	0.005	0.0007197	0.006	0.015	0.004	0.0018	0.701	0.717
	Allowable Industrial Load	(lbs/day)	(Lind)	121.977	0.325	931.137	3.768	1.814	51.850	-	1871.318	16.896	2.035	1.134	31.495	95.0
	Local Limit (Cind)	(mg/l)	(Cind)	17.200	0.046	131.297	0.531	0.256	7.311	-	263.8694	2.382	0.287	0.160	4.441	13.4
USEPA 503 Sludge Regulations	Allowable Headworks Load	(lbs/day)	(Lhw)	0.027	0.046	-	1.014	-	0.232	0.01747	0.268	0.617	0.256	-	2.018	-
	Domestic/Commercial	(lbs/day)	(Ldom)	0.007	0.001	0.007	0.501	0.018	0.005	0.0007197	0.006	0.015	0.004	0.0018	0.701	0.717
	Allowable Industrial Load	(lbs/day)	(Lind)	0.017	0.041	-	0.412	-	0.204	0.01501	0.236	0.540	0.226	-	1.115	-
	Local Limit (Cind)	(mg/l)	(Cind)	0.002	0.006	-	0.058	-	0.029	0.00212	0.0333	0.076	0.032	-	0.157	-

TABLE A10
Allowable Headworks Load

			Arsenic	Cadmium	Chromium	Copper	Cyanide	Lead	Mercury	Molybdenum	Nickel	Selenium	Silver	Zinc	Phenol
NPDES Monthly Limit ML	(lbs/day)	(Lhw)	-	-	-	1.799	-	-	0.0001	-	-	**0.108**	-	-	-
Secondary Treatment Inhibition SI	(lbs/day)	(Lhw)	1.42	7.14	4.02	6.369	1.434	8.604	0.453	-	7.2	-	1.920	36.61	248
Nitrification Inhibition NI	(lbs/day)	(Lhw)	298.98	156.59	795.85	78.510	23.128	35.132	-	-	12.4	-	-	49.69	62.59
Anaerobic Digester Inhibition DI	(lbs/day)	(Lhw)	0.629	1.114	**2.971**	25.587	25.373	36.661	-	-	2.779	-	10.116	27.28	-
Chronic Water Quality Standards C	(lbs/day)	(Lhw)	3.19	0.123	74.81	1.799	**0.273**	6.605	**0.00004**	129	1.162	**0.108**	**0.080**	20.9	**7.98**
Acute Water Quality Standards A	(lbs/day)	(Lhw)	135.54	0.361	1034.6	4.743	2.035	57.617	-	2079	19	2.266	1.262	35.77	106
USEPA 503 Sludge Regulations BS	(lbs/day)	(Lhw)	**0.027**	**0.046**	-	**1.014**	-	**0.232**	0.017	**0.268**	**0.617**	0.256	-	**2.02**	-
MAHL	(lbs/day)		0.027	0.046	2.971	1.014	0.273	0.232	0.00004	0.268	0.617	0.108	0.080	2.018	7.980
Basis			BS	BS	DI	BS	C	BS	C	BS	BS	C/ML	C	BS	C

TABLE A11
Allowable Industrial Load

			Arsenic	Cadmium	Chromium	Copper	Cyanide	Lead	Mercury	Molybdenum	Nickel	Selenium	Silver	Zinc	Phenol
NPDES Monthly Limit ML	(lbs/day)	(Lhw)	-	-	-	0.849	-	-	**-0.001**	-	-	**0.093**	-	-	-
Secondary Treatment Inhibition SI	(lbs/day)	(Lind)	1.269	6.427	3.613	4.276	1.273	7.738	0.407	-	6.438	-	1.727	32.252	222.5
Nitrification Inhibition NI	(lbs/day)	(Lind)	269.076	140.929	716.258	70.159	20.798	31.614	-	-	11.111	-	-	44.016	55.6
Anaerobic Digester Inhibition DI	(lbs/day)	(Lind)	0.559	1.002	**2.667**	22.528	22.818	32.990	-	-	2.486	-	9.102	23.851	-
Chronic Water Quality Standards C	(lbs/day)	(Lind)	2.863	0.110	67.322	0.849	**0.228**	5.939	**-0.001**	116.146	1.031	**0.093**	**0.070**	18.080	**6.46**
Acute Water Quality Standards A	(lbs/day)	(Lind)	121.977	0.325	931.137	3.768	1.814	51.850	-	1871.318	16.896	2.035	1.134	31.495	95.0
USEPA 503 Sludge Regulations BS	(lbs/day)	(Lind)	**0.017**	**0.041**	-	**0.412**	-	**0.204**	0.015	**0.236**	**0.540**	0.226	-	**1.115**	-
MAIL	(lbs/day)		0.017	0.041	2.667	0.412	0.228	0.204	-0.001	0.236	0.540	0.093	0.070	1.115	6.46
Basis			BS	BS	DI	BS	C	BS	C	BS	BS	C/ML	C	BS	C

TABLE A12
Calculated Standard Local Limits - Uniform Allocation

		Arsenic	Cadmium	Chromium	Copper	Cyanide	Lead	Mercury†	Molybdenum	Nickel	Selenium	Silver	Zinc	Phenol	Bis(2-ethyl hexylphthalate) *
Daily Maximum	(mg/L)	0.002	0.006	0.37	0.058	0.03	0.029	0.000	0.033	0.076	0.013	0.010	0.156	0.912	0.025
	(ug/L)	2	6	373	58	32	29	0	33	76	13	10	156	912	25
Basis		BS	BS	DI	BS	C	BS	C	BS	BS	C/ML	C	BS	C	ML

†Mercury Local Limit will be Non-Detect

*Bis(2-ethylhexylphthalate) Local Limit is based on dividing the total MAHL by the Total Influent Flow

TABLE A13 C_{crit} VALUES

Highlighted Values Were Used In This Local Limits Determination

The following tables are taken from EPA's Local Limits Development Guidance Appendixes, and Guidance Manual for Preventing Interference at POTW

	Appendix R tables				Appendix V
	Removal Efficiencies				Domestic
	Primary	Activated sludge	Trickling Fltr	Tertiary	Level
Ag	20%	75%	66%	62%	0.019
As		45%			0.007
Cd*	15%	67%	68%	50%	0.008
CN	27%	69%	59%	66%	0.082
Cr	27%	82%	55%	72%	0.034
Cu	22%	86%	55%	85%	0.061
Hg	10%	60%	50%	67%	0.002
Mo					
Ni	14%	42%	29%	17%	0.047
Pb	57%	61%	55%	52%	0.058
Se		33%			
Zn	27%	79%	67%	78%	0.231
Phenol					

Appendix G and Table 2-1 tables			
Inhibition Levels			
Activated sludge	Trickling Fltr	Nitrification	Digestion
0.25			13
0.1		1.5	1
1		5.2	1
0.1	30	0.5	30
1		1	5
1		0.48	40
0.1			
1		0.5	2
1		0.5	50
5		0.5	40
50		4	

Site Specific Data from Cadillac WWTP Sampling		
WWTF % Removals		
	Primary	Total Plant
Ag	-87%	0%
As	-112%	93%
Cd	-101%	52%
CN	0%	-590%
Cr	-256%	98%
Cu	-125%	91%
Hg	-217%	0%
Mo	-24%	17%
Ni	-100%	-8%
Pb	-67%	80%
Se	-44%	24%
Zn	-96%	86%
Phenol	-189%	8%

	Part 503 / 24 Land Application of Sludge			
	"Table 3"	Ceiling Concentration	Cumulative Pollutant Loading Rates	Annual Pollutant Loading Rate
As	41	75	37	1.8
Cd*	39	85	35	1.7
Cu*	1500	4300	1338	67
Hg	17	57	15	0.76
Mo	-	75	-	-
Ni*	420	420	375	19
Pb*	300	840	268	13
Se	100	100	89	4.5
Zn*	2800	7500	2498	125

City of Cadillac WWTP
Compatibles MAHL Calculations

Flows Summary

$Q_{\text{City domestic, est.}}$	0.86	MGD
$Q_{\text{industrial}}$	0.857	MGD
$Q_{\text{POTW, Total}}$	1.72	MGD

TABLE A14 - BOD MAHL CALCULATIONS

Allowable BOD Loading - NPDES Permit Basis

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW}}{1 - R_{Avg}}$$

$C_{\text{NPDES, Monthly avg BOD5 (Jun-Sept)}}$ =	7	mg/L
$R_{\text{Annual Avg, BOD5 (Jun-Sept)}}$ =	97.7%	
Q_{POTW} =	1.72	MGD
$L_{\text{NPDES, BOD5 (Monthly Avg, Jun-Sept)}}$ =	4,366	lb/day
$C_{\text{NPDES, monthly avg BOD5 (Oct-Nov)}}$ =	11	mg/L
$R_{\text{Avg, BOD5 (Oct-Nov) =}}$	97.5%	
Q_{POTW} =	1.72	MGD
$L_{\text{NPDES, BOD5 (Monthly Avg, Oct-Nov)}}$ =	6,312	lb/day
$C_{\text{NPDES, monthly avg BOD5 (Dec-Mar)}}$ =	25	mg/L
$R_{\text{Avg, BOD5 (Dec-Mar) =}}$	97.2%	
Q_{POTW} =	1.72	MGD
$L_{\text{NPDES, BOD5 (Monthly Avg, Dec-Mar)}}$ =	12,808	lb/day
$C_{\text{NPDES, monthly avg BOD5 (Apr-May)}}$ =	17	mg/L
$R_{\text{Avg, BOD5 (Apr-May) =}}$	96.8%	
Q_{POTW} =	1.72	MGD
$L_{\text{NPDES, BOD5 (Monthly Avg, Apr-May)}}$ =	7,621	lb/day

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW,Max}}{1 - R_{Min}}$$

$C_{NPDES,\ Daily\ Max\ BOD5\ (Jun\text{-}Sept)}$ =	10	mg/L
$R_{Min\ BOD5\ (Jun\text{-}Sept)}$ =	91.6%	
$Q_{POTW,\ Max}$ =	4.50	MGD
$L_{NPDES,\ BOD5\ (Daily\ Max,\ Jun\text{-}Sept)}$ =	4,468	lb/day
$C_{NPDES,\ Daily\ Max\ BOD5\ (Oct\text{-}Nov)}$ =	17	mg/L
$R_{Min,\ BOD5\ (Oct\text{-}Nov)}$ =	91.1%	
$Q_{POTW,\ Max}$ =	4.50	MGD
$L_{NPDES,\ BOD5\ (Daily\ Max,\ Oct\text{-}Nov)}$ =	7,169	lb/day
$C_{NPDES,\ Daily\ Max\ BOD5\ (Dec\text{-}Mar)}$ =	34	mg/L
$R_{Min,\ BOD5\ (Dec\text{-}Mar)}$ =	91.5%	
$Q_{POTW,\ Max}$ =	4.50	MGD
$L_{NPDES,\ BOD5\ (Daily\ Max,\ Dec\text{-}Mar)}$ =	15,012	lb/day
$C_{NPDES,\ Daily\ Max\ BOD5\ (Apr\text{-}May)}$ =	25	mg/L
$R_{Min,\ BOD5\ (Apr\text{-}May)}$ =	90.6%	
$Q_{POTW,\ Max}$ =	4.50	MGD
$L_{NPDES,\ BOD5\ (Daily\ Max,\ Apr\text{-}May)}$ =	9,981	lb/day

Allowable BOD Loading - Design Capacity Basis

1975 O&M Manual - Average WWTP BOD Capacity, prior to fine bubble addition in	3,300	lb BOD/day
Design basis check after 1996 improvements:		
Aeration Volume	330,000	gals
	118,705	cf
BOD loading (per 10 States Standards)	40	lbs/1000 cf
	4,748	lb BOD/day to secondary treatment
Maximum Allowable Headworks Loading =	**4,366**	**lb/day based on NPDES Monthly Avg (Jun-Sept)**

Determine Domestic Loading & MAIL

Average domestic BOD concentration 202 mg/L

Total Maximum Allowable Headworks Loading	4366	lb BOD5/day
Domestic Loading	1452	lb BOD5/day
Reserve Loading	50	lb BOD5/day
MAIL BOD	2864	lb BOD5/day

Uniform Allocation of BOD MAIL to Industrial Users	401	mg/L
Proposed Local Limit	**400**	**mg/L**

TABLE A15 - TSS MAHL CALCULATIONS

TSS Loading - NPDES Permit Basis

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW}}{1 - R_{Avg}}$$

$C_{NPDES, \text{ Monthly avg TSS (Jun-Sept)}}$ =	20	mg/L
$R_{\text{Avg, TSS (June-Sept)}}$ =	97.0%	
Q_{POTW} =	1.72	MGD
$L_{NPDES, \text{ TSS (Jun-Sept)}}$ =	9,563	lb TSS/day
$C_{NPDES, \text{ monthly avg TSS (Oct-May)}}$ =	30	mg/L
$R_{\text{Avg, TSS (Oct-May)}}$ =	96.3%	
Q_{POTW} =	1.72	MGD
$L_{NPDES, \text{ TSS (Monthly Avg, Oct-May)}}$ =	11,631	lb TSS/day

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW,Max}}{1 - R_{Min}}$$

$C_{NPDES, \text{ 7-day Max TSS (Jun-Sept)}}$ =	30	mg/L
$R_{\text{Min 7-day (Jun-Sept)}}$ =	84.5%	
$Q_{POTW, \text{ Max}}$ =	4.50	MGD
$L_{NPDES, \text{ BOD5 (7-day Max, Jun-Sept)}}$ =	7,264	lb TSS/day
$C_{NPDES, \text{ 7-day avg TSS (Oct-May)}}$ =	45	mg/L
$R_{\text{Min 7-day TSS (Oct-May)}}$ =	88.2%	
$Q_{POTW, \text{ Max (Oct-May)}}$ =	4.50	MGD
$L_{NPDES, \text{ TSS 7-day, Oct-May}}$ =	14,312	lb TSS/day

TSS Capacity - Design Capacity Basis

1972 Design Basis - Average TSS Capacity prior to adding additional primary and secondary clarification and upgrading tertiary treatment	4200	lb TSS/day; not applicable today
Maximum Allowable Headworks Loading TSS =	**7,264**	**lb/day based on 7-day Max Jun-Sept NPDES**

Determine Domestic Loading & MAIL

Average domestic TSS concentration 199 mg/L

Total Maximum Allowable Headworks Loading (with 10% SF)	6537	lb TSS/day
Domestic Loading	1430	lb TSS/day
Reserve Loading	100	lb TSS/day
MAIL TSS	5007	lb TSS/day

Uniform Allocation of TSS MAIL to Industrial Users	701	mg/L
Proposed Local Limit	**700**	**mg/L**

TABLE A16 - AMMONIA NITROGEN MAHL CALCULATIONS

Ammonia Loading - NPDES Permit Basis

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW}}{1 - R_{Avg}}$$

$C_{NPDES,\ Monthly\ Avg\ Ammonia\ (Jun\text{-}Sept)}$ =	0.9	mg/L
$R_{Avg,\ Ammonia\ (Jun\text{-}Sept)}$ =	98.5%	
$Q_{POTW,\ Avg}$ =	1.72	MGD
$L_{NPDES,\ Ammonia\ (Jun\text{-}Sept)}$ =	861	lb NH3/day
$C_{NPDES,\ Monthly\ Avg\ Ammonia\ (Oct\text{-}Nov)}$ =	1.6	mg/L
$R_{Avg,\ Ammonia\ (Oct\text{-}Nov)}$ =	97.7%	
$Q_{POTW,\ Avg}$ =	1.72	MGD
$L_{NPDES,\ Ammonia\ (Oct\text{-}Nov)}$ =	998	lb NH3/day
$C_{NPDES,\ Monthly\ Avg\ Ammonia\ (Dec\text{-}Mar)}$ =	3.7	mg/L
$R_{Avg,\ Ammonia\ (Dec\text{-}Mar)}$ =	96.3%	
$Q_{POTW,\ Avg}$ =	1.72	MGD
$L_{NPDES,\ Ammonia\ (Dec\text{-}Mar)}$ =	1434	lb NH3/day
$C_{NPDES,\ Monthly\ Avg\ Ammonia\ (Apr\text{-}May)}$ =	4.8	mg/L
$R_{Avg,\ Ammonia\ (Apr\text{-}May)}$ =	97.0%	
$Q_{POTW,\ Avg}$ =	1.72	MGD
$L_{NPDES,\ Ammonia\ (Apr\text{-}May)}$ =	2295	lb NH3/day

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW}}{1 - R_{Min}}$$

$C_{NPDES,\ Daily\ Max\ Ammonia\ (June\text{-}Sept)}$ =	2.0	mg/L
$R_{Min,\ Ammonia\ (June\text{-}Sept)}$ =	96.1%	
$Q_{POTW,\ Max}$ =	4.50	MGD
$L_{NPDES,\ Ammonia\ (June\text{-}Sept)}$ =	1925	lb NH3/day
$C_{NPDES,\ Daily\ Max\ Ammonia\ (Oct\text{-}Nov)}$ =	2.8	mg/L
$R_{Min,\ Ammonia\ (Oct\text{-}Nov)}$ =	78.8%	
$Q_{POTW,\ Max}$ =	4.50	MGD
$L_{NPDES,\ Ammonia\ (Oct\text{-}Nov)}$ =	496	lb NH3/day

Ammonia Design Capacity Analysis

1972 O&M Manual WWTP design basis for ammonia	751	lb/day, max day
Ammonia Maximum Allowable Headworks Loading =	**496**	**lb NH3/day based on max day NPDES permit, Oct-Nov**

Determine Domestic Loading & MAIL

Average domestic Ammonia concentration	33	mg/L

Total Maximum Allowable Headworks Loading	471	lb NH3/day
Domestic Loading	238	lb NH3/day
Reserve Loading	0	lb NH3/day
MAIL Ammonia	233	lb NH3/day

Uniform Allocation of Ammonia MAIL to Industrial Users	33	mg/L
Proposed Local Limit	**33**	**mg/L**

TABLE A17 - TOTAL PHOSPHORUS MAHL CALCULATIONS

Total Phosphorus Loading - NPDES Permit Basis

$$L_{NPDES} = \frac{8.34 * C_{NPDES} * Q_{POTW}}{1 - R_{Avg}}$$

$C_{NPDES,\ Monthly\ Avg\ (May\text{-}March)}$ =	0.5	mg/L
$R_{Avg,\ Total\ Phosphorus\ (May\text{-}March)}$ =	94.7%	
Q_{POTW} =	1.72	MGD
$L_{NPDES,\ Total\ Phosphorus}$ =	135	lb TP/day
$C_{NPDES,\ Monthly\ Avg\ (Apr)}$ =	0.65	mg/L
$R_{Avg,\ Total\ Phosphorus\ (Apr)}$ =	95.2%	
Q_{POTW} =	1.72	MGD
$L_{NPDES,\ Total\ Phosphorus}$ =	194	lb TP/day

Allowable Phosphorus Loading - Design Capacity Basis

1972 O&M Manual Design Basis for Phosphorus 667.2 lb TP/day

Maximum Allowable Headworks Loading for Total Phosphorus = 135 lbs TP/day, based on NPDES

Determine Domestic Loading & MAIL

Average domestic Total Phosphorus concentration 5.5 mg/L

Total Maximum Allowable Headworks Loading	129	lb TP/day
Domestic Loading	40	lb TP/day
Reserve Loading	0	lb TP/day
Total Phosphorus MAIL	89	lb TP/day

Uniform Allocation of Phosphorus MAIL to Industrial Users 12 mg/L

Proposed Local Limit 12 mg/L

Cadillac Castings Inc
Categorical Limit Calculations
Table A18

Regulated under 40 CFR Part 464.36

Processes include:

Dust Collection Scrubbing Operations (c), primarily ductile iron castings

Categorical Limits:

Pollutant	Max Day	Monthly Avg
	lbs/ billion SCF of air scrubbed	
Copper	0.218	0.12
Lead	0.398	0.195
Zinc	0.736	0.278
Total Phenols	0.646	0.225
TTO	2.04	0.664
FOG	22.5	7.51

Dust Scrubber Flow	52,000	scfm, each
Number of scrubbers	2	
Total flow, SCF/day	149,760,000	
Total flow, billion SCF/day	0.150	

Categorical Limits for CCI, based on scrubber flow:

Pollutant	**Max Day**	**Monthly Avg**
	lbs/ day	
Copper	**0.03**	**0.02**
Lead	**0.06**	**0.03**
Zinc	**0.11**	**0.04**
Total Phenols	**0.10**	**0.03**
TTO	**0.31**	**0.10**
FOG	**3.37**	**1.12**

Melting Furnace Scrubbing Operations (f), primarily ductile iron

Categorical Limits:

Pollutant	Max Day	Monthly Avg
	lbs/ billion SCF of air scrubbed	
Copper	1.02	0.561
Lead	1.86	0.911
Zinc	3.44	1.3
Total Phenols	3.01	1.05
TTO	8.34	2.73
FOG	105	35

Melt Furnace Scrubber Flow	30,000
Number of scrubbers	1
Total flow, SCF/day	43,200,000
Total flow, billion SCF/day	0.043

Categorical Limits for CCI, based on scrubber flow:

Pollutant	**Max Day**	**Monthly Avg**
	lbs/ day	
Copper	**0.15**	**0.08**
Lead	**0.28**	**0.14**
Zinc	**0.52**	**0.19**
Total Phenols	**0.45**	**0.16**
TTO	**1.25**	**0.41**
FOG	**15.72**	**5.24**

Domestic/Background Samples

		Grab Sample Parameters		Metals (Total)											Compatible Pollutants				Organic Pollutants								
Collection		**Cyanide, Total**	**Cyanide, Available**	**Mercury**	**Arsenic**	**Cadmium**	**Chromium**	**Copper**	**Lead**	**Molybdenum**	**Nickel**	**Selenium**	**Silver**	**Zinc**	**NH_3-N**	**BOD_5**	**TSS**	**Phosphorus**	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**Location**	**mg/L**	**µg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	Domestic #1	0.0052	16.8	0.00045	0.0054	0.0016	0.0151	0.694	<0.001	0.0055	0.0192	<0.005	0.0038	1.37	47.4	620	4920	24.5	<0.0001	<0.0048	<0.005	<0.005	<0.005	0.0067	<0.010	0.0416	<0.00051
10/5/2022	Domestic #2	<0.0050	4.6	<0.0002	0.0013	<0.0002	<0.002	0.0962	0.0013	0.0012	0.0027	<0.001	<0.0005	0.125	38.8	186	400	7.0	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.143	<0.00051
10/13/2022	Domestic #1	<0.0050	3.4	0.00074	0.0043	0.00052	0.0158	0.0497	0.13	0.0011	0.0131	<0.001	<0.0005	0.239	10.1	141	414	4.7	<0.000096	<0.0048	<0.005	<0.005	<0.005	0.0054	<0.010	<0.0667	<0.000048
10/13/2022	Domestic #2	<0.0050	4.3	<0.0002	0.0016	<0.00020	<0.002	0.0892	0.001	0.0014	0.0024	<0.001	<0.0005	0.141	34.6	254	216	6.3	<0.000096	<0.0048	<0.005	<0.005	<0.005	0.0054	<0.010	0.149	<0.000048
10/14/2022	Domestic #1	<0.0050	3.5	0.0002	<0.002	0.00049	0.0062	0.255	0.0089	0.0025	0.0064	<0.002	0.0023	0.608	41	410	1280	14.7	<0.00011	<0.0048	<0.005	<0.005	<0.005	0.0066	<0.010	<0.0667	<0.000056
10/14/2022	Domestic #2	<0.0050	4.1	<0.0002	0.0013	<0.00020	<0.002	0.0638	<0.001	<0.001	0.0019	<0.001	<0.0005	0.0809	29.5	116	176	4.4	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/21/2022	Domestic #1	<0.0050	5.3	0.00037	0.0016	0.0005	0.0055	0.249	0.0124	0.0025	0.0064	0.0011	0.0013	0.568	35.9	440	1320	13.2	<0.000098	<0.0048	<0.005	<0.005	<0.005	0.0056	<0.010	<0.0667	<0.000049
10/21/2022	Domestic #2	<0.0050	4.5	<0.0002	0.001	<0.0002	<0.002	0.0681	<0.001	<0.001	0.002	0.0011	<0.0005	0.0897	25.8	138	184	4.8	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.200	<0.000050
10/22/2022	Domestic #1	<0.0050	3.2	<0.0002	<0.001	0.00025	<0.002	0.078	0.0021	0.0017	0.0055	<0.001	<0.0005	0.268	37.8	>750	440	17.6	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0683	<0.000049
10/22/2022	Domestic #2	<0.0050	4.6	<0.0002	<0.001	<0.0002	<0.002	0.0496	<0.001	<0.001	0.002	<0.001	<0.0005	0.0856	30.2	353	75	5.4	<0.00011	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000053
10/25/2022	Domestic #1	<0.0050	4.5	<0.0002	0.0018	0.00052	0.0056	0.263	0.0123	0.0025	0.0056	0.0014	0.0026	0.584	32.5	530	1220	13	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.000049
10/25/2022	Domestic #2	<0.0050	5.9	<0.0002	<0.001	<0.0002	<0.002	0.0504	<0.001	<0.001	0.0014	<0.001	<0.0005	0.0623	39.2	163	141	5.2	<0.000099	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.239	<0.000050
Average - DS 1		0.0030	0.006	0.00034	0.0024	0.0006	0.0082	0.2648	0.0278	0.0026	0.0094	0.0012	0.00175	0.606	34	481.8	1599	15	0.00020	0.0024	0.0025	0.0025	0.0025	0.0049	0.005	0.043	0.00006
Average - DS 2		0.0025	0.005	0.00010	0.0010	0.0001	0.0010	0.070	0.0007	0.0008	0.0021	0.0006	0.00025	0.097	33	201.7	199	6	0.00005	0.0024	0.0025	0.0025	0.0025	0.0029	0.005	0.100	0.00006
Average		**0.0027**	**0.005**	**0.00022**	**0.0017**	**0.0004**	**0.0046**	**0.167**	**0.0143**	**0.0017**	**0.0057**	**0.0009**	**0.00100**	**0.352**	**34**	**341.8**	**899**	**10**	**0.00012**	**0.00240**	**0.00250**	**0.00250**	**0.00250**	**0.00390**	**0.005**	**0.071**	**0.00006**

WWTP Influent

8389800	Grab Sample Parameters		Metals (Total)											Compatible Pollutants				Organic Pollutants								
Collection	Cyanide, Total	Cyanide, Available	Mercury	Arsenic	Cadmium	Chromium	Copper	Lead	Molybdenum	Nickel	Selenium	Silver	Zinc	NH_3-N	BOD_5	TSS	Phosphorus	PCB	Chloroform	1,2-Dichloro benzene	Ethylbenzene	Styrene	Toluene	Xylene	Phenolics	Lindane
Date	mg/L	µg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L
10/5/2022	<0.0050	<10.0	<0.0002	0.0027	0.00038	0.008	0.0757	0.002	0.0047	0.0036	0.0018	<0.0005	0.122	19.7	203	162	4.5	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.144	<0.00051
10/13/2022	<0.0050	3	<0.0002	0.0022	0.00022	0.0049	0.0646	0.0029	0.0077	0.0032	<0.001	<0.0005	0.107	20.6	205	162	3.9	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000048
10/14/2022	<0.0050	2.3	<0.0002	0.0021	0.00025	0.0054	0.0928	0.0045	0.0098	0.0037	0.0016	<0.0005	0.127	23.4	258	180	4.5	<0.00012	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000062
10/21/2022	<0.0050	3.3	<0.0002	0.0025	0.00021	0.0062	0.0626	0.0029	0.0078	0.0033	<0.001	<0.0005	0.114	21	190	156	3.9	<0.00011	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000053
10/22/2022	<0.0050	2.6	<0.0002	0.0023	<0.0002	0.005	0.0531	0.0013	0.0031	0.0032	<0.001	<0.0005	0.0787	24	228	63	4.4	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/25/2022	<0.0050	2.5	<0.0002	0.0021	<0.0002	0.0038	0.047	0.0011	0.0052	0.0028	<0.001	<0.0005	0.0729	24.6	283	70	3	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.000050
Average	0.0025	0.0031	0.0001	0.0023	0.0002	0.0056	0.0660	0.0025	0.0064	0.0033	0.0009	######	0.104	22	228	132	4.0	0.00005	0.0024	0.0025	0.0025	0.0025	0.0025	0.005	0.0546	0.00003

WWTP Primary Effluent

	Grab Sample Parameters		Metals (Total)											Compatible Pollutants			
Collection	**Cyanide, Total**	**Cyanide, Available**	**Mercury**	**Arsenic**	**Cadmium**	**Chromium**	**Copper**	**Lead**	**Molybdenum**	**Nickel**	**Selenium**	**Silver**	**Zinc**	**NH_3-N**	**BOD_5**	**TSS**	**Phosphorus**
Date	**mg/L**	**µg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0050	3.7	<0.0002	0.0042	0.00047	0.0169	0.13	0.0031	0.0064	0.0059	0.0016	0.00062	0.179	22.4	243	248	6.3
10/13/2022	<0.0050	5.1	<0.0002	0.004	0.00031	0.0161	0.114	0.0033	0.0096	0.0056	0.0011	<0.00050	0.154	22.1	218	232	5.2
10/14/2022	<0.0050	5.6	<0.0002	0.0056	0.00054	0.0254	0.208	0.0063	0.0101	0.0076	0.0022	0.00051	0.265	23.9	268	332	8.8
10/21/2022	<0.0050	5.3	0.0014	0.006	0.0006	0.0282	0.199	0.0053	0.0096	0.0084	0.0011	0.00063	0.285	22.8	175	428	13.2
10/22/2022	<0.0050	6.3	<0.0002	0.0033	<0.0002	0.0098	0.0622	0.0018	0.0046	0.0045	<0.001	<0.0005	0.0872	24.1	180	70	4.6
10/25/2022	<0.0050	4.4	<0.0002	0.0063	0.00051	0.0223	0.178	0.0047	0.0072	0.0076	0.0013	0.00054	0.247	26.4	253	380	11.6

Average	0.0025	0.005	0.00032	0.005	0.0004	0.0198	0.149	0.0041	0.0079	0.0066	0.0013	0.0005	0.203	24	223	282	8
Prim Eff % rem	0.0%	-63%	-217%	-112%	-101%	-256%	-125%	-67%	-24%	-100%	-44%	-87%	-96%	-6%	2%	-113%	-105%

Organic Pollutants								
PCB	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
mg/L	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
<0.0001	<0.0048	<0.005	<0.005	<0.005	0.118	<0.010	0.136	<0.00051
<0.000096	<0.0048	<0.005	<0.005	<0.005	0.103	<0.010	0.184	<0.000048
<0.000099	<0.0048	<0.005	<0.005	<0.005	0.127	<0.010	<0.0667	<0.000050
<0.00011	<0.0048	<0.005	<0.005	<0.005	0.0346	<0.010	0.178	<0.000053
<0.000049	<0.0048	<0.005	<0.005	<0.005	0.0218	<0.010	<0.100	<0.000005
<0.000099	<0.0048	<0.005	<0.005	<0.005	0.0199	<0.010	0.365	<0.000005
0.00004	0.0024	0.0025	0.0025	0.0025	0.071	0.005	0.1577	0.00003
19%	0%	0%	0%	0%	-2729%	0%	-189%	0%

WWTP Effluent

	Grab Sample Parameters		Metals (Total)										
Collection	Cyanide, Total	Cyanide, Available	Mercury	Arsenic	Cadmium	Chromium	Copper	Lead	Molybdenum	Nickel	Selenium	Silver	Zinc
Date	mg/L	µg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L
10/5/2022	<0.0050	<0.002	<0.0002	<0.001	<0.0002	<0.002	0.0064	<0.001	0.0042	0.0036	0.001	<0.0005	0.0132
10/13/2022	0.091	<0.002	<0.0002	<0.001	<0.0002	<0.002	0.006	<0.001	0.0055	0.0039	<0.001	<0.0005	0.0159
10/14/2022	<0.0050	0.0027	<0.0002	<0.001	<0.0002	<0.002	0.0067	<0.001	0.0064	0.0036	0.0011	<0.0005	0.0164
10/21/2022	<0.0050	<0.002	<0.0002	<0.001	<0.0002	<0.002	0.0052	<0.001	0.0057	0.0033	<0.001	<0.0005	0.0138
10/22/2022	<0.0050	0.0028	<0.0002	<0.001	<0.0002	<0.002	0.0051	<0.001	0.0063	0.0035	<0.001	<0.0005	0.015
10/25/2022	<0.0050	<0.002	<0.0002	<0.001	<0.0002	<0.002	0.0053	<0.001	0.0036	0.0035	<0.001	<0.0005	0.0154
Average	**0.01725**	**0.0016**	**0.0001**	**0.0002**	**0.0001**	**0.0001**	**0.0058**	**0.0005**	**0.0053**	**0.0036**	**0.0007**	**0.00025**	**0.015**

	Compatible Pollutants				Organic Pollutants								
Collection	NH_3-N	BOD_5	TSS	Phosphorus	PCB	Chloroform	1,2-Dichloro benzene	Ethylbenzene	Styrene	Toluene	Xylene	Phenolics	Lindane
Date	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L
10/5/2022	<0.10	7.5	<8.3	0.22	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00051
10/13/2022	<0.10	9.5	12.5	0.59	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000048
10/14/2022	0.23	11.2	14	0.42	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.00005
10/21/2022	<0.10	5.8	8.6	0.41	<0.000099	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00005
10/22/2022	<0.10	7.5	7	0.35	<0.000097	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.000049
10/25/2022	<0.10	4.6	6.5	0.4	<0.000099	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.200	<0.0000050
Average	**0.08**	**8**	**9**	**0.40**	**0.000049**	**0.0024**	**0.0025**	**0.0025**	**0.0025**	**0.0025**	**0.005**	**0.0500**	**0.00006**

Akwel 5th St

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**:hylbenzei**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/3/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.02	<0.00051
10/11/2022	<0.000097	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.000049
10/20/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000049
10/26/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.02	<0.000049

Akwel

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/3/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00051
10/11/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	0.0051	<0.010	<0.100	<0.00048
10/20/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	0.0262	<0.010	<0.0667	<0.000049
10/26/2022	<0.000095	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000048

Borg Warner

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/3/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0209	<0.00051
10/11/2022	<0.000097	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.00049
10/20/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000048
10/26/2022	<0.000095	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000048

AAR 6th St

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/4/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00052
10/12/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00048
10/20/2022	<0.000098	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	<0.100	<0.000049
10/24/2022	<0.00010	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050

AAR Facility

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/4/2022	<0.0001	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	<0.0667	<0.00052
10/14/2022	<0.00012	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.067	<0.000062
10/17/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0255	<0.000049
10/27/2022	<0.000096	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	<0.200	<0.000048

AAR Rinse Tank

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/4/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.00052
10/12/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.00048
10/17/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000050
10/28/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000049

CCI

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/4/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0993	<0.00052
10/10/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.573	<0.000049
10/19/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.278	<0.000049
10/28/2022	<0.000098	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.233	<0.000049

CRE

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000052
10/13/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000050
10/21/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/24/2022	<0.00012	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000059

FIAMM

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.151	<0.00052
10/10/2022	<0.000099	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	1.57	<0.00050
10/21/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.108	<0.000049
10/27/2022	<0.000096	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.332	<0.000048

HUTCHINSON

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.377	<0.00052
10/10/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	1.7	<0.00050
10/21/2022	<0.000097	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.099	<0.000048
10/27/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.689	<0.000048

MI RUBBER

	Organic Pollutants									
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0703	<0.00052
10/10/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00050
10/21/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/27/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.200	<0.000048

MUNSON

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/6/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.457	<0.00051
10/14/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.071	<0.00005
10/18/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.234	<0.000049
10/24/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.00005

SPENCER PLASTICS

	Organic Pollutants									
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/6/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.679	<0.00051
10/14/2022	<0.00014	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.02	<0.000071
10/18/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0362	<0.000048
10/24/2022	<0.00011	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000056

ARVCO

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/6/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.35	<0.00051
10/12/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.64	<0.00048
10/18/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000049
10/28/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000049

AVON PROTECTION

	Organic Pollutants									
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/5/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.689	<0.000051
10/12/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.306	<0.000048
10/18/2022	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.203	<0.000049
10/28/2022	<0.000098	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.474	<0.000049

PIRANHA HOSE

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/7/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000050
10/13/2022	<0.0001	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.715	<0.000050
10/19/2022	<0.000096	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.000048
10/25/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.020	<0.00005

REC BOATS - TRAILER

	Organic Pollutants								
Collection	**PCB**	**Chloroform**	**1,2-Dichloro benzene**	**Ethylbenzene**	**Styrene**	**Toluene**	**Xylene**	**Phenolics**	**Lindane**
Date	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**	**mg/L**
10/7/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.329	<0.000050
10/13/2022	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/19/2022	<0.000098	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.361	<0.000049
10/25/2022	<0.00011	<0.024	<0.025	<0.025	<0.025	<0.025	<0.050	0.27	<0.000053

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
3	1/1/2021	1.27	248		29	88.3%	0.3		39.6	99.2%	0.658		5.833	88.7%	14.15		284.15	95.0%	63360
4	1/2/2021	1.2																	63740
5	1/3/2021	1.34																	54280
6	1/4/2021	1.44	232		15.33	93.4%	0.3		37.7	99.2%	0.395		5.388	92.7%	11.7		315	96.3%	44110
7	1/5/2021	1.74	356		6.8	98.1%	0.3		40.4	99.3%	0.315		2.695	88.3%	19.5		240	91.9%	38730
8	1/6/2021	1.7	244		4.7	98.1%	0.3		34.7	99.1%	0.25		3.697	93.2%	27		350	92.3%	33900
9	1/7/2021	1.8	232		6.7	97.1%	0.3		38.7	99.2%	0.167		2.829	94.1%	25		355	93.0%	33310
10	1/8/2021	1.82	280		4	98.6%	0.3		36.8	99.2%	0.158		4.653	96.6%	25		360	93.1%	33500
11	1/9/2021	1.66																	38460
12	1/10/2021	1.55																	44150
13	1/11/2021	1.49	384		9.5	97.5%	0.3		26.2	98.9%	0.199		5.855	96.6%	7.6		280	97.3%	45370
14	1/12/2021	1.79	284		4.5	98.4%	0.667		21.7	96.9%	0.207		4.075	94.9%	5.85		340	98.3%	54180
15	1/13/2021	1.75	304		8.5	97.2%	0.3		16.4	98.2%	0.192		3.452	94.4%	8.15		320	97.5%	54640
16	1/14/2021	1.87	364		13.3	96.3%	0.3		23.9	98.7%	0.247		5.432	95.5%	11.3		320	96.5%	56290
17	1/15/2021	1.72	236		15.3	93.5%	0.3		27.4	98.9%	0.231		3.741	93.8%	11.1		310	96.4%	61020
18	1/16/2021	1.7																	61650
19	1/17/2021	1.49																	61360
20	1/18/2021	1.48	204		8.7	95.7%	0.3		26.4	98.9%	0.206		4.92	95.8%	7.6		245	96.9%	66010
21	1/19/2021	1.84	264		9.3	96.5%	0.896		19.9	95.5%	0.236		4.787	95.1%	6.8		193	96.5%	68280
22	1/20/2021	1.77	252		10.7	95.8%	0.3		19.1	98.4%	0.186		4.787	96.1%	6.3		255	97.5%	68420
23	1/21/2021	1.67	308		8.7	97.2%	0.3		19.7	98.5%	0.158		3.029	94.8%	6.1		279	97.8%	68620
24	1/22/2021	1.72	280		4.7	98.3%	0.3		16.3	98.2%	0.175		4.52	96.1%	6		335	98.2%	69360
25	1/23/2021	1.53																	69510
26	1/24/2021	1.35																	69440
27	1/25/2021	1.53	340		6.7	98.0%	0.3		23.9	98.7%	0.128		4.297	97.0%	5.6		290	98.1%	69370
28	1/26/2021	1.79	260		4.5	98.3%	0.472		29.7	98.4%	0.12		4.676	97.4%	4.55		275	98.3%	67660
29	1/27/2021	1.69	212		3	98.6%	0.3		22	98.6%	0.127		4.542	97.2%	3.5		280	98.8%	65670
30	1/28/2021	1.63	236		4	98.3%	0.3		29.6	99.0%	0.124		5.121	97.6%	3.9		315	98.8%	63400
31	1/29/2021	1.74	304		4	98.7%	0.3		24.4	98.8%	0.132		3.585	96.3%	6.55		265	97.5%	63470
32	1/30/2021	1.64																	63470
33	1/31/2021	1.47																	63370
34	2/1/2021	1.56	296		4.4	98.5%	0.3		30.8	99.0%	0.17		5.254	96.8%	4.2		223.4	98.1%	63530
35	2/2/2021	1.8	164		5.2	96.8%	0.3		30.5	99.0%	0.186		3.452	94.6%	5.1		233.4	97.8%	61560
36	2/3/2021	1.65	268		4	98.5%	0.3		26	98.8%	0.159		5.145	96.9%	6.1		315	98.1%	61500
37	2/4/2021	1.67	264		5.6	97.9%	0.3		32.1	99.1%	0.186		4.787	96.1%	6.2		310	98.0%	61360
38	2/5/2021	1.7	216		4.8	97.8%	0.3		32.9	99.1%	0.194		5.143	96.2%	7.4		290	97.4%	61530
39	2/6/2021	1.58																	61440
40	2/7/2021	1.46																	61530
41	2/8/2021	1.57	264		4	98.5%	0.3		25.7	98.8%	0.187		4.676	96.0%	4.2		241	98.3%	61580
42	2/9/2021	1.8	188		4	97.9%	0.3		25.9	98.8%	0.144		4.275	96.6%	4.2		185	97.7%	61470
43	2/10/2021	1.81	140		2	98.6%	0.3		36.7	99.2%	0.132		3.163	95.8%	6		290	97.9%	60590
44	2/11/2021	1.74	184		4.4	97.6%	0.3		29.5	99.0%	0.12		4.386	97.3%	4.7		290	98.4%	60410
45	2/12/2021	1.71	228		2.4	98.9%	0.3		33.1	99.1%	0.128		4.676	97.3%	4.8		193.3	97.5%	60580
46	2/13/2021	1.62																	60410
47	2/14/2021	1.44																	60490
48	2/15/2021	1.53	288		4.4	98.5%	0.3		23.8	98.7%	0.154		4.609	96.7%	4.5		285	98.4%	61450
49	2/16/2021	1.76	232		3.6	98.4%	0.3		23.5	98.7%	0.194		3.897	95.0%	4.5		290	98.4%	60490
50	2/17/2021	1.8	292		5.6	98.1%	0.3		23.3	98.7%	0.147		5.254	97.2%	5.65		310	98.2%	60570
51	2/18/2021	1.73	304		4	98.7%	0.3		26.7	98.9%	0.162		3.808	95.7%	7		330	97.9%	60510

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
52	2/19/2021	1.76	200		4.4	97.8%	0.3		25.6	98.8%	0.151		3.875	96.1%	5.4		305	98.2%	60500
53	2/20/2021	1.68																	60470
54	2/21/2021	1.48																	60480
55	2/22/2021	1.67	308		4	98.7%	0.3		25.9	98.8%	0.206		5.833	96.5%	3.9		229.2	98.3%	60550
56	2/23/2021	1.84	136		5.6	95.9%	0.3		25.8	98.8%	0.207		4.364	95.3%	3.9		279	98.6%	60730
57	2/24/2021	1.76	208		5.2	97.5%	0.3		35	99.1%	0.158		3.14	95.0%	6.9		295	97.7%	60370
58	2/25/2021	1.58	208		5.2	97.5%	0.3		32	99.1%	0.184		5.855	96.9%	7.4		365	98.0%	60610
59	2/26/2021	1.64	212		4.4	97.9%	0.3		42.2	99.3%	0.183		3.852	95.2%	7.6		295	97.4%	60350
60	2/27/2021	1.63																	60470
61	2/28/2021	1.41																	60480
62	3/1/2021	1.46	212		8	96.2%	0.3		34.1	99.1%	0.267		3.852	93.1%	7.6		375	98.0%	60500
63	3/2/2021	1.65	368		6.8	98.2%	0.3		35	99.1%	0.206		4.898	95.8%	7.56		295	97.4%	61450
64	3/3/2021	1.65	248		8	96.8%	0.3		36.1	99.2%	0.246		4.275	94.2%	8.4		380	97.8%	63390
65	3/4/2021	1.62	264		8	97.0%	0.3		32.7	99.1%	0.227		3.964	94.3%	9.9		325	97.0%	65570
66	3/5/2021	1.5	240		9.6	96.0%	0.3		36.3	99.2%	0.246		4.275	94.2%	11.1		305	96.4%	67330
67	3/6/2021	1.45																	69400
68	3/7/2021	1.41																	69260
69	3/8/2021	1.47	280		8	97.1%	0.3		28.6	99.0%	0.337		4.52	92.5%	9		330	97.3%	66440
70	3/9/2021	1.68	204		6.8	96.7%	0.3		26.6	98.9%	0.321		4.142	92.3%	7.21		275	97.4%	66440
71	3/10/2021	1.66	208		8	96.2%	0.3		32.3	99.1%	0.267		4.075	93.4%	6.5		305	97.9%	66600
72	3/11/2021	1.72	196		6.5	96.7%	0.3		23.2	98.7%	0.232		4.03	94.2%	6.3		270	97.7%	65500
73	3/12/2021	1.66	232		6	97.4%	0.3		21.8	98.6%	0.243		3.407	92.9%	5.6		260	97.8%	65450
74	3/13/2021	1.48																	65420
75	3/14/2021	1.35																	62750
76	3/15/2021	1.39	300		8	97.3%	0.3		28.1	98.9%	0.289		5.21	94.5%	9.5		305	96.9%	65520
77	3/16/2021	1.6	272		8.7	96.8%	0.3		23.8	98.7%	0.235		4.854	95.2%	9		315	97.1%	65740
78	3/17/2021	1.62	208		8.7	95.8%	0.3		22.9	98.7%	0.267		3.83	93.0%	9.23		265	96.5%	65280
79	3/18/2021	1.56	224		5.3	97.6%	0.3		28.1	98.9%	0.238		3.919	93.9%	8.1		300	97.3%	65390
80	3/19/2021	1.64	288		8.7	97.0%	0.3		27.9	98.9%	0.248		4.342	94.3%	7.8		300	97.4%	63710
81	3/20/2021	1.47																	63480
82	3/21/2021	1.33																	63400
83	3/22/2021	1.73	92	128	9.3	89.9%	2.7	27.8	26.1	89.7%	0.254		3.83	93.4%	9.9	235	213.5	95.4%	63740
84	3/23/2021	1.71	252	184	8.5	96.6%	9.95	31.3	30.5	67.4%	0.244		3.919	93.8%	8.6	285	310	97.2%	63290
85	3/24/2021	1.71	180	164	5	97.2%	3.54	25.1	25.8	86.3%	0.176	3.5	3.452	94.9%	12.15	295	290	95.8%	63670
86	3/25/2021	1.59	72	124	6.25	91.3%	0.664	24.4	21.4	96.9%	0.171		3.541	95.2%	11.8	265	265	95.5%	63430
87	3/26/2021	1.6	108	140	5	95.4%	2.56	27.4	26	90.2%	0.184		4.32	95.7%	15	250	290	94.8%	63410
88	3/27/2021	1.36																	63390
89	3/28/2021	1.24																	63390
90	3/29/2021	1.26	476	212	4.5	99.1%	5.13	27.9	24.9	79.4%	0.23		5.076	95.5%	9.8	243.4	340	97.1%	63830
91	3/30/2021	1.6	160	80	5	96.9%	9.3	26.5	24.6	62.2%	0.225		4.142	94.6%	11.2	210	270	95.9%	63460
92	3/31/2021	1.57	148	124	4.5	97.0%	2.58	27.1	22	88.3%	0.154		3.608	95.7%	13	236.7	275	95.3%	63370
93	4/1/2021	1.49	168	100	4.75	97.2%	0.7445	25.5	27.1	97.3%	0.135		3.85	96.5%	14.3	250	325	95.6%	63710
94	4/2/2021	1.45	96		5.6	94.2%	0.596		41.7	98.6%	0.135		3.229	95.8%	14.6		250	94.2%	63690
95	4/3/2021	1.44																	62820
96	4/4/2021	1.29																	62040
97	4/5/2021	1.37	300	212	6.5	97.8%	1.05	30.6	37.5	97.2%	0.178		4.898	96.4%	8.7	237.5	345	97.5%	59220
98	4/6/2021	1.63	224	184	6.5	97.1%	2.21	29.3	18.4	88.0%	0.186		3.919	95.3%	8.6	270	325	97.4%	58720
99	4/7/2021	1.69	260	120	7.5	97.1%	0.573	20	14.3	96.0%	0.174		3.964	95.6%	8.6	250	315	97.3%	59630
100	4/8/2021	1.68	296	260	11.5	96.1%	0.3	23.8	19.9	98.5%	0.171		4.653	96.3%	9.6	295	360	97.3%	59340

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
101	4/9/2021	1.6	368	324	7	98.1%	0.3	23.2	24.1	98.8%	0.167	3.82	3.852	95.7%	9.5	275	360	97.4%	59570
102	4/10/2021	1.45																	59470
103	4/11/2021	1.38																	59450
104	4/12/2021	1.42	296	400	7	97.6%	0.427	29.7	23.7	98.2%	0.175		4.698	96.3%	8.28	310	360	97.7%	59510
105	4/13/2021	1.64	268	308	6.5	97.6%	1.11	29.5	26.1	95.7%	0.207		3.897	94.7%	6.6	270	335	98.0%	59540
106	4/14/2021	1.72	172	356	7	95.9%	0.763	30.7	22.8	96.7%	0.203		3.608	94.4%	6.9	285	330	97.9%	59560
107	4/15/2021	1.58	188	168	9.5	94.9%	0.452	37.9	31.9	98.6%	0.194		4.164	95.3%	7.4	315	365	98.0%	59480
108	4/16/2021	1.56	212	296	8.5		0.3	33.5	27.6	98.9%	0.179		3.763	95.2%	8.2	280	325	97.5%	59500
109	4/17/2021	1.51																	59400
110	4/18/2021	1.35																	59610
111	4/19/2021	1.36	260	276	11	95.8%	0.3	40.9	27.9	98.9%	0.228	5.128	3.786	94.0%	6.3	305	290	97.8%	59510
112	4/20/2021	1.72	288	164	16	94.4%	0.582	28.2	30.9	98.1%	0.357		4.542	92.1%	9.13	248.34	330	97.2%	59520
113	4/21/2021	1.59	444	304	12	97.3%	0.391	35.3	35.9	98.9%	0.322		5.098	93.7%	10.1	260	365	97.2%	59560
114	4/22/2021	1.58	380	356	12.7	96.7%	0.377	33	31.5	98.8%	0.29		4.297	93.3%	6.7	245	325	97.9%	59440
115	4/23/2021	1.56	244	232	6.7	97.3%	0.427	37	25.2	98.3%	0.234		3.741	93.7%	5.3	240	310	98.3%	59690
116	4/24/2021	1.48																	59310
117	4/25/2021	1.28																	59540
118	4/26/2021	1.36	364	400	9.3	97.4%	0.3	27.5	30.5	99.0%	0.309		6.011	94.9%	5.2	360	370	98.6%	59530
119	4/27/2021	1.61	420	244	9.3	97.8%	0.3	31.3	32.6	99.1%	0.306	4.754	5.21	94.1%	4.9	325	365	98.7%	59440
120	4/28/2021	1.73	248	208	7	97.2%	0.3	29.1	25	98.8%	0.268		3.832	93.0%	8.9	315	320	97.2%	59520
121	4/29/2021	1.65	388	240	12.7	96.7%	0.3	21	21.9	98.6%	0.231		4.655	95.0%	10	325	365	97.3%	59380
122	4/30/2021	1.6	252	336	8	96.8%	0.3	29.5	26.9	98.9%	0.236		4.4998	94.8%	11.2	345	360	96.9%	59620
123	5/1/2021	1.78																	59420
124	5/2/2021	1.36																	59530
125	5/3/2021	1.6	412	420	25.3	93.9%	0.3	24.2	28.3	98.9%	0.508		4.851	89.5%	12.5	340	370	96.6%	59350
126	5/4/2021	2	460	272	27	94.1%	0.3	29	22	98.6%	0.456		4.449	89.8%	9.4	305	315	97.0%	60440
127	5/5/2021	1.79	208	260	7	96.6%	0.3	26.8	18.9	98.4%	0.249	3.651	3.049	91.8%	10.5	330	285	96.3%	60470
128	5/6/2021	1.71	372	268	14	96.2%	0.3	23.2	24.8	98.8%	0.219		4.361	95.0%	15.6	350	310	95.0%	61720
129	5/7/2021	1.75	260	284	7	97.3%	0.3	36.1	30.5	99.0%	0.272		4.165	93.5%	13	320	275	95.3%	63940
130	5/8/2021	1.59																	64640
131	5/9/2021	1.39																	64670
132	5/10/2021	1.44	568	280	12	97.9%	0.3	36.2	25.9	98.8%	0.305		5.311	94.3%	6.1	295	330	98.2%	64430
133	5/11/2021	1.68	292	220	9	96.9%	0.3	31.4	27.2	98.9%	0.36		4.704	92.3%	5.7	315	355	98.4%	67110
134	5/12/2021	1.76	244	164	10	95.9%	0.3	21	22.4	98.7%	0.326		4.302	92.4%	7.4	315	320	97.7%	69020
135	5/13/2021	1.77	228	252	7	96.9%	0.3	25.7	22	98.6%	0.259	4.649	4.058	93.6%	8.2	320	275	97.0%	69980
136	5/14/2021	1.73	244	204	6	97.5%	0.3	26.9	19.1	98.4%	0.222		4.635	95.2%	10.7	320	240.9	95.6%	69510
137	5/15/2021	1.7																	69250
138	5/16/2021	1.53																	69730
139	5/17/2021	1.51	552	260	9	98.4%	0.3	36.5	36.9	99.2%	0.256		4.792	94.7%	5.4	230	275	98.0%	69280
140	5/18/2021	1.88	276	224	10	96.4%	0.3	25	45.6	99.3%	0.294		5.428	94.6%	5.75	320	325	98.2%	69680
141	5/19/2021	1.89	300	232	6	98.0%	0.3	27.5	30.3	99.0%	0.281		4.156	93.2%	5.9	340	300	98.0%	69500
142	5/20/2021	1.79	220	212	7.3	96.7%	0.3	27.1	28.2	98.9%	0.247		3.803	93.5%	7.8	365	335	97.7%	69260
143	5/21/2021	1.96	332	376	12.7	96.2%	0.3	24.5	17.7	98.3%	0.251	4.403	3.441	92.7%	11.3	375	325	96.5%	69420
144	5/22/2021	1.76																	69400
145	5/23/2021	1.64																	69410
146	5/24/2021	1.53	244	304	4	98.4%	0.3	30.5	30.4	99.0%	0.343		4.714	92.7%	4.95	280	295	98.3%	69770
147	5/25/2021	1.97	352	368	6.7	98.1%	0.3	22.8	19.7	98.5%	0.343		4.312	92.0%	6.2	370	270	97.7%	69320
148	5/26/2021	2.04	240	324	6	97.5%	0.3	26.8	20.5	98.5%	0.285		3.921	92.7%	8.6	370	330	97.4%	69390
149	5/27/2021	1.82	200	256	4	98.0%	0.3	22.5	24.4	98.8%	0.266		3.725	92.9%	10	370	330	97.0%	69370

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
150	5/28/2021	1.77	328	388	7.3	97.8%	0.3	23.4	24.1	98.8%	0.291		4.302	93.2%	10.9	370	305	96.4%	69500
151	5/29/2021	1.75																	69400
152	5/30/2021	1.47																	69430
153	5/31/2021	1.48	188		3.3	98.2%	0.3		34.3	99.1%	0.386		5.35	92.8%	8.1		280	97.1%	69630
154	6/1/2021	1.58	260	400	32	87.7%	0.468	30.8	36.8	98.7%	0.572	8.098	5.164	88.9%	23	370	275	91.6%	69340
155	6/2/2021	1.94	272	316	6.67	97.5%	0.3	27.8	26.8	98.9%	0.414		4.312	90.4%	7.1	375	280	97.5%	69520
156	6/3/2021	1.92	236	256	11.33	95.2%	0.3	27	22.5	98.7%	0.331		4.077	91.9%	9.5	350	280	96.6%	71050
157	6/4/2021	1.92	220	260	8.67	96.1%	0.3	26.2	20.4	98.5%	0.333		3.568	90.7%	8.9	300	200.9	95.6%	71790
158	6/5/2021	1.78																	71310
159	6/6/2021	1.57																	71350
160	6/7/2021	1.67	240	244	8	96.7%	0.3	33.9	25.4	98.8%	0.343		3.098	88.9%	5.1	360	290	98.2%	71630
161	6/8/2021	2.03	196	236	8.67	95.6%	0.3	25.3	19.8	98.5%	0.357		3.921	90.9%	8.2	305	205	96.0%	69330
162	6/9/2021	1.97	176	228	3.33	98.1%	0.3	27	20	98.5%	0.329	5.854	3.95	91.7%	9	295	365	97.5%	69170
163	6/10/2021	1.98	224	248	5.33	97.6%	0.3	30.2	20.1	98.5%	0.313		3.891	92.0%	6.4	305	210.9	97.0%	69450
164	6/11/2021	1.85	352	48	4.7	98.7%	0.3	20	17.2	98.3%	0.328		3.343	90.2%	11.7	220.9	152.5	92.3%	56790
165	6/12/2021	1.85																	67250
166	6/13/2021	1.59																	66600
167	6/14/2021	1.63	316	300	13.33	95.8%	0.3	30.3	27.7	98.9%	0.447		5.246	91.5%	7.7	325	260	97.0%	65380
168	6/15/2021	1.96	196	252	5.33	97.3%	0.3	28.9	26.1	98.9%	0.366		3.666	90.0%	5.7	340	255	97.8%	61810
169	6/16/2021	1.84	240	304	6.67	97.2%	0.3	25.5	23.3	98.7%	0.317		4.547	93.0%	6.8	350	241.7	97.2%	61480
170	6/17/2021	1.8	208	372	12	94.2%	0.3	26.7	26.4	98.9%	0.265	7.211	3.774	93.0%	7.5	365	227	96.7%	60620
171	6/18/2021	1.9	228	288	4.5	98.0%	0.3	23.5	18.9	98.4%	0.36		4.175	91.4%	8.1	345	226	96.4%	60380
172	6/19/2021	1.61																	60430
173	6/20/2021	1.47																	60390
174	6/21/2021	1.59	280	292	3.3	98.8%	0.3	31.4	38.2	99.2%	0.431		5.595	92.3%	3	365	310	99.0%	59650
175	6/22/2021	1.82	372	308	3.2	99.1%	0.3	30.1	29.1	99.0%	0.396		4.723	91.6%	3	355	270	98.9%	59630
176	6/23/2021	1.79	336	252	3.6	98.9%	0.3	27.8	28.6	99.0%	0.281		3.842	92.7%	3.8	360	315	98.8%	59420
177	6/24/2021	1.82	236	272	4	98.3%	0.3	27.8	27.6	98.9%	0.288		4.302	93.3%	3.6	355	285	98.7%	59440
178	6/25/2021	1.91	312	284	2.4	99.2%	0.3	28	20.3	98.5%	0.271	5.695	4.41	93.9%	3	340	230	98.7%	59710
179	6/26/2021	1.92																	59290
180	6/27/2021	1.59																	59400
181	6/28/2021	1.51	292	412	4.8	98.4%	0.3	34	30.6	99.0%	0.288		5.693	94.9%	3	370	238.34	98.7%	59700
182	6/29/2021	1.7	288	308	2.8	99.0%	0.3	31.4	31.3	99.0%	0.285		4.978	94.3%	3	290	176.67	98.3%	59760
183	6/30/2021	1.79	248	340	4	98.4%	0.3	28	31.5	99.0%	0.234		4.968	95.3%	3	315	182.5	98.4%	59120
184	7/1/2021	1.72	204	288	2.4	98.8%	0.3	35.5	41.6	99.3%	0.251		5.379	95.3%	3	340	217.5	98.6%	59570
185	7/2/2021	1.7	212	292	4.4	97.9%	0.3	33.3	36.1	99.2%	0.299		4.674	93.6%	3	340	188.4	98.4%	59310
186	7/3/2021	1.69																	58720
187	7/4/2021	1.67																	58400
188	7/5/2021	1.62	244		4.4	98.2%	0.3		22.9	98.7%	0.403		6.065	93.4%	5.9		290	98.0%	58580
189	7/6/2021	2.13	92	104	6.4	93.0%	0.3	20	16.7	98.2%	0.3985	3.551	3.539	88.7%	8.5	260	232.5	96.3%	57380
190	7/7/2021	2.15	60	120	2.8	95.3%	0.3	22.3	19.9	98.5%	0.305		3.069	90.1%	6.6	251	120	94.5%	57580
191	7/8/2021	2.06	216	168	4	98.1%	0.3	24.9	17.9	98.3%	0.268		4.626	94.2%	5.25	270	185.85	97.2%	57220
192	7/9/2021	1.91	224	204	3.2	98.6%	0.3	22.6	25.1	98.8%	0.236		4.273	94.5%	8.65	320	225	96.2%	57160
193	7/10/2021	1.91																	56610
194	7/11/2021	1.77																	56260
195	7/12/2021	1.69	240	208	2.8	98.8%	0.3	19.6	18.8	98.4%	0.24		4.93	95.1%	3	305	204.15	98.5%	56650
196	7/13/2021	2.3	252	188	4	98.4%	0.3	19.6	19	98.4%	0.273		6.192	95.6%	3.45	285	263.35	98.7%	55550
197	7/14/2021	2.19	80	128	2	97.5%	0.3	13.2	13.6	97.8%	0.2145	4.262	5.419	96.0%	4.58	249.15	138.35	96.7%	55120
198	7/15/2021	2.31	132	268	4	97.0%	0.3	21.6	23.8	98.7%	0.206		3.832	94.6%	4.9	295	220	97.8%	55570

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
199	7/16/2021	2.31	116	252	2.8	97.6%	0.3	17.1	17.7	98.3%	0.158		2.922	94.6%	4.05	295	144	97.2%	54760
200	7/17/2021	2.02																	55710
201	7/18/2021	1.82																	55180
202	7/19/2021	1.88	232	260	6.8	97.1%	0.3	19.1	23.2	98.7%	0.211		4.714	95.5%	6.6	330	234.2	97.2%	55770
203	7/20/2021	2.22	100	200	2.8	97.2%	0.3	15.5	16.7	98.2%	0.1945		3.225	94.0%	3.3	235	175	98.1%	55400
204	7/21/2021	2.07	244	244	3.6	98.5%	0.3	15	16	98.1%	0.174		4.684	96.3%	3.9	275	230	98.3%	55340
205	7/22/2021	2.08	220	220	2.8	98.7%	0.3	15.9	16	98.1%	0.152	5.143	3.97	96.2%	8.2	290	218.4	96.2%	54600
206	7/23/2021	1.82	272	692	2	99.3%	0.3	33.3	32.9	99.1%	0.187		5.722	96.7%	7.7	207	200.5	96.2%	54450
207	7/24/2021	1.99																	54440
208	7/25/2021	1.9																	54470
209	7/26/2021	1.82	264	144	3.6	98.6%	0.3	20.1	17.8	98.3%	0.223		4.851	95.4%	4.4	225.9	255	98.3%	54520
210	7/27/2021	2.2	204	104	2	99.0%	0.3	17.1	16	98.1%	0.238		3.783	93.7%	3.45	200	181.5	98.1%	54570
211	7/28/2021	2.04	132	104	3.2	97.6%	0.3	17.4	16.8	98.2%	0.19		3.78	95.0%	4	209.2	214.2	98.1%	54360
212	7/29/2021	2.06	120	140	3	97.5%	0.3	14.7	15.1	98.0%	0.176		4.087	95.7%	6.8	228.3	221.7	96.9%	54480
213	7/30/2021	2.08	276	100	1.6	99.4%	0.3	15	18.1	98.3%	0.1485	2.963	5.027	97.0%	8.1	270	239.2	96.6%	54530
214	7/31/2021	1.84																	53510
215	8/1/2021	1.84																	53330
216	8/2/2021	1.77	220	308	4.8	97.8%	0.3	18.7	18.4	98.4%	0.199		4.146	95.2%	3	365	265	98.9%	53710
217	8/3/2021	2.2	80	236	2.8	96.5%	0.3	17.2	19.6	98.5%	0.184		3.372	94.5%	4.2	325	189.2	97.8%	53500
218	8/4/2021	2.22	144	308	8.4	94.2%	0.3	20.9	15.5	98.1%	0.211		3.411	93.8%	6.1	355	230.8	97.4%	53470
219	8/5/2021	1.73	336	560	10.4	96.9%	0.3	18.8	18.2	98.4%	0.24		3.842	93.8%	6.3	370	253.3	97.5%	52440
220	8/6/2021	2.08	432	780	10.8	97.5%	0.3	23	15.4	98.1%	0.281		4.391	93.6%	4.9	365	231.7	97.9%	53440
221	8/7/2021	1.88																	53350
222	8/8/2021	1.7																	53560
223	8/9/2021	1.81	256	324	5.6	97.8%	0.3	27.9	23.9	98.7%	0.517	7.152	4.958	89.6%	3	340	275	98.9%	53520
224	8/10/2021	2.75	80	932	10	87.5%	0.3	25.8	20	98.5%	0.608		3.313	81.6%	3.6	365	160	97.8%	53530
225	8/11/2021	2.85	116	392	9.5	91.8%	0.3	24.1	18.7	98.4%	0.497		3.362	85.2%	5.4	370	169.2	96.8%	53460
226	8/12/2021	2.36	152	468	9	94.1%	0.3	20.3	18.1	98.3%	0.41		3.803	89.2%	9.3	365	194.2	95.2%	53490
227	8/13/2021	2.3	288	440	8.5	97.0%	0.3	22.9	18.9	98.4%	0.332		3.196	89.6%	8.1	370	226.7	96.4%	53700
228	8/14/2021	2.09																	53450
229	8/15/2021	1.82																	53340
230	8/16/2021	1.93	288	524	5.5	98.1%	0.3	29.5	17.6	98.3%	0.409		5.703	92.8%	3.15	370	275	98.9%	53590
231	8/17/2021	2.2	72	372	7.6	89.4%	0.3	25.5	20.7	98.6%	0.412	4.614	3.078	86.6%	6.05	370	173.5	96.5%	53450
232	8/18/2021	2.1	212	352	8.4	96.0%	0.3	21	19.7	98.5%	0.379		3.597	89.5%	6.2	360	202.5	96.9%	53480
233	8/19/2021	2.02	444	436	8.8	98.0%	0.3	21.8	17.9	98.3%	0.335		4.185	92.0%	9	370	186.7	95.2%	57080
234	8/20/2021	1.99	144	260	6.4	95.6%	0.3	21.1	21.8	98.6%	0.371		2.99	87.6%	6.3	315	179.2	96.5%	57490
235	8/21/2021	1.9																	57220
236	8/22/2021	1.8																	57600
237	8/23/2021	1.73	176	304	4.4	97.5%	0.3	22.8	21.7	98.6%	0.273		4.253	93.6%	4.25	360	265	98.4%	57450
238	8/24/2021	2.05	168	200	6	96.4%	0.3	23.6	22	98.6%	0.314		4.107	92.4%	4.55	305	223.35	98.0%	57400
239	8/25/2021	2.07	284	296	8.4	97.0%	0.3	25.5	20.7	98.6%	0.233	4.291	4.518	94.8%	4.7	350	250	98.1%	57000
240	8/26/2021	2	228	288	5.2	97.7%	0.3	25.2	21.5	98.6%	0.22		4.694	95.3%	8.3	365	315	97.4%	57570
241	8/27/2021	2.06	312	288	6.4	97.9%	0.3	25.6	22.7	98.7%	0.198		5.82	96.6%	10.5	335	220	95.2%	57380
242	8/28/2021	2.16																	57460
243	8/29/2021	2.04																	57490
244	8/30/2021	2.01	260	280	3.2	98.8%	0.3	24.2	17.7	98.3%	0.235		4.528	94.8%	3	350	190	98.4%	57320
245	8/31/2021	2.28	260	224	2.8	98.9%	0.3	19.2	12.1	97.5%	0.195		3.225	94.0%	3.75	320	212.5	98.2%	48000
246	9/1/2021	1.83	252	248	4.8	98.1%	0.3	20.5	13.5	97.8%	0.164		4.253	96.1%	5.9	360	210.9	97.2%	50810
247	9/2/2021	1.97	212	212	2	99.1%	0.3	21.8	14.7	98.0%	0.162	4.367	3.46	95.3%	4.9	330	214	97.7%	57730

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
248	9/3/2021	1.84	316	224	2	99.4%	0.3	13.1	12.4	97.6%	0.169		4.949	96.6%	3.9	325	245.9	98.4%	57340
249	9/4/2021	1.75																	57280
250	9/5/2021	1.64																	57580
251	9/6/2021	1.61	160		2.8	98.3%	0.3		30.2	99.0%	0.227		6.222	96.4%	3		345	99.1%	57660
252	9/7/2021	1.71	260	260	3.2	98.8%	0.3	26.2	24.4	98.8%	0.268		4.009	93.3%	3	370	275	98.9%	57350
253	9/8/2021	2.41	276	280	2	99.3%	0.3	17.9	14.2	97.9%	0.231		5.066	95.4%	3	345	290	99.0%	55950
254	9/9/2021	2.1	576	256	4.8	99.2%	0.3	18.6	14.8	98.0%	0.194	4.626	3.235	94.0%	4.6	330	212.5	97.8%	55600
255	9/10/2021	2	288	248	3.2	98.9%	0.3	17.7	13.9	97.8%	0.148		4.351	96.6%	9.8	350	245	96.0%	55160
256	9/11/2021	1.95																	55730
257	9/12/2021	2.05																	55470
258	9/13/2021	1.63	544	276	2.4	99.6%	0.3	20.6	33.5	99.1%	0.2		11.059	98.2%	3	290	67	95.5%	55570
259	9/14/2021	1.94	348	212	2	99.4%	0.3	14.1	14.7	98.0%	0.162		5.125	96.8%	3.5	315	265	98.7%	55420
260	9/15/2021	1.91	264	220	2.8	98.9%	0.3	16.2	13.9	97.8%	0.121		4.165	97.1%	6.2	325	305	98.0%	55510
261	9/16/2021	1.81	280	180	3.6	98.7%	0.3	17.7	14	97.9%	0.119		3.725	96.8%	8.2	305	220.85	96.3%	54660
262	9/17/2021	1.9	420	232	3.2	99.2%	0.3	18.5	16.2	98.1%	0.166	3.968	3.94	95.8%	9.3	320	280	96.7%	54510
263	9/18/2021	1.7																	53390
264	9/19/2021	1.45																	53210
265	9/20/2021	1.61	268	372	2.8	99.0%	0.3	20.7	17.9	98.3%	0.164		4.988	96.7%	3	370	275	98.9%	52050
266	9/21/2021	2	260	192	2.4	99.1%	0.3	15.2	11.7	97.4%	0.159		4.479	96.5%	4.25	365	270	98.4%	51500
267	9/22/2021	1.81	204	284	2.4	98.8%	0.3	19	18.5	98.4%	0.132		3.979	96.7%	5.2	360	300	98.3%	51430
268	9/23/2021	1.74	416	228	3.2	99.2%	0.3	21.8	18.5	98.4%	0.104		4.694	97.8%	6.7	370	370	98.2%	51590
269	9/24/2021	1.67	364	176	2.4	99.3%	0.3	20.5	20.7	98.6%	0.124		5.438	97.7%	5.35	365	310	98.3%	51470
270	9/25/2021	1.55																	51590
271	9/26/2021	1.47																	52120
272	9/27/2021	1.56	312	284	1.2	99.6%	0.3	27.7	20.5	98.5%	0.164	2.846	4.44	96.3%	3	360	295	99.0%	50880
273	9/28/2021	1.84	268	256	2	99.3%	0.3	13.9	11	97.3%	0.172		3.901	95.6%	3	365	250	98.8%	51510
274	9/29/2021	1.78	264	236	5.2	98.0%	0.3	18.1	14.8	98.0%	0.172		4.205	95.9%	3.3	370	270	98.8%	51700
275	9/30/2021	1.74	292	268	4	98.6%	0.3	17.4	13.8	97.8%	0.152		4.586	96.7%	5.4	370	370	98.5%	51290
276	10/1/2021	1.75	240	184	3.6	98.5%	0.3	22.8	17.5	98.3%	0.127		3.872	96.7%	5.5	350	256.7	97.9%	51730
277	10/2/2021	1.6																	51300
278	10/3/2021	1.44																	51410
279	10/4/2021	1.65	260	224	2.4	99.1%	0.3		10.4	97.1%	0.154		5.556	97.2%	3	355	320	99.1%	51830
280	10/5/2021	1.82	248	232	2.8	98.9%	0.3	26	28	98.9%	0.166	5.454	3.823	95.7%	3	330	265	98.9%	51430
281	10/6/2021	1.78	608	208	3.2	99.5%	0.3	33.5	30	99.0%	0.139		5.321	97.4%	4.5	345	340	98.7%	51420
282	10/7/2021	1.84	384	252	4.8	98.8%	0.3	32.3	34.6	99.1%	0.179		3.881	95.4%	5.9	370	365	98.4%	51300
283	10/8/2021	1.79	372	228	2.8	99.2%	0.3	37.2	36.2	99.2%	0.174		4.077	95.7%	5.4	355	320	98.3%	53590
284	10/9/2021	1.75																	53590
285	10/10/2021	1.55																	53590
286	10/11/2021	1.66	256	260	0.8	99.7%	0.3	33.9	27.5	98.9%	0.17		4.048	95.8%	3	375	295	99.0%	53400
287	10/12/2021	2.03	276	260	2	99.3%	0.319	26.4	24.5	98.7%	0.204		4.577	95.5%	3.8	370	340	98.9%	52800
288	10/13/2021	1.99	240	204	4	98.3%	0.3	30.6	29.7	99.0%	0.182	4.802	3.48	94.8%	7.7	365	330	97.7%	52310
289	10/14/2021	1.91	256	296	4.4	98.3%	0.3	27.1	22.4	98.7%	0.152		4.371	96.5%	9.2	365	305	97.0%	52470
290	10/15/2021	1.88	248	312	4	98.4%	0.3	20.4	24.1	98.8%	0.137		4.939	97.2%	10.1	370	295	96.6%	56410
291	10/16/2021	1.73																	52670
292	10/17/2021	1.5																	52460
293	10/18/2021	1.64	296	268	2.4	99.2%	0.3	31.8	29.7	99.0%	0.162		5.614	97.1%	4.05	375	355	98.9%	52780
294	10/19/2021	1.84	296	224	3.6	98.8%	0.3	34.5	47.1	99.4%	0.157		4.988	96.9%	3	375	300	99.0%	52310
295	10/20/2021	1.88	208	212	3.6	98.3%	0.3	34.5	31.1	99.0%	0.127		3.705	96.6%	4.4	365	280	98.4%	52500
296	10/21/2021	1.84	208	352	6.4	96.9%	0.3	31.6	22.4	98.7%	0.116	4.555	3.313	96.5%	4.8	300	310	98.5%	52660

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
297	10/22/2021	1.89	324	416	3.2	99.0%	0.3	40.4	38.4	99.2%	0.144		4.577	96.9%	3	360	300	99.0%	52450
298	10/23/2021	1.88																	52650
299	10/24/2021	1.67																	52510
300	10/25/2021	1.7	276	308	3.2	98.8%	0.3	24	21.9	98.6%	0.173		4.616	96.3%	3	360	305	99.0%	53470
301	10/26/2021	1.9	244	396	3.6	98.5%	0.3	40.3	33.4	99.1%	0.188		3.392	94.5%	3	370	360	99.2%	53480
302	10/27/2021	1.85	212	264	3.2	98.5%	0.3	28.1	25.2	98.8%	0.144		2.795	94.8%	3	370	305	99.0%	53680
303	10/28/2021	1.85	308	408	3.2	99.0%	0.3	24.2	29.9	99.0%	0.135		5.967	97.7%	4.05	370	300	98.7%	53410
304	10/29/2021	1.87	228	368	3.6	98.4%	0.3	30.3	24.9	98.8%	0.153	4.832	8.111	98.1%	3	365	275	98.9%	53470
305	10/30/2021	1.84																	53580
306	10/31/2021	1.55																	53660
307	11/1/2021	1.56	288	432	4.8	98.3%	0.3	22	24.9	98.8%	0.179		4.283	95.8%	3	370	335	99.1%	53430
308	11/2/2021	1.87	308	312	4.4	98.6%	0.3	18.7	19	98.4%	0.213		5.722	96.3%	3	375	355	99.2%	56940
309	11/3/2021	1.82	176	384	3.2	98.2%	0.3	24.2	22.4	98.7%	0.161		5.174	96.9%	3	375	280	98.9%	66380
310	11/4/2021	1.82	212		3.6	98.3%	0.3		18.5	98.4%	0.165		3.519	95.3%	3.3		205.9	98.4%	59480
311	11/5/2021	1.77	192	228	4	97.9%	0.3	20.9	17.3	98.3%	0.158		3.823	95.9%	3.5	385	280	98.8%	59330
312	11/6/2021	1.7																	59530
313	11/7/2021	1.67																	61920
314	11/8/2021	1.68	312	228	2	99.4%	0.3	18.6	22	98.6%	0.148	3.087	4.567	96.8%	3	355	260	98.8%	59600
315	11/9/2021	1.91	252	292	1.6	99.4%	0.3	15.4	20.2	98.5%	0.133		3.999	96.7%	3	365	265	98.9%	59360
316	11/10/2021	1.8	200	276	0.4	99.8%	0.3	18.8	25.5	98.8%	0.138		3.5	96.1%	4.1	360	285	98.6%	59620
317	11/11/2021	1.77	292		2.8	99.0%	0.3		27.3	98.9%	0.151		4.214	96.4%	6.2		360	98.3%	59530
318	11/12/2021	1.77	184	384	2.4	98.7%	0.3	30	18.2	98.4%	0.134		3.715	96.4%	8.69	224.15	212.5	95.9%	59430
319	11/13/2021	1.68																	59580
320	11/14/2021	1.47																	59430
321	11/15/2021	1.54	288	308	2.8	99.0%	0.3	44.3	33.2	99.1%	0.148		4.665	96.8%	3	350	260	98.8%	59800
322	11/16/2021	1.87	248	296	3.2	98.7%	0.3	23.6	20.3	98.5%	0.169	4.221	4.528	96.3%	3	360	243.34	98.8%	59490
323	11/17/2021	1.76	224	264	5.2	97.7%	0.3	23.4	24.7	98.8%	0.141		3.775	96.3%	3	340	215.85	98.6%	59340
324	11/18/2021	1.74	136	124	1.8	98.7%	0.445	27.6	20.2	97.8%	0.129		4.479	97.1%	4.85	290	235	97.9%	59670
325	11/19/2021	1.69	260	288	2.8	98.9%	0.3	26	18.1	98.3%	0.131		4.626	97.2%	5.6	345	213.3	97.4%	59500
326	11/20/2021	1.69																	59280
327	11/21/2021	1.7																	59550
328	11/22/2021	1.61	220	276	2.4	98.9%	0.3	20.4	14.6	97.9%	0.16		0.45	64.4%	3	360	240.9	98.8%	59720
329	11/23/2021	1.72	240	188	6	97.5%	0.3	20.4	19.2	98.4%	0.164		4.224	96.1%	3	350	245	98.8%	59540
330	11/24/2021	1.66	324	256	1.6	99.5%	0.3	23.9	32.3	99.1%	0.15	4.409	4.116	96.4%	3.8	360	320	98.8%	59390
331	11/25/2021	1.54	176		2.4	98.6%	19.8		42.8	53.7%	0.119		4.684	97.5%	3.3		196.7	98.3%	59340
332	11/26/2021	1.36	236		3.6	98.5%	15.5		55.5	72.1%	0.141		5.321	97.4%	3		270	98.9%	60210
333	11/27/2021	1.34																	57180
334	11/28/2021	1.34																	55460
335	11/29/2021	1.5	212	220	4.4	97.9%	0.3	31.5	29.6	99.0%	0.212		3.5	93.9%	3	350	295	99.0%	54570
336	11/30/2021	1.76	196	196	5.6	97.1%	0.3	17.4	22.6	98.7%	0.199		4.342	95.4%	4.55	360	280	98.4%	50950
337	12/1/2021	1.72	156	224	5.6	96.4%	0.3	20.9	25.1	98.8%	0.155		3.93	96.1%	3.8	225	250.8	98.5%	49840
338	12/2/2021	1.73	64	212	5.2	91.9%	0.3	13.6	20.2	98.5%	0.157	3.739	3.167	95.0%	3.6	350	300	98.8%	50630
339	12/3/2021	1.69	132	192	7.2	94.5%	0.3	14.2	17.3	98.3%	0.162		3.813	95.8%	5.4	280	197.5	97.3%	50300
340	12/4/2021	1.47																	50560
341	12/5/2021	1.37																	50620
342	12/6/2021	1.47	236	264	3.2	98.6%	0.3	20.5	31	99.0%	0.182		4.645	96.1%	3	350	325	99.1%	50590
343	12/7/2021	1.79	160	264	5.6	96.5%	0.3	17.3	18.2	98.4%	0.189		3.529	94.6%	3.6	305	252.5	98.6%	50490
344	12/8/2021	1.65	164	236	3.6	97.8%	0.3	18.5	13.6	97.8%	0.12		3.088	96.1%	5.1	300	234.2	97.8%	50570
345	12/9/2021	1.63	188	252	4	97.9%	0.3	21.8	22.3	98.7%	0.105		3.686	97.2%	7	365	>380	#VALUE!	50500

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
346	12/10/2021	1.68	228	288	6.4	97.2%	0.3	19.6	19.2	98.4%	0.101	3.833	3.646	97.2%	7	290	315	97.8%	50640
347	12/11/2021	1.72																	49390
348	12/12/2021	1.63																	49780
349	12/13/2021	1.55	152	232	2	98.7%	0.3	21.4	19.4	98.5%	0.112		3.764	97.0%	7.55	355	265	97.2%	48390
350	12/14/2021	1.82	176	244	5.6	96.8%	0.3	21.5	15.6	98.1%	0.124		3.225	96.2%	6.3	320	270	97.7%	47730
351	12/15/2021	1.74	72	212	4.8	93.3%	0.3	19.4	25.2	98.8%	0.11		2.599	95.8%	6.5	310	236.65	97.3%	47530
352	12/16/2021	1.79	164	204	1.2	99.3%	0.3	19.7	21.3	98.6%	0.103		3.176	96.8%	6.45	305	280	97.7%	47620
353	12/17/2021	1.53	268	212	1.6	99.4%	0.3	18.4	17.6	98.3%	0.1		4.616	97.8%	6.9	236.65	171.65	96.0%	47510
354	12/18/2021	1.54																	46590
355	12/19/2021	1.42																	46820
356	12/20/2021	1.39	232	208	5.6	97.6%	0.3	14.7	16.1	98.1%	0.1805	5.795	3.803	95.3%	3.6	325	325	98.9%	46430
357	12/21/2021	1.54	224	216	5.2	97.7%	0.3	14.5	13.2	97.7%	0.207		3.803	94.6%	3.6	247.5	190	98.1%	44710
358	12/22/2021	1.46	216	176	4	98.1%	0.3	13.1	17	98.2%	0.231		4.812	95.2%	3.6	223.4	145	97.5%	44150
359	12/23/2021	1.61	180		4.8	97.3%	2.43		20.3	88.0%	0.228		3.362	93.2%	4.1		137.5	97.0%	41130
360	12/24/2021	1.41	160		5.2	96.8%	1.89		20.5	90.8%	0.221		4.44	95.0%	4.2		231.7	98.2%	38800
361	12/25/2021	1.37																	36470
362	12/26/2021	1.23																	36560
363	12/27/2021	1.34	172	180	4.8	97.2%	10.4	134	129	91.9%	0.154		4.067	96.2%	3.45	270	290	98.8%	35560
364	12/28/2021	1.42	244	168	4	98.4%	11.5	73.4	113	89.8%	0.168		4.567	96.3%	3	204.17	255	98.8%	35570
365	12/29/2021	1.45	264	152	4	98.5%	7.75	79.8	77	89.9%	0.184	3.927	4.567	96.0%	3	183.35	239.15	98.7%	35600
366	12/30/2021	1.49	168		2.8	98.3%	6.96		129	94.6%	0.178		4.518	96.1%	3		172.5	98.3%	34600
367	12/31/2021	1.45	88		2.4	97.3%	4.14		74.3	94.4%	0.165		2.697	93.9%	3		227.5	98.7%	34640
368	1/1/2022	1.36																	34610
369	1/2/2022	1.28																	33740
370	1/3/2022	1.51	236	128	2	99.2%	5.97	128	141	95.8%	0.364		4.645	92.2%	3.3	208.4	255	98.7%	32790
371	1/4/2022	1.68	236	132	2.8	98.8%	6.33	141	127	95.0%	0.258		4.018	93.6%	12.4	218.4	285	95.6%	32700
372	1/5/2022	1.79	208	120	0.8	99.6%	0.3	15.7	9.08	96.7%	0.232		3.686	93.7%	19.6	216.65	230.85	91.5%	32590
373	1/6/2022	1.65	220	124	2.8	98.7%	0.3	18.1	15.2	98.0%	0.157	3.263	4.273	96.3%	18.9	239.15	240	92.1%	32520
374	1/7/2022	1.63	172	120	4.8	97.2%	0.3	16.3	14.1	97.9%	0.15		3.832	96.1%	12.75	255	270	95.3%	30820
375	1/8/2022	1.55																	30420
376	1/9/2022	1.49																	30660
377	1/10/2022	1.4	260	188	8.4	96.8%	0.3	27.1	24.2	98.8%	0.111		4.518	97.5%	6.7	233.35	310	97.8%	30650
378	1/11/2022	1.69	176	152	3.6	98.0%	0.3	19.6	11.2	97.3%	0.152		3.48	95.6%	5.6	255	250	97.8%	30570
379	1/12/2022	1.76	152	108	5.6	96.3%	0.3	21.1	20.9	98.6%	0.151		3.382	95.5%	6.4	260	290	97.8%	30530
380	1/13/2022	1.71	304	132	5.2	98.3%	0.3	22.5	21.7	98.6%	0.136		3.096	95.6%	5.4	260	255	97.9%	32270
381	1/14/2022	1.69	196	128	2.4	98.8%	0.3	33.3	30.5	99.0%	0.115	3.551	3.803	97.0%	5.4	275	295	98.2%	32580
382	1/15/2022	1.55																	32620
383	1/16/2022	1.43																	32500
384	1/17/2022	1.51	236		4.4	98.1%	1.12		18.7	94.0%	0.152		5.242	97.1%	6.3		285	97.8%	32590
385	1/18/2022	1.77	232	116	3.2	98.6%	2.01	17	13.5	85.1%	0.165		3.911	95.8%	5.7	265	250	97.7%	32670
386	1/19/2022	1.84	260	132	2.8	98.9%	0.38	14.5	14.5	97.4%	0.118		3.842	96.9%	6.2	280	285	97.8%	34290
387	1/20/2022	1.68	244	132	3.6	98.5%	0.3	24.7	30.1	99.0%	0.101		3.48	97.1%	8.8	270	315	97.2%	34510
388	1/21/2022	1.64	272	136	2.4	99.1%	0.3	17.1	11.5	97.4%	0.117		2.706	95.7%	10.5	275	275	96.2%	34590
389	1/22/2022	1.62																	34550
390	1/23/2022	1.41																	34440
391	1/24/2022	1.42	440	176	5.2	98.8%	1.5	30.4	34	95.6%	0.16	4.949	5.83	97.3%	7.2	237.5	380	98.1%	34820
392	1/25/2022	1.75	236	136	3.6	98.5%	2.49	22.6	20	87.6%	0.168		3.96	95.8%	6.75	265	223	97.0%	34590
393	1/26/2022	1.64	232	140	5.2	97.8%	0.3	16	17.6	98.3%	0.164		3.617	95.5%	0.65	253.4	285	99.8%	34690
394	1/27/2022	1.67	188	168	4.4	97.7%	0.3	26.3	17.1	98.2%	0.13		3.782	96.6%	7	245	270	97.4%	34420

City of Cadillac
WWTP MOR Data

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1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
395	1/28/2022	1.65	240	172	5.6	97.7%	0.3	20.7	21.3	98.6%	0.114		3.588	96.8%	6.9	390	315	97.8%	34560
396	1/29/2022	1.55																	34510
397	1/30/2022	1.43																	34540
398	1/31/2022	1.48	200	204	4	98.0%	0.336	34.5	36.7	99.1%	0.141		4.596	96.9%	5.9	242.5	260	97.7%	34360
399	2/1/2022	1.81	188	144	13.2	93.0%	0.51	21.9	21.5	97.6%	0.151	3.504	3.774	96.0%	6.1	244	255	97.6%	34650
400	2/2/2022	1.73	108	168	6.3	94.2%	0.3	23.4	23.8	98.7%	0.164		3.617	95.5%	6.6	290	300	97.8%	34470
401	2/3/2022	1.62	164	160	3.5	97.9%	0.3	26.1	18.6	98.4%	0.129		3.97	96.8%	6.2	300	340	98.2%	34600
402	2/4/2022	1.6	216	144	4.4	98.0%	0.3	19.8	17.8	98.3%	0.164		4.136	96.0%	7.6	270	315	97.6%	34630
403	2/5/2022	1.55																	34560
404	2/6/2022	1.46																	34560
405	2/7/2022	1.52	220	188	7.5	96.6%	0.3	18.1	19	98.4%	0.182		3.744	95.1%	6.15	238.34	250	97.5%	34520
406	2/8/2022	1.66	224	144	9	96.0%	0.451	18.1	23.7	98.1%	0.241		4.205	94.3%	6.67	220.84	270	97.5%	32780
407	2/9/2022	1.7	212	148	9.5	95.5%	0.498	16.9	18.9	97.4%	0.246	3.31	3.529	93.0%	5.9	184.2	305	98.1%	29760
408	2/10/2022	1.72	192	252	12.5	93.5%	0.3	17.7	19.5	98.5%	0.333		4.42	92.5%	10.8	305	345	96.9%	27950
409	2/11/2022	1.71	248	184	21	91.5%	0.3	16.2	18.2	98.4%	0.319		4.606	93.1%	8.8	248.4	355	97.5%	27540
410	2/12/2022	1.88																	25690
411	2/13/2022	1.37																	25610
412	2/14/2022	1.5	280	256	26	90.7%	1	25.1	25.2	96.0%	0.436		4.498	90.3%	7.4	275	300	97.5%	25670
413	2/15/2022	1.86	312	228	47	84.9%	0.817	21.3	19	95.7%	0.558		4.165	86.6%	12.9	239.2	340	96.2%	25420
414	2/16/2022	1.86	224	160	17	92.4%	0.412	18	15.6	97.4%	0.333		3.95	91.6%	7.2	247.5	295	97.6%	25530
415	2/17/2022	1.91	256	328	16	93.8%	0.364	21.6	19.6	98.1%	0.238	3.692	3.588	93.4%	7.3	285	315	97.7%	25630
416	2/18/2022	1.71	204	220	9	95.6%	0.3	22.3	20.9	98.6%	0.192		4.41	95.6%	6.7	260	310	97.8%	25640
417	2/19/2022	1.76																	25600
418	2/20/2022	1.59																	25380
419	2/21/2022	1.7	288	236	16	94.4%	0.895	24.9	24.4	96.3%	0.481		4.919	90.2%	10.6	225	310	96.6%	27020
420	2/22/2022	1.76	180	268	13	92.8%	0.569	21.9	19.5	97.1%	0.335		3.705	91.0%	6.9	253.35	235	97.1%	28660
421	2/23/2022	1.78	276	196	23	91.7%	0.48	21.9	22.8	97.9%	0.44		4.009	89.0%	10.08	237.5	325	96.9%	28520
422	2/24/2022	1.77	208	252	18	91.3%	0.368	20.6	20.5	98.2%	0.398		4.087	90.3%	9.6	240	310	96.9%	28460
423	2/25/2022	1.74	212	216	13	93.9%	0.363	18.5	18.8	98.1%	0.289	3.762	3.803	92.4%	8.38	246.65	295	97.2%	28820
424	2/26/2022	1.66																	28380
425	2/27/2022	1.72																	28550
426	2/28/2022	1.67	112	180	10	91.1%	0.388	31.6	23.2	98.3%	0.272		3.715	92.7%	5.8	202.5	186.7	96.9%	28490
427	3/1/2022	1.8	176	188	10	94.3%	0.581	26.4	22.8	97.5%	0.252		3.921	93.6%	9.3	330	265	96.5%	28590
428	3/2/2022	1.78	196	244	8	95.9%	0.3	27.2	29.8	99.0%	0.202		4.185	95.2%	6.5	300	305	97.9%	28400
429	3/3/2022	1.75	160	108	8.5	94.7%	0.3	25.8	24.4	98.8%	0.188		3.313	94.3%	6.58	240	244.15	97.3%	29280
430	3/4/2022	1.73	244	204	6	97.5%	0.3	25.6	25.5	98.8%	0.175		3.353	94.8%	7.6	305	340	97.8%	30810
431	3/5/2022	1.61																	30980
432	3/6/2022	1.71																	31270
433	3/7/2022	1.71	280	360	7	97.5%	0.36	25	20.5	98.2%	0.181	3.516	3.578	94.9%	5.025	320	385	98.7%	31270
434	3/8/2022	1.89	292	164	10	96.6%	0.877	23.3	21.7	96.0%	0.185		2.452	92.5%	6.2	241.7	265	97.7%	31090
435	3/9/2022	1.88	196	232	6	96.9%	0.392	21.6	20.9	98.1%	0.142		3.294	95.7%	5.375	233.35	234.15	97.7%	31060
436	3/10/2022	1.82	224	168	7	96.9%	0.627	24.1	24.4	97.4%	0.161		3.627	95.6%	6.92	385	390	98.2%	31080
437	3/11/2022	1.75	272	164	5	98.2%	0.85	23.2	23.5	96.4%	0.14		2.706	94.8%	16.5	385	390	95.8%	31150
438	3/12/2022	1.59																	32910
439	3/13/2022	1.39																	33490
440	3/14/2022	1.56	192	312	10	94.8%	0.3	14.6	17.6	98.3%	0.115		4.136	97.2%	9.22	350	325	97.2%	34900
441	3/15/2022	1.9	116	336	7	94.0%	0.3	18.7	18.1	98.3%	0.107	3.386	3.47	96.9%	7.3	275	255	97.1%	34860
442	3/16/2022	1.84	176	244	6	96.6%	0.329	16.6	16.5	98.0%	0.082		3.695	97.8%	6.55	241.7	260	97.5%	34950
443	3/17/2022	1.86	156	276	8	94.9%	0.86	18.9	18.1	95.2%	0.099		3.539	97.2%	6.7	370	340	98.0%	34740

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
444	3/18/2022	1.79	192	300	5	97.4%	0.562	20.5	19.4	97.1%	0.087		3.94	97.8%	8.22	265	270	97.0%	34870
445	3/19/2022	1.59																	34140
446	3/20/2022	1.49																	33820
447	3/21/2022	1.76	164	256	14	91.5%	3.04	15.5	11.7	74.0%	0.237		3.774	93.7%	7.73	255	200	96.1%	33880
448	3/22/2022	1.82	172	260	8	95.3%	1.7	18.1	17.5	90.3%	0.221		4.058	94.6%	8.35	247.5	240	96.5%	32820
449	3/23/2022	1.81	240	280	10	95.8%	1.04	19.8	18.9	94.5%	0.175	3.944	3.705	95.3%	7.4	225	229.15	96.8%	32940
450	3/24/2022	1.8	160	228	12	92.5%	0.7505	17.6	14.8	94.9%	0.174		4.126	95.8%	6.33	233.35	220.85	97.1%	32820
451	3/25/2022	1.75	212	324	6	97.2%	0.345	18.3	15.3	97.7%	0.153		4.508	96.6%	6.15	214.15	199.15	96.9%	31840
452	3/26/2022	1.6																	30950
453	3/27/2022	1.45																	29440
454	3/28/2022	1.51	196	292	11	94.4%	2.7	19.7	15.9	83.0%	0.272		4.42	93.8%	7.8	225.9	252.5	96.9%	29490
455	3/29/2022	1.83	288	320	13	95.5%	3.71	21.5	18.3	79.7%	0.221		4.067	94.6%	10.9	247.5	240	95.5%	29410
456	3/30/2022	1.87	176	196	11	93.8%	1.09	18.2	16.9	93.6%	0.16		3.431	95.3%	10.62	197.5	186.65	94.3%	29490
457	3/31/2022	1.97	180	284	9.5	94.7%	0.508	13.3	12.8	96.0%	0.132	2.887	2.863	95.4%	6.5	235	315	97.9%	29360
458	4/1/2022	2.12	104	272	9	91.3%	1.46	17.1	15.5	90.6%	0.158		3.294	95.2%	10.1	265	198.3	94.9%	29320
459	4/2/2022	1.956																	31240
460	4/3/2022	1.787																	31330
461	4/4/2022	1.763	348	332	11	96.8%	0.472	19.5	18.8	97.5%	0.206		4.655	95.6%	6.86	209	189	96.4%	31460
462	4/5/2022	1.964	168	300	10	94.0%	1.18	21.5	16.4	92.8%	0.227		3.891	94.2%	6.53	240.5	241	97.3%	31280
463	4/6/2022	2.031	228	224	10	95.6%	0.643	25.7	19.7	96.7%	0.171		3.343	94.9%	8.6	237.5	240	96.4%	33150
464	4/7/2022	2.16	208	196	8	96.2%	0.3	15.8	14.9	98.0%	0.144		3.457	95.8%	7.3	245	220.8	96.7%	36910
465	4/8/2022	2.002	212	292	9	95.8%	0.3	25.6	19.7	98.5%	0.141	2.758	2.93	95.2%	7.6	270	222.5	96.6%	41030
466	4/9/2022	1.963																	43410
467	4/10/2022	1.863																	43290
468	4/11/2022	1.933	432	200	8	98.1%	OOR	21.1	10.5		0.174		5.972		6	205.7	365		43270
469	4/12/2022	1.992	352	204	7	98.0%	0.335	29.3	25	98.7%	0.167		5.077	96.7%	5.8	247.5	305	98.1%	46870
470	4/13/2022	1.966	280	300	7	97.5%	0.3	21.5	31.9	99.1%	0.153		4.569	96.7%	5.7	280	310	98.2%	52950
471	4/14/2022	1.967	196	292	11	94.4%	0.3	15.2	20.6	98.5%	0.183		3.391	94.6%	6.7	265	310	97.8%	58000
472	4/15/2022	1.851	224		6	97.3%	0.3		16.6	98.2%	0.142		3.994	96.4%	6.6		189.2	96.5%	62850
473	4/16/2022	1.594																	63500
474	4/17/2022	1.532																	63180
475	4/18/2022	1.622	212	280	8	96.2%	0.3	17.5	15.1	98.0%	0.185		3.759	95.1%	5.65	265	260	97.8%	63120
476	4/19/2022	1.88	484	328	10	97.9%	0.3	17	27.6	98.9%	0.207	2.261	4.993	95.9%	7.1	320	340	97.9%	63290
477	4/20/2022	1.892	244	236	10	95.9%	0.316	16.1	20.5	98.5%	0.17		4.173	95.9%	10.1	320	275	96.3%	64120
478	4/21/2022	1.953	244	300	8	96.7%	0.3	17.1	25.2	98.8%	0.183		3.834	95.2%	9.4	315	330	97.2%	68820
479	4/22/2022	1.963	212	384	15	92.9%	0.379	17.3	23.9	98.4%	0.202		3.57	94.3%	10.9	325	375	97.1%	68610
480	4/23/2022	2.077																	69190
481	4/24/2022	2.101																	69320
482	4/25/2022	2.136	176	304	12	93.2%	1.79	13.9	17.1	89.5%	0.202		3.532	94.3%	6.45	213.3	216	97.0%	69170
483	4/26/2022	2.139	248	240	12	95.2%	2.25	18	22.5	90.0%	0.183		3.683	95.0%	10.22	280	290	96.5%	69250
484	4/27/2022	2.196	148	176	8	94.6%	2.43	16.9	23.8	89.8%	0.158	2.289	3.278	95.2%	21	255	265	92.1%	67450
485	4/28/2022	2.103	184	412	12	93.5%	1.6	16.9	21.4	92.5%	0.144		3.73	96.1%	22.5	310	238.4	90.6%	67140
486	4/29/2022	1.939	280	328	12	95.7%	1.29	15.6	25.5	94.9%	0.137		4.682	97.1%	15.4	275	305	95.0%	67280
487	4/30/2022	1.962																	64670
488	5/1/2022	1.856																	64250
489	5/2/2022	1.817	284	264	13	95.4%	2.93	16.8	21	86.0%	0.15		5.539	97.3%	5.9	255	270	97.8%	64260
490	5/3/2022	2.006	320	160	10.7	96.7%	3.14	17.7	25.3	87.6%	0.161		3.41	95.3%	8.4	255	340	97.5%	62210
491	5/4/2022	1.967	256	164	9.3	96.4%	1.13	14.5	20.9	94.6%	0.113		5.106	97.8%	9.6	239.15	335	97.1%	61180
492	5/5/2022	1.91	220	172	6	97.3%	0.774	15	20.7	96.3%	0.127	3.029	4.032	96.9%	7.3	265	265	97.2%	61090

City of Cadillac
WWTP MOR Data

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1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
493	5/6/2022	1.957	176	216	6.4	96.4%	0.521	16.6	21.2	97.5%	0.129		4.06	96.8%	8.6	223.35	225	96.2%	60950
494	5/7/2022	1.937																	61090
495	5/8/2022	1.717																	61020
496	5/9/2022	1.682	224	264	9.6	95.7%	1.18	18.4	31.5	96.3%	0.162		5.068	96.8%	6.3	239.17	260	97.6%	61270
497	5/10/2022	2.027	252	184	7.3	97.1%	1.55	13.7	21.2	92.7%	0.198		4.117	95.2%	6.6	275	285	97.7%	60910
498	5/11/2022	2.1	200	164	8	96.0%	0.87	18	22.6	96.2%	0.168		3.41	95.1%	7.5	230	264.15	97.2%	59090
499	5/12/2022	2.165	240	280	10	95.8%	0.959	19.5	21.7	95.6%	0.17		3.768	95.5%	9.6	305	265	96.4%	59100
500	5/13/2022	2.089	264	1168	12	95.5%	0.706	21.4	17.6	96.0%	0.228	10.575	4.522	95.0%	9.3	350	290	96.8%	59200
501	5/14/2022	2.014																	58990
502	5/15/2022	1.829																	59170
503	5/16/2022	2.003	320	316	62	80.6%	1.18	14.2	14.5	91.9%	0.787		6.547	88.0%	17.7	285	224.15	92.1%	59380
504	5/17/2022	2.056	324	196	50	84.6%	0.92	14.8	14	93.4%	0.753		5.972	87.4%	17.4	260	285	93.9%	58900
505	5/18/2022	2.063	280	280	26	90.7%	0.215	10.2	12.7	98.3%	0.51		4.832	89.4%	11.2	255	260	95.7%	59140
506	5/19/2022	1.948	152	236	14	90.8%	0.135	13.3	15.38	99.1%	0.373		4.22	91.2%	10.7	320	305	96.5%	57390
507	5/20/2022	1.972	232	520	13.3	94.3%	0.982	13	18.6	94.7%	0.379		4.663	91.9%	13.8	350	270	94.9%	59070
508	5/21/2022	1.908																	59130
509	5/22/2022	1.775																	59160
510	5/23/2022	1.786	200	748	10	95.0%	0.63	23.5	24.9	97.5%	0.235	10.751	4.201	94.4%	12.6	375	245	94.9%	58870
511	5/24/2022	1.912	320	308	6.7	97.9%	0.3	12.4	23.8	98.7%	0.248		5.096	95.1%	8.8	375	345	97.4%	59150
512	5/25/2022	1.959	268	220	8.7	96.8%	0.3	18.4	28.7	99.0%	0.251		4.38	94.3%	11.13	365	285	96.1%	59120
513	5/26/2022	1.948	176	516	6	96.6%	0.3	15.4	19.4	98.5%	0.227		3.787	94.0%	12.4	360	270	95.4%	59070
514	5/27/2022	1.996	220	460	7.3	96.7%	0.3	14.3	13.1	97.7%	0.227		3.391	93.3%	16.4	350	245	93.3%	59000
515	5/28/2022	1.907																	59000
516	5/29/2022	1.76																	59080
517	5/30/2022	1.656	356		11.3	96.8%	0.3		22.1	98.6%	0.318		5.727	94.4%	8.7		252.5	96.6%	59200
518	5/31/2022	1.832	232	632	10	95.7%	0.3	15.7	24.3	98.8%	0.328		4.644	92.9%	5.6	360	250	97.8%	58790
519	6/1/2022	2.15	340	352	62	81.8%	0.3	14	17.9	98.3%	0.873	5.516	5.2	83.2%	15	239.2	265	94.3%	59290
520	6/2/2022	1.96	280	548	15	94.6%	0.3	14.9	21.4	98.6%	0.331		4.107	91.9%	8.9	330	365	97.6%	58960
521	6/3/2022	1.97	260	320	9	96.5%	0.3	14.2	18.3	98.4%	0.286		3.9	92.7%	10.4	365	315	96.7%	58990
522	6/4/2022	1.96																	59110
523	6/5/2022	1.75																	58940
524	6/6/2022	1.73	268	488	6	97.8%	0.3	16.7	27.2	98.9%	0.345		4.851	92.9%	5.85	360	242.5	97.6%	59080
525	6/7/2022	1.9	308	436	9	97.1%	0.3	17.9	24.9	98.8%	0.417		4.324	90.4%	7.38	365	290	97.5%	59180
526	6/8/2022	2.05	240	836	14	94.2%	0.3	41	27	98.9%	0.362		5.002	92.8%	11.8	360	305	96.1%	60850
527	6/9/2022	2.06	108	508	49	54.6%	0.3	22.2	23.1	98.7%	0.691	6.783	3.523	80.4%	15.5	365	315	95.1%	61230
528	6/10/2022	2.01	292	168	21	92.8%	0.3	24.4	19.7	98.5%	0.679		4.889	86.1%	12.8	285	350	96.3%	65720
529	6/11/2022	1.95																	66150
530	6/12/2022	1.68																	65910
531	6/13/2022	1.78	268	472	7	97.4%	0.3	20.3	19	98.4%	0.397		4.041	90.2%	3.6	340	251.67	98.6%	68030
532	6/14/2022	1.97	64	492	10	84.4%	0.3	22.2	16.6	98.2%	0.37		2.6	85.8%	5.675	365	275	97.9%	68070
533	6/15/2022	2.05	120	392	8	93.3%	0.3	18.6	17.3	98.3%	0.299		3.222	90.7%	5.4	360	335	98.4%	69980
534	6/16/2022	2.28	168	424	8	95.2%	0.3	19.9	15.6	98.1%	0.297		3.58	91.7%	5.6	355	245	97.7%	71190
535	6/17/2022	2.12	144	536	10	93.1%	0.3	17.6	20.5	98.5%	0.317	6.297	3.523	91.0%	5.4	355	220.8	97.6%	71980
536	6/18/2022	1.89																	72920
537	6/19/2022	1.58																	73010
538	6/20/2022	1.68	264	624	9	96.6%	0.3	27.4	27.9	98.9%	0.343		4.691	92.7%	3	365	320	99.1%	73240
539	6/21/2022	2.11	244	552	7.3	97.0%	0.3	25.7	19.3	98.4%	0.31		3.636	91.5%	3.3	365	290	98.9%	74000
540	6/22/2022	2.08	224	448	4	98.2%	0.3	19.1	18.8	98.4%	0.262		3.288	92.0%	4.6	365	290	98.4%	74220
541	6/23/2022	2.02	272	528	10	96.3%	0.3	19	20.2	98.5%	0.266		3.608	92.6%	6.2	360	255	97.6%	74020

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2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
542	6/24/2022	2.06	256	472	10	96.1%	0.3	20.4	20.1	98.5%	0.274		3.824	92.8%	5.3	355	250	97.9%	74920
543	6/25/2022	1.96																	75160
544	6/26/2022	1.71																	74980
545	6/27/2022	1.56	268	596	2.7	99.0%	0.3	23.5	25.2	98.8%	0.289	2.826	4.399	93.4%	5.4	365	275	98.0%	75170
546	6/28/2022	1.79	380	444	6	98.4%	0.3	23.3	22.2	98.6%	0.233		4.785	95.1%	3.2	335	199.2	98.4%	75040
547	6/29/2022	1.84	248	576	4.8	98.1%	0.3	26.2	23	98.7%	0.283		3.919	92.8%	3	167	191.5	98.4%	76160
548	6/30/2022	1.84	232	616	6.4	97.2%	0.3	19.3	21.7	98.6%	0.29		4.946	94.1%	3	365	199.15	98.5%	77050
549	7/1/2022	1.91	192	492	4.4	97.7%	0.3	23.8	16.6	98.2%	0.245		3.768	93.5%	3.3	370	196	98.3%	77230
550	7/2/2022	1.86																	76250
551	7/3/2022	1.69																	75130
552	7/4/2022	1.69	300		4.4	98.5%					0.231		5.859		3		330		74290
553	7/5/2022	1.75	232	468	3.6	98.4%	0.3	23.7	24.5	98.8%	0.213		4.484	95.2%	3	325	167.5	98.2%	72860
554	7/6/2022	2.15	180	508	2.8	98.4%	0.3	16.6	16.1	98.1%	0.201	6.042	2.986	93.3%	3	335	155.85	98.1%	73190
555	7/7/2022	2.08	200	440	2.8	98.6%	0.3	15.6	18.5	98.4%	0.2		3.985	95.0%	3	330	190.85	98.4%	71360
556	7/8/2022	2.09	136	344	3.2	97.6%	0.3	19.1	20.7	98.6%	0.227		3.532	93.6%	4.6	345	193.35	97.6%	71800
557	7/9/2022	2																	72040
558	7/10/2022	1.79																	72080
559	7/11/2022	1.83	336	432	2.4	99.3%	0.3	20.5	27.9	98.9%	0.255		4.663	94.5%	3	320	225	98.7%	72510
560	7/12/2022	2.07	216	300	3.6	98.3%	0.3	19.3	20.3	98.5%	0.207		3.909	94.7%	3.6	370	370	99.0%	71990
561	7/13/2022	2.12	152	300	2.4	98.4%	0.3	18.9	19.4	98.5%	0.195		3.448	94.3%	8	365	210	96.2%	72140
562	7/14/2022	2.17	132	288	3.4	97.4%	0.3	18.9	19.3	98.4%	0.1655	4.414	3.25	94.9%	10.3	340	212.5	95.2%	72010
563	7/15/2022	1.97	164	208	6.8	95.9%	0.3	16.7	18.1	98.3%	0.153		3.193	95.2%	8.9	320	227.5	96.1%	72000
564	7/16/2022	2.03																	72300
565	7/17/2022	1.98																	72710
566	7/18/2022	1.89	204	248	2.4	98.8%	0.3	18.6	18.1	98.3%	0.166		4.474	96.3%	3.3	300	205	98.4%	71910
567	7/19/2022	2.13	252	280	2.8	98.9%	0.3	18	15.3	98.0%	0.196		3.429	94.3%	3	270	202.5	98.5%	71870
568	7/20/2022	2.11	344	344	2.4	99.3%	0.3	21.9	19.6	98.5%	0.192		2.713	92.9%	5	397	280	98.2%	72250
569	7/21/2022	2.15	136	252	2.8	97.9%	0.3	21.2	16.8	98.2%	0.187		3.514	94.7%	3.88	350	270	98.6%	72140
570	7/22/2022	2.12	144	452	2	98.6%	0.3	15	20.6	98.5%	0.155	2.877	3.419	95.5%	5.3	355	240	97.8%	73000
571	7/23/2022	2.05																	73800
572	7/24/2022	1.89																	75170
573	7/25/2022	1.93	220	432	3.2	98.5%	0.3	25.9	20	98.5%	0.188		3.777	95.0%	3	360	265	98.9%	75160
574	7/26/2022	2.12	164	224	3.2	98.0%	0.3	24.6	21.2	98.6%	0.209		3.231	93.5%	3	350	224.17	98.7%	74930
575	7/27/2022	2.16	156	340	5.2	96.7%	0.3	20.9	18.6	98.4%	0.224		3.184	93.0%	3.58	370	223.5	98.4%	77370
576	7/28/2022	2.09	116	244	5.2	95.5%	0.3	17.3	18.6	98.4%	0.199		3.109	93.6%	6.4	340	244.2	97.4%	77810
577	7/29/2022	2.09	124	128	3.6	97.1%	0.3	18.3	20.1	98.5%	0.166		3.193	94.8%	7	240	148.3	95.3%	79080
578	7/30/2022	2.03																	78960
579	7/31/2022	1.88																	78990
580	8/1/2022	1.95	224	280	7.2	96.8%	0.3	20.5	16.9	98.2%	0.205	5.528	4.267	95.2%	3.3	375	232.5	98.6%	79200
581	8/2/2022	1.93	196	296	4.8	97.6%	0.3	24.7	22	98.6%	0.19		3.514	94.6%	3	345	310	99.0%	78900
582	8/3/2022	2.05	260	1028	6	97.7%	0.3	21.2	22.3	98.7%	0.21		4.013	94.8%	4.4	355	231.7	98.1%	79130
583	8/4/2022	2.4	244	328	5.6	97.7%	0.3	19.9	22	98.6%	0.202		3.806	94.7%	5.5	360	325	98.3%	78950
584	8/5/2022	2.2	372	508	7.6	98.0%	0.3	17.3	21.2	98.6%	0.17		4.343	96.1%	9.9	360	285	96.5%	80150
585	8/6/2022	2.18																	80020
586	8/7/2022	1.99																	80090
587	8/8/2022	1.98	292	560	8.8	97.0%	0.3	32.6	29	99.0%	0.219		4.644	95.3%	4.8	370	335	98.6%	80090
588	8/9/2022	2.2	276	496	4.8	98.3%	0.3	13.8	13.1	97.7%	0.226	6.183	3.768	94.0%	3	370	265	98.9%	77360
589	8/10/2022	2.11	212	380	7.2	96.6%	0.3	16.3	20	98.5%	0.214		3.58	94.0%	5.3	365	265	98.0%	77140
590	8/11/2022	2.05	164	364	7.6	95.4%	0.3	15.9	16.4	98.2%	0.198		3.024	93.5%	6.4	370	275	97.7%	75950

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flov	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
591	8/12/2022	2.04	208	308	11.2	94.6%	0.3	18.4	17.2	98.3%	0.203		3.052	93.3%	9.4	360	212.5	95.6%	75460
592	8/13/2022	2.07																	75810
593	8/14/2022	1.91																	76080
594	8/15/2022	1.85	248	468	9.6	96.1%	0.3	31.3	35.1	99.1%	0.14		3.947	96.5%	5.25	365	228.5	97.7%	75910
595	8/16/2022	2.02	296	372	7.2	97.6%	0.3	14.9	7.78	96.1%	0.196		3.589	94.5%	3.6	360	275	98.7%	76050
596	8/17/2022	2.08	232	376	8	96.6%	0.3	9.55	11	97.3%	0.179	4.833	3.938	95.5%	5.4	370	320	98.3%	75660
597	8/18/2022	2.08	212	352	10.4	95.1%	0.3	28.3	23.3	98.7%	0.169		3.824	95.6%	6.4	370	253.3	97.5%	76010
598	8/19/2022	2.09	164	412	6.4	96.1%	0.3	19.1	22.5	98.7%	0.166		3.655	95.5%	8.5	340	221.7	96.2%	75760
599	8/20/2022	2.04																	75800
600	8/21/2022	1.7																	76100
601	8/22/2022	1.82	196	548	4	98.0%	0.3	25.7	21.9	98.6%	0.145		3.57	95.9%	3	380	223.5	98.7%	75730
602	8/23/2022	2.03	240	428	3.6	98.5%	0.3	21.9	21.4	98.6%	0.154		3.824	96.0%	3	375	275	98.9%	75770
603	8/24/2022	2.07	148	460	3.6	97.6%	0.3	18.7	17.7	98.3%	0.145		3.145	95.4%	3.2	370	230	98.6%	73890
604	8/25/2022	2.08	184	332	3.6	98.0%	0.3	15.3	15.4	98.1%	0.122	4.691	3.363	96.4%	6.9	350	290	97.6%	73750
605	8/26/2022	2.2	416	404	2.8	99.3%	0.3	17.4	19.4	98.5%	0.127		5.096	97.5%	6.9	370	315	97.8%	73840
606	8/27/2022	1.91																	73150
607	8/28/2022	1.79																	72590
608	8/29/2022	1.92	3344	476	3.6	99.9%	0.3	25.8	25.1	98.8%	0.158		7.894	98.0%	3	355	270	98.9%	73100
609	8/30/2022	2.14	604	476	5.2	99.1%	0.3	32.2	34.7	99.1%	0.19		7.649	97.5%	3	365	305	99.0%	72830
610	8/31/2022	2.02	184	472	4	97.8%	0.3	17.6	28.2	98.9%	0.199		3.617	94.5%	3.9	365	216.5	98.2%	63490
611	9/1/2022	1.892	136	384	4.4	96.8%	0.3	13	17.5	98.3%	0.203		3.429	94.1%	3.2	345	203.4	98.4%	63200
612	9/2/2022	2.207	308	372	12.8	95.8%	0.3	19.6	10.8	97.2%	0.222	5.997	3.796	94.2%	7.7	370	375	97.9%	72950
613	9/3/2022	2.077																	72810
614	9/4/2022	1.781																	75000
615	9/5/2022	1.546	248		6.6	97.3%	0.3		27.3	98.9%	0.211		5.661	96.3%	4.2		230.84	98.2%	73870
616	9/6/2022	1.757	508	628	5.3	99.0%	0.3	21.9	24.5	98.8%	0.263		7.319	96.4%	3.3	375	239.17	98.6%	72960
617	9/7/2022	2.045	532	488	8.4	98.4%	0.3	23.4	26.2	98.9%	0.288		7.687	96.3%	4.6	375	290	98.4%	72980
618	9/8/2022	2.042	724	448	10.7	98.5%	0.3	20.8	19	98.4%	0.266		3.89	93.2%	5.9	340	231.7	97.5%	72780
619	9/9/2022	2.094	256	424	7.3	97.1%	0.3	25.2	16.8	98.2%	0.225		4.352	94.8%	9.1	370	255	96.4%	72740
620	9/10/2022	1.967																	74150
621	9/11/2022	1.825																	73860
622	9/12/2022	1.826	488	420	5.3	98.9%	0.3	29.1	23.9	98.7%	0.188	7.551	5.944	96.8%	4.5	355	360	98.8%	73850
623	9/13/2022	2.033	512	356	6	98.8%	0.3	23.7	20.9	98.6%	0.24		5.793	95.9%	8.5	350	370	97.7%	74100
624	9/14/2022	2.122	184	344	5.2	97.2%	0.3	27.9	23.7	98.7%	0.176		3.664	95.2%	8.5	345	255	96.7%	73750
625	9/15/2022	2.009	172	320	3.6	97.9%	0.3	29.7	25	98.8%	0.198		4.022	95.1%	10.7	295	239.15	95.5%	74050
626	9/16/2022	2.024	280	204	4.4	98.4%	0.3	25.3	20.4	98.5%	0.145		3.975	96.4%	15.2	335	255	94.0%	73060
627	9/17/2022	1.902																	72190
628	9/18/2022	1.609																	71370
629	9/19/2022	1.99	284	384	3.6	98.7%	0.3	27.1	27.6	98.9%	0.164		4.701	96.5%	4.1	280	315	98.7%	70320
630	9/20/2022	1.977	208	300	4.4	97.9%	0.3	22.9	19.9	98.5%	0.154	5.409	3.843	96.0%	3.8	320	247.5	98.5%	68260
631	9/21/2022	1.963	252	320	8	96.8%	0.3	29.6	23	98.7%	0.187		4.927	96.2%	6.1	335	235	97.4%	67910
632	9/22/2022	1.889	300	296	10.8	96.4%	0.3	28.9	26.6	98.9%	0.151		5.878	97.4%	6.4	317.5	290	97.8%	66130
633	9/23/2022	1.836	308	296	4.4	98.6%	0.3	23.5	22.9	98.7%	0.117		5.727	98.0%	8.1	357.5	290	97.2%	63380
634	9/24/2022	1.736																	62550
635	9/25/2022	1.596																	62980
636	9/26/2022	1.623	152	408	10.4	93.2%	0.3	26.6	24.3	98.8%	0.163		4.474	96.4%	5.1	430	327.5	98.4%	63090
637	9/27/2022	1.905	264	292	6.4	97.6%	0.3	23	21.8	98.6%	0.177		4.936	96.4%	4.8	350	215	97.8%	61190
638	9/28/2022	1.945	256	248	7.6	97.0%	0.3	20.5	19.5	98.5%	0.149	5.194	4.606	96.8%	6.1	340	262.5	97.7%	60140
639	9/29/2022	1.882	204	320	6.4	96.9%	0.3	29.6	24.3	98.8%	0.151		4.361	96.5%	6.5	370	260	97.5%	60090

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
640	9/30/2022	1.899	264	244	8.4	96.8%	0.3	27.3	22.8	98.7%	0.196		5.473	96.4%	7.6	342.5	255	97.0%	59930
641	10/1/2022	1.84																	60060
642	10/2/2022	1.575																	60110
643	10/3/2022	1.557	700	444	10	98.6%	0.3	27.2	27.2	98.9%	0.225		6.368	96.5%	6.1	390	380	98.4%	60040
644	10/4/2022	1.9	224	330	12	94.6%	0.3	21.7	18.5	98.4%	0.2		3.165	93.7%	6.08	352.5	272.5	97.8%	60220
645	10/5/2022	1.922	172	432	9.5	94.5%	0.3	20.6	13.2	97.7%	0.187		3.862	95.2%	7.5	242.5	202.5	96.3%	59760
646	10/6/2022	1.918	188	344	13.5	92.8%	0.3	21.2	19.8	98.5%	0.244		4.06	94.0%	6.3	387.5	285	97.8%	61070
647	10/7/2022	1.934	208	396	11	94.7%	0.3	18	16.7	98.2%	0.234	5.415	3.853	93.9%	6.6	362.5	275	97.6%	61760
648	10/8/2022	1.811																	62070
649	10/9/2022	1.676																	62040
650	10/10/2022	1.638	540	416	15.5	97.1%	0.3	23.5	19.9	98.5%	0.251		6.632	96.2%	4.68	407.5	350	98.7%	63040
651	10/11/2022	1.923	300	396	17.5	94.2%	0.3	16.3	15	98.0%	0.323		5.464	94.1%	6.05	350	232.5	97.4%	63030
652	10/12/2022	1.971	196	444	18	90.8%	0.3	20.5	16.7	98.2%	0.331		4.098	91.9%	7.4	405	257.5	97.1%	63060
653	10/13/2022	1.98	168	368	12	92.9%	0.3	21.7	19.6	98.5%	0.337		3.809	91.2%	9.5	217.5	205	95.4%	62830
654	10/14/2022	1.873	212	400	21	90.1%	0.3	19.5	17.9	98.3%	0.277		4.192	93.4%	11.2	267.5	257.5	95.7%	64200
655	10/15/2022	1.838																	64850
656	10/16/2022	1.62																	65860
657	10/17/2022	1.653	256	292	14	94.5%	0.3	19.8	15	98.0%	0.385	3.708	4.465	91.4%	5.45	392.5	295	98.2%	67090
658	10/18/2022	1.879	180	548	12	93.3%	0.3	23.6	16.7	98.2%	0.293		3.721	92.1%	6.35	480	232.5	97.3%	67920
659	10/19/2022	2.055	204	580	11	94.6%	0.3	24.9	22.2	98.6%	0.293		3.787	92.3%	6.6	560	257.5	97.4%	68040
660	10/20/2022	1.992	160	180	13	91.9%	0.3	24	20	98.5%	0.278		3.523	92.1%	5.6	440	215	97.4%	70730
661	10/21/2022	1.938	180	216	10	94.4%	0.3	22.1	21.1	98.6%	0.253		3.457	92.7%	5.8	175	190	96.9%	70910
662	10/22/2022	1.848																	71120
663	10/23/2022	1.588																	70960
664	10/24/2022	1.698	244	480	10	95.9%	0.3	23.5	16.7	98.2%	0.321		4.682	93.1%	4.3	405	270	98.4%	70870
665	10/25/2022	1.94	252	440	11	95.6%	0.3	24.3	24	98.8%	0.362	5.33	4.032	91.0%	4.6	252.5	282.5	98.4%	70840
666	10/26/2022	1.906	208	370	8	96.2%	0.3	26.3	21.7	98.6%	0.312		3.655	91.5%	6.3	377.5	270	97.7%	72010
667	10/27/2022	1.883	152	410	8	94.7%	0.3	25.9	23.2	98.7%	0.307		4.107	92.5%	5	377.5	237.5	97.9%	71800
668	10/28/2022	1.725	224	244	9	96.0%	0.3	19.9	23.6	98.7%	0.276		3.956	93.0%	5.9	290	290	98.0%	71930
669	10/29/2022	1.627																	70050
670	10/30/2022	1.45																	68060
671	10/31/2022	1.651	148	200	4	97.3%	0.3	18.7	21	98.6%	0.254		4.211	94.0%	4.9	240	267.5	98.2%	67880
672	11/1/2022	1.56	160	172	4.8	97.0%	0.3	17.7	21.8	98.6%	0.223		3.429	93.5%	4.15	247.5	217.5	98.1%	67200
673	11/2/2022	1.87	100	164	4	96.0%	0.3	18.3	18.4	98.4%	0.192	3.674	3.306	94.2%	5.2	267.5	240	97.8%	65030
674	11/3/2022	1.82	120	180	5.6	95.3%	0.3	16.4	16.2	98.1%	0.174		3.853	95.5%	6.95	287.5	260	97.3%	65040
675	11/4/2022	1.82	160	124	7.2	95.5%	0.3	18.6	19.6	98.5%	0.192		3.768	94.9%	9.6	280	250	96.2%	64980
676	11/5/2022	1.77																	63890
677	11/6/2022	1.7																	66490
678	11/7/2022	1.67	208	264	7.2	96.5%	0.3	30.5	28.4	98.9%	0.189		3.58	94.7%	4.6	312.5	277.5	98.3%	63610
679	11/8/2022	1.68	360	180	4.8	98.7%	0.3	21.9	23.2	98.7%	0.164		3.193	94.9%	5.6	212.5	202.5	97.2%	61000
680	11/9/2022	1.91	228	200	5.6	97.5%	0.3	18.3	20.2	98.5%	0.126		3.316	96.2%	6.65	267.5	247.5	97.3%	61600
681	11/10/2022	1.8	108	168	6	94.4%	0.3	17.7	21.2	98.6%	0.137	3.64	3.542	96.1%	9.3	255	257.5	96.4%	61570
682	11/11/2022	1.77	264		8.4	96.8%	0.3		25.3	98.8%	0.147		3.297	95.5%	10.1		247.5	95.9%	61430
683	11/12/2022	1.77																	61260
684	11/13/2022	1.68																	61680
685	11/14/2022	1.47	280	240	6.8	97.6%	0.3	20.6	20.8	98.6%	0.178		3.975	95.5%	12.7	320	232.5	94.5%	61650
686	11/15/2022	1.54	168	282	12	92.9%	0.3	18.5	22.8	98.7%	0.18		3.137	94.3%	11.25	302.5	267.5	95.8%	61460
687	11/16/2022	1.87	488	340	13	97.3%	0.3	24	23.1	98.7%	0.179		3.457	94.8%	14.95	300	275	94.6%	61600
688	11/17/2022	1.76	328	296	19	94.2%	0.3	16.6	22.2	98.6%	0.159		3.222	95.1%	14.05	277.5	245	94.3%	61430

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	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
689	11/18/2022	1.74	276	340	14	94.9%	0.461	26.7	26.9	98.3%	0.185	3.487	3.344	94.5%	15.9	302.5	240	93.4%	61630
690	11/19/2022	1.69																	61370
691	11/20/2022	1.69																	61470
692	11/21/2022	1.7	200	176	17	91.5%	0.3	12.7	14	97.9%	0.209		4.135	94.9%	13.45	260	200	93.3%	61600
693	11/22/2022	1.61	256	232	14	94.5%	0.3	20.1	22	98.6%	0.222		3.589	93.8%	11.9	292.5	252.5	95.3%	61650
694	11/23/2022	1.72	224	296	22	90.2%	0.804	22.3	23.9	96.6%	0.228		3.514	93.5%	17.4	255	195	91.1%	60580
695	11/24/2022	1.66	204		16	92.2%	0.706		20.3	96.5%	0.21		3.834	94.5%	13.6		215	93.7%	60280
696	11/25/2022	1.54	252		13	94.8%	0.178		21.9	99.2%	0.218		4.371	95.0%	11.07		242.5	95.4%	59470
697	11/26/2022	1.36																	58710
698	11/27/2022	1.34																	58630
699	11/28/2022	1.34	320	320	20	93.8%	0.3	20	24.3	98.8%	0.321		4.87	93.4%	9.3	345	320	97.1%	58690
700	11/29/2022	1.5	196	216	14	92.9%	0.3	28.4	26.9	98.9%	0.358		3.843	90.7%	13.55	312.5	247.5	94.5%	56470
701	11/30/2022	1.76	172	256	12	93.0%	0.3	21.6	23	98.7%	0.226	4.064	3.466	93.5%	16.5	380	270	93.9%	54610
702	12/1/2022	1.72	264	224	8	97.0%	0.476	25.6	27.3	98.3%	0.18		4.889	96.3%	12.1	305	265	95.4%	54720
703	12/2/2022	1.73	304	332	13	95.7%	0.3	25.8	20.4	98.5%	0.193		4.201	95.4%	8.7	335	277.5	96.9%	54450

City of Cadillac
WWTP MOR Data

	A	B	D	E	H	I	P	S	T	U	V	Y	Z	AA	AB	AE	AF	AG	AN
1		Effluent Flow	RAW	PRIMARY	TER	Final SS													WASTE SLUDGE
2	Date	MGD	S.S.	S.S.	S.S.	% Rem	NH3 TER	NH3 PRI	NH3 RAW	NH3 %	PO4 TER	PO4 PRI	PO4 RAW	PO4 %	BOD TER	BOD PRI	BOD RAW	BOD %	
704	12/3/2022	1.69																	54610
705	12/4/2022	1.47																	53550
706	12/5/2022	1.37	372	340	12	96.8%	0.3	30.4	30.4	99.0%	0.171		4.418	96.1%	5.7	280	265	97.8%	52640
707	12/6/2022	1.47	284	356	7	97.5%	0.3	29.4	25.6	98.8%	0.193		3.645	94.7%	9.2	357.5	310	97.0%	52550
708	12/7/2022	1.79	220	392	5	97.7%	0.3	30.3	28.5	98.9%	0.156		3.608	95.7%	12.3	320	257.5	95.2%	52540
709	12/8/2022	1.65	292	292	6	97.9%	0.3	33	31.4	99.0%	0.137	4.047	4.569	97.0%	7.5	342.5	317.5	97.6%	52570
710	12/9/2022	1.63	524	328	8	98.5%	0.3	25.1	28.1	98.9%	0.137		6.877	98.0%	8.1	335	325	97.5%	52370
711	12/10/2022	1.68																	51550
712	12/11/2022	1.72																	51680
713	12/12/2022	1.63	332	296	7	97.9%	0.388	32.9	30.5	98.7%	0.165		4.437	96.3%	10.8	325	367.5	97.1%	50460
714	12/13/2022	1.55	280	200	6	97.9%	0.314	23.6	22.2	98.6%	0.168		5.096	96.7%	9.1	290	265	96.6%	50550
715	12/14/2022	1.82	260	210	4.7	98.2%	0.3	23.9	21	98.6%	0.136		3.862	96.5%	10.2	302.5	247.5	95.9%	50630
716	12/15/2022	1.74	150	180	4.7	96.9%	0.3	20.8	17.7	98.3%	0.123		4.295	97.1%	10.3	350	285	96.4%	49690
717	12/16/2022	1.79	210	230	4	98.1%	0.3	27.1	25.3	98.8%	0.115	3.08	4.333	97.3%	10.5	287.5	340	96.9%	49790
718	12/17/2022	1.53																	47940
719	12/18/2022	1.54																	46770
720	12/19/2022	1.42	212	196	13.2	93.8%	0.768	27.1	22.8	96.6%	0.163		4.305	96.2%	6	270	260	97.7%	45760
721	12/20/2022	1.39	250	290	3.6	98.6%	0.3	29.7	26.7	98.9%	0.158		4.098	96.1%	3.6	312.5	222.5	98.4%	44080
722	12/21/2022	1.54	190	130	3.6	98.1%	0.3	21.7	22	98.6%	0.146		3.702	96.1%	3.8	220	217.5	98.3%	42760
723	12/22/2022	1.46	300	170	2.8	99.1%	0.3	20.6	20.9	98.6%	0.138		5.482	97.5%	4.4	242.5	235	98.1%	42770
724	12/23/2022	1.61	230		2	99.1%	0.809		32.4	97.5%	0.133		4.814	97.2%	4.4		205	97.9%	42830
725	12/24/2022	1.41																	42530
726	12/25/2022	1.37																	43710
727	12/26/2022	1.23	150		2	98.7%	0.3		23.9	98.7%	0.154		5.238	97.1%	7.3		295	97.5%	41950
728	12/27/2022	1.34	540	420	4.4	99.2%	0.3	24	18.7	98.4%	0.151		4.333	96.5%	3	280	182.5	98.4%	42970
729	12/28/2022	1.42	240	144	2.4	99.0%	0.505	25.2	25.1	98.0%	0.155	3.578	4.71	96.7%	3	165	223	98.7%	42610
730	12/29/2022	1.45	360	148	3.2	99.1%	0.3	24.6	23.6	98.7%	0.18		6.905	97.4%	3	175	225	98.7%	42660
731	12/30/2022	1.49	180		4	97.8%	0.3		24.8	98.8%	0.166		5.303	96.9%	3		225	98.7%	42830
732	12/31/2022	1.45																	42840
733																			
734																			
735	AVERAGE	1.766	252	293	7.5		0.6	24	24		0.22	4.6	4.22		7	313	270		56984
736	JUNE-SEPT		266	346	6		0.3	22	21		0.3	5	4		6	337	249		
737	OCT-NOV		248	294	7		1	24	24		0.2	4	4		6	338	271		
738	DEC-MARCH		230	209	7		1	26	27		0.2	4	4		8	270	278		
739	APR-MAY		272	294	11		1	23	24		0.2	5	4		9	296	297		
740	MAY-MARCH										0.2	4.7	4.2						
741	APRIL										0.2	3.5	4.1						
742																			
743	OVERALL AVG			-16.4%	97.0%		97.3%	1.1%			94.7%	-9%			97.4%	-16%			
744																			
745	**AVERAGES**																		
746	JUNE-SEPT			-30%	97.6%	97.0%	98.6%	-6%		98.5%	94.2%	-20%		93.9%	97.7%	-35%		97.6%	
747	OCT-NOV			-18%	97.0%	96.7%	97.0%	-3%		97.7%	95.2%	-5%		94.7%	97.6%	-25%		97.5%	
748	DEC-MARCH			9%	96.9%	96.5%	96.5%	4%		96.3%	95.3%	9%		95.0%	97.3%	3%		97.2%	
749	APR-MAY			-8%	96.1%	95.9%	97.3%	3%		97.0%	94.3%	-12%		94.3%	96.9%	1%		96.8%	
750	MAY-MARCH										94.7%	-11%							
751	APRIL										95.2%	15%							
752	OCT-MAY			-6%	96.7%	96.3%													

INDUSTRIAL SAMPLES
January 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER	
14266	Hutchinson	1/4/2022	< 0.20	41.2	< 2.0	207	--	--	--	--	--	
14267	Munson	1/4/2022	--	--	--	--	--	--	--	< 1.0	Hg < 0.20	ug/L
14250	AAR 6th Ave	1/5/2022	2.0	73.7	< 2.0	119	--	< 20.0	34.0	--	--	
14251	AAR RT	1/5/2022	< 0.20	< 25.0	< 2.0	< 25.0	--	< 20.0	4.9	--	--	
14252	MI Rubber	1/6/2022	0.29	68.7	< 2.0	65.0	--	--	--	--	--	
14253	AAR RT	1/6/2022	--	--	--	--	--	< 20.0	6.7	--	--	
14254	FIAMM	1/7/2022	< 0.20	**182**	2.7	366	--	--	--	< 1.0	--	
14255	AAR FAC	1/7/2022	< 0.20	37.6	2.1	208	--	--	--	--	--	
14256	CCI	1/11/2022	**2.8**	47.2	23.7	**990**	100	< 20.0	--	1.3	--	
14257	CCI	1/12/2022	--	--	11.7	373	**190**	--	--	--	--	
14258	Arvco (8 hr)	1/12/2022	**4.7**	**223**	< 2.0	193	--	--	< 3.0	--	--	
14259	AKWEL	1/13/2022	< 0.20	< 25.0	< 2.0	71.2	--	< 20.0	--	--	--	
14260	Arvco (8 hr)	1/13/2022	0.76	**293**	7.5	260	--	--	< 3.0	--	--	
14261	AKWEL	1/14/2022	< 0.20	< 25.0	< 2.0	49.3	--	< 20.0	--	--	--	
14262	Rec Boat Holdings (8hr)	1/18/2022	0.31	**524**	7.7	**30700**	< 10.0	89.6	40.7	--	--	
14263	AKWEL 5th Street	1/19/2022	< 0.20	< 25.0	< 2.0	< 25.0	--	< 20.0	--	--	--	
14264	AKWEL 5th Street	1/20/2022	< 0.20	< 25.0	< 2.0	63.7	--	< 20.0	--	--	--	
14265	Piranha	1/21/2022	--	< 25.0	--	--	--	--	--	--	--	
	Reporting Limit		< 0.20	< 25.0	< 2.0	< 25.0	< 10.0	< 20.0	< 3.0	< 1.0		

Reported By: **Date:** 2/16/2022

INDUSTRIAL SAMPLES
February 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)
14268	CRE	2/1/2022	< 0.20	< 20.0	< 2.0	84.4	< 10.0	< 20.0	10.4
14269	Arvco (8 hr)	2/1/2022	0.44	**185**	2.4	203	--	--	3.5
14270	Spencer Plastics	2/2/2022	< 0.20	22.0	2.3	67.0	--	--	--
14271	Rec Boat Holdings (8hr)	2/2/2022	--	**368**	--	--	16.8	--	--
14272	Piranha	2/3/2022	--	21.5	--	--	--	--	--
14273	AAR 6th Ave	2/4/2022	0.88	40.1	< 2.0	57.0	--	< 20.0	29.5
14274	AKWEL 5th Street	2/8/2022	< 0.20	27.5	< 2.0	60.3	--	< 20.0	--
14275	AAR RT	2/8/2022	< 0.20	< 20.0	2.6	128	--	< 20.0	16.8
14276	AKWEL 5th Street	2/9/2022	0.21	24.2	< 2.0	30.0	--	< 20.0	--
14277	AAR RT	2/9/2022	--	--	--	--	--	< 20.0	16.2
14278	AKWEL	2/10/2022	< 0.20	< 20.0	< 2.0	160	--	< 20.0	--
14279	AAR FAC	2/10/2022	< 0.20	35.4	< 2.0	265	--	--	--
14280	AKWEL	2/11/2022	0.32	65.0	2.5	179	--	< 20.0	--
14281	CCI	2/15/2022	--	--	**140**	**7089**	**244**	--	--
14282	CCI	2/16/2022	--	--	**319**	**20866**	**181**	--	--
14283	Borg Warner	2/17/2022	< 0.20	30.6	4.4	192	--	--	--
14284	Arvco (8 hr)	2/21/2022	1.1	**292**	8.8	290	--	--	< 3.0
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0

Reported By:

Date: 3/14/2022

INDUSTRIAL SAMPLES
March 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
14285	CCI	3/1/2022	--	--	**116**	**4440**	**306**	--	--	--	--
14286	AAR RT	3/1/2022	< 0.20	< 20.0	2.0	< 20.0	--	< 20.0	8.6	--	--
14287	CCI	3/2/2022	--	--	**106**	**4490**	**167**	--	--	--	--
14288	AAR RT	3/2/2022	--	--	--	--	--	< 20.0	19.6	--	--
14289	AAR 6th Ave	3/3/2022	1.5	39.1	< 2.0	79.4	--	< 20.0	56.9	--	--
14290	AAR FAC	3/3/2022	< 0.20	37.8	3.1	250	--	--	--	--	--
14291	Piranha	3/4/2022	--	< 20.0	--	--	--	--	--	--	--
14292	AKWEL 5th Street	3/8/2022	< 0.20	54.6	2.1	47.0	--	< 20.0	--	--	--
14293	AKWEL 5th Street	3/9/2022	0.25	37.7	2.1	37.5	--	< 20.0	--	--	--
14294	AKWEL	3/10/2022	< 0.20	< 20.0	< 2.0	92.5	--	< 20.0	--	--	--
14295	AKWEL	3/11/2022	0.25	< 20.0	2.1	294	--	< 20.0	--	--	--
14296	Arvco (8 hr)	3/14/2022	< 0.20	77.4	< 2.0	82.3	--	--	< 3.0	--	--
14297	Arvco (8 hr)	3/15/2022	0.23	46.5	< 2.0	106	--	--	< 3.0	--	--
14298	Rec Boat Holdings (8hr)	3/16/2022	--	**747**	--	--	15.7	--	--	--	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50	

Reported By:

Date: 4/25/2022

INDUSTRIAL SAMPLES
April 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
14301	CCI	4/5/2022	1.7	40.2	14.4	138	**157**	< 20.0	--	< 3.0
14302	AAR RT	4/5/2022	< 0.20	20.3	4.2	77.5	--	< 20.0	23.8	--
14303	CCI	4/6/2022	--	--	39.5	103	**263**	--	--	--
14304	AAR RT	4/6/2022	--	--	--	--	--	< 20.0	14.3	--
14305	AKWEL	4/7/2022	< 0.20	< 20.0	2.1	83.5	--	< 20.0	--	--
14306	AAR FAC	4/7/2022	0.38	36.4	10.1	163	--	--	--	--
14307	AKWEL	4/8/2022	< 0.20	< 20.0	< 2.0	126	--	< 20.0	--	--
14308	Arvco (8 hr)	4/11/2022	< 0.20	23.6	2.2	37.6	--	--	< 3.0	--
14309	AKWEL 5th Street	4/12/2022	< 0.20	< 20.0	4.9	53.6	--	< 20.0	--	--
14310	Rec Boat Holdings (8hr)	4/12/2022	--	**604**	--	**--**	15.3	75.8	42.7	--
14311	AKWEL 5th Street	4/13/2022	< 0.20	< 20.0	2.3	35.3	--	< 20.0	--	--
14312	Arvco (8 hr)	4/13/2022	0.30	76.4	6.5	176	--	--	7.2	--
14313	AAR 6th Ave	4/14/2022	0.38	29.9	< 2.0	44.3	--	< 20.0	7.6	--
14314	Piranha	4/19/2022	--	24.8	--	--	--	--	--	--
14315	Borg Warner	4/20/2022	< 0.20	40.0	3.4	166	--	--	--	< 3.0
14316	FIAMM	4/21/2022	< 0.20	**224**	5.7	561	--	--	--	--
14317	MI Rubber	4/22/2022	0.29	**192**	3.7	177	--	--	--	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 3.0

Reported By:

Date: 6/22/2022

INDUSTRIAL SAMPLES
May 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
14318	AKWEL 5th Street	5/3/2022	< 0.20	21.6	2.1	48.8	--	< 20.0	--	--
14319	AAR RT	5/3/2022	< 0.20	< 20.0	< 2.0	< 20.0	--	< 20.0	< 3.0	--
14320	AKWEL 5th Street	5/4/2022	< 0.20	30.5	2.2	27.7	--	< 20.0	--	--
14321	AAR RT	5/4/2022	--	--	--	--	--	< 20.0	< 3.0	--
14322	CCI	5/4/2022	--	--	3.1	**1530**	**168**	--	--	--
14323	AKWEL	5/5/2022	< 0.20	< 20.0	< 2.0	90.1	--	< 20.0	--	--
14324	AAR FAC	5/5/2022	0.38	59.7	2.2	229	--	--	--	--
14325	AKWEL	5/6/2022	< 0.20	< 20.0	< 2.0	63.2	--	< 20.0	--	--
14326	Arvco (8 hr)	5/10/2022	< 0.20	42.0	< 2.0	100	--	--	< 3.0	--
14327	Spencer Plastics	5/10/2022	0.89	41.3	2.2	186	--	--	--	--
14328	Rec Boat Holdings (8hr)	5/10/2022	0.64	**578**	4.8	**6804**	13.4	--	--	--
14329	Piranha	5/11/2022	--	25.1	--	--	--	--	--	--
14330	Arvco (8 hr)	5/11/2022	< 0.20	47.8	< 2.0	174	--	--	< 3.0	--
14331	CRE	5/12/2022	0.22	26.5	< 2.0	< 20.0	< 10.0	< 20.0	3.6	--
14332	CCI	5/13/2022	--	--	2.4	90.5	**285**	--	--	--
14333	Hutchinson	5/17/2022	< 0.20	55.4	< 2.0	186	--	--	--	--
14334	AAR 6th Ave	5/18/2022	0.96	51.2	< 2.0	47.5	--	< 20.0	109	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50

Reported By:

Date: 6/24/2022

INDUSTRIAL SAMPLES
June 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
14335	AAR 6th Ave	6/2/2022	< 0.50	26.7	< 2.0	54.7	--	< 20.0	20.0	--
14336	Arvco (8 hr)	6/2/2022	0.56	327	< 2.0	409	--	--	< 3.0	--
14337	Piranha	6/3/2022	--	< 20.0	--	--	--	--	--	--
14338	AKWEL	6/7/2022	< 0.50	< 20.0	< 2.0	69.2	--	< 20.0	--	--
14339	AAR RT	6/7/2022	< 0.50	< 20.0	< 2.0	< 50.0	--	< 20.0	4.6	--
14340	AKWEL	6/8/2022	< 0.50	< 20.0	< 2.0	91.8	--	< 20.0	--	--
14341	AAR RT	6/8/2022	--	--	--	--	--	< 20.0	5.7	--
14342	AKWEL 5th Street	6/9/2022	< 0.50	< 20.0	< 2.0	< 50.0	--	< 20.0	--	--
14343	AAR FAC	6/9/2022	< 0.50	**112**	6.0	587	--	--	--	--
14344	AKWEL 5th Street	6/10/2022	< 0.50	< 20.0	< 2.0	< 50.0	--	< 20.0	--	--
14345	CCI	6/14/2022	--	--	4.4	184	74.9	--	--	--
14346	Arvco (8 hr)	6/14/2022	0.50	**315**	< 2.0	517	--	--	3.2	--
14347	CCI	6/15/2022	--	--	< 2.0	658	97.8	--	--	--
14348	Rec Boat Holdings (8hr)	6/15/2022	--	**979**	--	--	22.1	--	--	--
	Reporting Limit		< 0.50	< 20.0	< 2.0	< 50.0	< 10.0	< 20.0	< 3.0	< 0.50

Reported By:

Date: 7/27/2022

INDUSTRIAL SAMPLES
July 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
14349	CCI	7/6/2022	--	--	7.2	300	11.1	--	--	--
14350	AAR RT	7/6/2022	< 0.20	< 20.0	< 2.0	53.6	--	< 20.0	10.1	--
14351	CCI	7/7/2022	--	--	2.9	114	< 10.0	--	--	--
14352	AAR RT	7/7/2022	--	--	--	--	--	< 20.0	13.6	--
14353	AAR 6th Ave	7/8/2022	0.37	30.3	< 2.0	85.6	--	< 20.0	10.5	--
14354	AAR FAC	7/8/2022	1.9	**104**	< 2.0	327	--	--	--	--
14355	AKWEL	7/12/2022	< 0.20	< 20.0	< 2.0	30.1	--	< 20.0	--	--
14356	Arvco (8 hr)	7/12/2022	< 0.20	**1090.0**	8.0	44.1	--	--	< 3.0	--
14357	AKWEL	7/13/2022	< 0.20	< 20.0	< 2.0	74.3	--	< 20.0	--	--
14358	Rec Boat Holdings (8hr)	7/13/2022	**5.1**	**736**	4.6	**9150**	14.9	--	--	--
14359	AKWEL 5th Street	7/14/2022	< 0.20	29.5	< 2.0	34.3	--	< 20.0	--	--
14360	AKWEL 5th Street	7/15/2022	< 0.20	< 20.0	< 2.0	< 20.0	--	< 20.0	--	--
14361	Munson	7/19/2022	--	--	--	--	--	--	--	< 0.50
14362	Piranha	7/19/2022	--	< 20.0	--	--	--	--	--	--
14363	Arvco (8 hr)	7/19/2022	0.43	**534**	8.4	432	--	--	3.5	--
14364	Spencer Plastics	7/20/2022	< 0.20	33.0	< 2.0	40.2	--	--	--	--
14365	CRE	7/21/2022	< 0.20	80.4	< 2.0	39.1	42.8	< 20.0	< 3.0	--
14366	Borg Warner	7/22/2022	< 0.20	< 20.0	< 2.0	34.8	--	--	--	--
14367	Hutchinson	7/26/2022	< 0.20	48.5	2.6	424	--	--	--	--
14368	MI Rubber	7/27/2022	0.21	97.0	< 2.0	101	--	--	--	--
14369	FIAMM	7/28/2022	< 0.20	**185**	8.3	695	--	--	--	< 0.50
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50

Reported By: **Date:** 8/24/2022

INDUSTRIAL SAMPLES
August 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
19374	AAR 6th Ave	8/4/2022	1.49	50.0	< 2.0	84.8	--	< 20.0	137	--
19375	AAR FAC	8/4/2022	< 0.20	33.3	< 2.0	114	--	--	--	--
19371	AAR RT	8/2/2022	< 0.20	< 20.0	< 2.0	105	--	< 20.0	7.7	--
19373	AAR RT	8/3/2022	--	--	--	--	--	< 20.0	11.3	--
19370	AKWEL	8/2/2022	< 0.20	< 20.0	< 2.0	62.3	--	< 20.0	--	--
19372	AKWEL	8/3/2022	< 0.20	< 20.0	< 2.0	51.8	--	< 20.0	--	--
19381	AKWEL 5th Street	8/11/2022	< 0.20	< 20.0	< 2.0	< 20.0	--	< 20.0	--	--
19383	AKWEL 5th Street	8/12/2022	< 0.20	< 20.0	< 2.0	< 20.0	--	< 20.0	--	--
19380	Arvco (8 hr)	8/10/2022	0.25	51.5	< 2.0	241	--	--	< 3.0	--
19382	Arvco (8 hr)	8/11/2022	< 0.20	50.9	< 2.0	258	--	--	< 3.0	--
19377	CCI	8/9/2022	< 0.20	< 20.0	< 2.0	75.6	23.8	< 20.0	--	0.76
19379	CCI	8/10/2022	--	--	7.0	138	< 10.0	--	--	--
19376	Piranha	8/5/2022	--	< 20.0	--	--	--	--	--	--
19378	Rec Boat Holdings (8hr)	8/9/2022	--	**340**	--	--	< 10.0	< 20.0	16.2	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50

Reported By:

Date: 9/29/2022

INDUSTRIAL SAMPLES
September 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
14384	CCI	9/1/2022	--	--	14.4	498	**408**	--	--	--
14385	CCI	9/2/2022	--	--	2.0	< 20.0	**217**	--	--	--
14386	Rec Boat Holdings (8hr)	9/6/2022	**2.0**	**378**	3.6	**4176**	16.2	--	--	--
14387	AKWEL	9/7/2022	< 0.50	< 20.0	< 2.0	40.9	--	< 20.0	--	--
14388	AAR RT	9/7/2022	< 0.50	< 20.0	< 2.0	54.8	--	< 20.0	9.1	--
14389	AKWEL	9/8/2022	< 0.50	30.2	< 2.0	82.6	--	< 20.0	--	--
14390	AAR 6th Ave	9/9/2022	0.66	31.9	< 2.0	57.4	--	< 20.0	45.8	--
14391	AAR FAC	9/9/2022	0.55	48.9	< 2.0	289	--	--	--	--
14392	Arvco (8 hr)	9/12/2022	**3.3**	**299**	< 2.0	**860**	--	--	5.3	--
14393	Arvco (8 hr)	9/13/2022	< 0.50	**209**	< 2.0	161	--	--	< 3.0	--
14394	AKWEL 5th Street	9/14/2022	< 0.50	< 20.0	< 2.0	< 20.0	--	< 20.0	--	--
14395	Piranha	9/16/2022	--	25.0	--	--	--	--	--	--
14396	AKWEL 5th Street	9/15/2022	< 0.50	25.4	< 2.0	37.9	--	< 20.0	--	--
14397	AAR RT	9/8/2022	--	--	--	--	--	< 20.0	9.9	--
	Reporting Limit		< 0.50	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50

Reported By: **Date:** 10/26/2022

INDUSTRIAL SAMPLES
October 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
14398	AKWEL	10/4/2022	< 0.50	< 20.0	< 2.0	70.7	--	< 20.0	--	--
14399	AAR RT	10/5/2022	< 0.50	< 20.0	< 2.0	< 50.0	--	< 20.0	5.7	--
14400	CCI	10/5/2022	**19.0**	22.8	5.9	678	**156**	< 20.0	--	< 3.0
14401	CRE	10/6/2022	< 0.50	74.3	< 2.0	< 50.0	11.8	< 20.0	5.3	--
14402	Arvco (8 hr)	10/6/2022	< 0.50	**134**	**DNR**	155	--	--	< 3.0	--
14403	Spencer Plastics	10/7/2022	< 0.50	35.1	2.3	63.4	--	--	--	--
14404	AKWEL	10/12/2022	< 0.50	< 20.0	< 2.0	50.1	--	< 20.0	--	--
14405	CCI	10/11/2022	--	--	< 2.0	59.3	**209**	--	--	--
14406	Arvco (8 hr)	10/12/2022	< 0.50	**360**	14.1	198	--	--	9.1	--
14407	AAR 6th Ave	10/13/2022	1.0	30.2	< 2.0	56.3	--	< 20.0	8.8	--
14408	Rec Boat Holdings (8hr)	10/13/2022	--	**533**	--	--	10.4	< 20.0	11.0	--
14409	Piranha	10/14/2022	--	< 20.0	--	--	--	--	--	--
14410	AAR RT	10/18/2022	--	--	--	--	--	< 20.0	4.1	--
14411	Hutchinson	10/19/2022	< 0.50	30.5	< 2.0	251	--	--	--	--
14412	AKWEL 5th Street	10/20/2022	< 0.50	< 20.0	< 2.0	< 50.0	--	< 20.0	--	--
14413	AKWEL 5th Street	10/21/2022	< 0.50	< 20.0	< 2.0	< 50.0	--	< 20.0	--	--
14414	AAR FAC	10/26/2022	< 0.50	42.1	< 2.0	115	--	--	--	--
14415	MI Rubber	10/26/2022	< 0.50	63.5	< 2.0	51.1	--	--	--	--
14416	Borg Warner	10/27/2022	< 0.50	< 20.0	< 2.0	< 50.0	--	--	--	--
14417	FIAMM	10/27/2022	0.54	**104**	3.0	464	--	--	--	< 3.0
	Reporting Limit		< 0.50	< 20.0	< 2.0	< 50.0	< 10.0	< 20.0	< 3.0	< 3.0

Reported By:

Date: 11/23/2022

INDUSTRIAL SAMPLES
November 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
14418	AKWEL 5th Street	11/1/2022	< 0.20	< 20.0	< 2.0	23.6	--	< 20.0	--	--
14419	AAR RT	11/1/2022	< 0.20	< 20.0	< 2.0	< 20.0	--	< 20.0	4.3	--
14420	AAR RT	11/2/2022	--	--	--	--	--	< 20.0	4.7	--
14421	AKWEL 5th Street	11/2/2022	< 0.20	21.0	< 2.0	32.5	--	< 20.0	--	--
14422	Arvco (8 hr)	11/2/2022	< 0.20	52.0	< 2.0	32.7	--	--	< 3.0	--
14423	AKWEL	11/3/2022	< 0.20	< 20.0	< 2.0	75.4	--	< 20.0	--	--
14424	AAR FAC	11/3/2022	0.22	34.4	< 2.0	212	--	--	--	--
14425	Rec Boat Holdings (8hr)	11/3/2022	1.39	**240**	< 2.0	**956**	< 10.0	--	--	--
14426	AKWEL	11/4/2022	< 0.20	< 20.0	< 2.0	76.1	--	< 20.0	--	--
14427	Arvco (8 hr)	11/4/2022	0.45	**128**	< 2.0	118	--	--	< 3.0	--
14428	CCI	11/8/2022	--	--	2.8	57.0	**165**	--	--	--
14429	CCI	11/9/2022	--	--	7.0	236	**281**	--	--	--
14430	Piranha	11/10/2022	--	< 20.0	--	--	--	--	--	--
14431	AAR 6th Ave	11/10/2022	**2.2**	37.5	< 2.0	75.0	--	< 20.0	51.5	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.5

Reported By:

Date: 12/28/2022

INDUSTRIAL SAMPLES
December 2022

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
14432	AAR RT	12/6/2022	< 1.0	< 20.0	< 2.0	< 50.0	--	< 20.0	5.0	--
14433	CCI	12/6/2022	--	--	9.1	213	108	--	--	--
14434	Arvco (8 hr)	12/6/2022	< 1.0	96.7	< 2.0	< 50.0	--	--	< 5.0	--
14435	AAR RT	12/7/2022	--	--	--	--	--	< 20.0	5.5	--
14436	CCI	12/7/2022	--	--	< 2.0	82.9	**132**	--	--	--
14437	Rec Boat Holdings (8hr)	12/7/2022	--	**521**	--	--	13.7	--	--	--
14438	AAR FAC	12/8/2022	< 1.0	29.3	< 2.0	102	--	--	--	--
14439	AKWEL	12/8/2022	< 1.0	< 20.0	< 2.0	72.5	--	< 20.0	--	--
14440	AKWEL	12/9/2022	< 1.0	< 20.0	< 2.0	88.4	--	< 20.0	--	--
14441	Arvco (8 hr)	12/12/2022	< 1.0	**146**	< 2.0	71.1	--	--	< 5.0	--
14442	AKWEL 5th Street	12/13/2022	< 1.0	27.5	< 2.0	< 50.0	--	< 20.0	--	--
14443	AKWEL 5th Street	12/14/2022	< 1.0	25.4	< 2.0	< 50.0	--	< 20.0	--	--
14444	Piranha	12/15/2022	--	24.0	--	--	--	--	--	--
14445	AAR 6th Ave	12/16/2022	< 1.0	33.6	< 2.0	64.2	--	< 20.0	6.8	--
	Reporting Limit		< 1.0	< 20.0	< 2.0	< 50.0	< 10.0	< 20.0	< 5.0	< 3.0

Reported By:

Date: 1/18/2023

INDUSTRIAL SAMPLES
January 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
13951	AAR 6th Ave	1/6/2021	0.63	48.4	< 2.0	36.1	--	< 20.0	62.6	--
13952	AAR FAC	1/6/2021	< 0.50	82.6	< 2.0	180	--	--	11.0	--
13950	AAR RT	1/5/2021	< 0.50	< 20.0	< 2.0	< 20.0	--	< 20.0	< 3.0	--
13954	AAR RT	1/7/2021	--	--	--	--	--	< 20.0	3.2	--
13955	AKWEL	1/8/2021	< 0.50	< 20.0	< 2.0	85.1	--	< 20.0	--	--
13964	AKWEL	1/21/2021	< 0.50	< 20.0	< 2.0	47.0	--	< 20.0	--	--
13953	AKWEL 5th Street	1/7/2021	< 0.50	< 20.0	< 2.0	44.1	--	< 20.0	--	--
13965	AKWEL 5th Street	1/22/2021	< 0.50	22.3	< 2.0	33.1	--	< 20.0	--	--
13956	Arvco (8 hr)	1/11/2021	0.63	**305**	< 2.0	154	--	--	< 3.0	--
13958	Arvco (8 hr)	1/13/2021	**5.30**	**400**	2.7	**864**	--	--	3.1	--
13966	Borg Warner	1/26/2021	< 0.50	25.6	< 2.0	68.3	--	--	--	--
13949	CCI	1/5/2021	< 0.50	23.4	4.6	148	11.2	< 20.0	--	1.3
13963	CCI	1/20/2021	--	--	2.5	**5410**	39.3	--	--	--
13957	CRE (MS/MSD)	1/13/2021	< 0.50	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	4.3	--
13968	FIAMM	1/28/2021	0.62	**195**	3.1	608	--	--	--	< 0.50
13969	Hutchinson	1/29/2021	< 0.50	44.5	< 2.0	312	--	--	--	--
13967	MI Rubber	1/27/2021	< 0.50	**99.4**	2.8	139	--	--	--	--
13962	Munson	1/15/2021	--	--	--	--	--	--	Hg < 0.20	< 0.50
13959	Piranha	1/12/2021	--	24.9	--	--	--	--	--	--
13960	Rec Boat Holdings (8hr)	1/12/2021	< 0.50	**351**	3.5	**3140**	21.8	< 20.0	7.3	--
13961	Spencer Plastics	1/14/2021	< 0.50	87.5	3.9	235	--	--	--	--
	Reporting Limit		< 0.50	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50

Reported By: **Date:** 2/26/2021

INDUSTRIAL SAMPLES
February 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
13974	AAR 6th Ave	2/4/2021	0.71	29.4	< 2.0	49.9	--	< 20.0	13.8	--	--
13973	AAR FAC	2/3/2021	< 0.50	31.7	< 2.0	205	--	--	--	--	--
13971	AAR RT	2/2/2021	< 0.50	20.0	< 2.0	124	--	31.1	15.6	--	--
13975	AAR RT	2/4/2021	--	--	--	--	--	168	17.7	--	--
13972	AKWEL	2/3/2021	< 0.50	42.4	< 2.0	433	--	< 20.0	--	--	--
13978	AKWEL	2/19/2021	< 0.50	16.2	< 2.0	81.8	--	< 20.0	--	--	--
13976	AKWEL 5th Street	2/5/2021	< 0.50	58.3	2.0	70.2	--	< 20.0	--	--	--
13983	AKWEL 5th Street	2/24/2021	< 0.50	27.7	< 2.0	60.4	--	< 20.0	--	--	--
13979	Arvco (8 hr) (MS/MSD)	2/19/2021	1.2	**243**	11.4	142	--	--	2.7	--	--
13982	Arvco (8 hr)	2/23/2021	0.91	**332**	1.8	207	--	--	2.3	--	--
13970	CCI	2/2/2021	--	--	< 2.0	52.3	44.0	--	--	--	--
13977	CCI	2/18/2021	--	--	< 2.0	81.6	86.6	--	--	--	--
13981	Piranha	2/23/2021	--	26.7	--	--	--	--	--	--	--
13980	Rec Boat Holdings (8hr)	2/22/2021	--	**472**	--	--	16.5	--	--	--	CN=14 ug/L
	Reporting Limit		< 0.50	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50	

Reported By: **Date:** 3/30/2021

batch:	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS

INDUSTRIAL SAMPLES
March 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
13990	AAR 6th Ave	3/5/2021	1.3	28.7	< 2.0	51.5	--	< 20.0	20.9	--	--
13985	AAR FAC	3/2/2021	< 0.50	44.2	< 2.0	102	--	--	--	--	--
13989	AAR RT	3/4/2021	--	--	--	--	--	< 20.0	3.3	--	--
13987	AAR RT (MS/MSD)	3/3/2021	< 0.50	< 20.0	< 2.0	33.1	--	< 20.0	< 3.0	--	--
13986	AKWEL	3/3/2021	< 0.50	< 20.0	< 2.0	73.5	--	< 20.0	--	--	--
13996	AKWEL	3/11/2021	< 0.50	< 20.0	< 2.0	100	--	< 20.0	--	--	--
13988	AKWEL 5th Street	3/4/2021	< 0.50	28.1	< 2.0	32.1	--	< 20.0	--	--	--
13997	AKWEL 5th Street	3/12/2021	< 0.50	35.3	< 2.0	106	--	< 20.0	--	--	--
13991	Arvco (8 hr)	3/5/2021	**2.3**	**333**	< 2.0	279	--	--	< 3.0	--	--
13993	Arvco (8 hr)	3/9/2021	< 0.50	72.8	< 2.0	85.2	--	--	< 3.0	--	--
13984	CCI	3/2/2021	--	--	< 2.0	92.9	**245**	--	--	--	--
13994	CCI	3/10/2021	--	--	2.3	52.7	**246**	< 20.0	--	--	--
13992	Piranha	3/9/2021	--	31.6	--	--	--	--	--	--	--
13995	Rec Boat Holdings (8hr)	3/10/2021	< 0.50	**435**	4.9	**4660**	12.1	--	--	--	--
	Reporting Limit		< 0.50	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50	

Reported By: **Date:** 4/21/2021

	batch:	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS

INDUSTRIAL SAMPLES
April 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
13998	CCI (MS/MSD)	4/6/2021	1.6	< 20.0	< 2.0	43.2	**172**	< 20.0	--	< 0.50
13999	AAR RT	4/6/2021	< 0.50	< 20.0	< 2.0	< 20.0	--	< 20.0	3.4	--
14000	AKWEL	4/7/2021	< 0.50	< 20.0	< 2.0	83.5	--	< 20.0	--	--
14001	AAR FAC	4/7/2021	< 0.50	42.4	< 2.0	529	--	--	--	--
14002	AKWEL 5th Street	4/8/2021	< 0.50	20.2	< 2.0	34.3	--	< 20.0	--	--
14003	AAR RT	4/8/2021	--	--	--	--	--	< 20.0	8.8	--
14004	AAR 6th Ave	4/9/2021	< 0.50	29.6	< 2.0	38.1	--	< 20.0	4.4	--
14005	Arvco (8 hr)	4/13/2021	1.8	**282**	< 2.0	311	--	--	< 3.0	--
14006	Piranha	4/13/2021	--	30.8	--	--	--	--	--	--
14007	Rec Boat Holdings (8hr)	4/13/2021	--	**499**	--	--	17.3	< 20.0	8.0	--
14008	Spencer Plastics	4/14/2021	< 0.50	**117**	5.6	375	--	--	--	--
14009	Arvco (8 hr)	4/14/2021	**2.2**	**211**	< 2.0	475	--	--	< 3.0	--
14010	CRE	4/15/2021	< 0.50	24.3	< 2.0	< 20.0	< 10.0	< 20.0	6.1	--
14011	Borg Warner	4/16/2021	< 0.50	29.4	< 2.0	92.3	--	--	--	--
14012	CCI	4/20/2021	--	--	3.5	122	80.0	--	--	--
14013	AKWEL	4/21/2021	< 0.50	< 20.0	< 2.0	65.1	--	< 20.0	--	--
14014	AKWEL 5th Street	4/22/2021	< 0.50	26.3	< 2.0	33.1	--	< 20.0	--	--
14015	MI Rubber	4/25/2021	< 0.50	**165**	5.3	439	--	--	--	--
14016	FIAMM	4/27/2021	< 0.50	**176**	< 2.0	335	--	--	--	< 0.50
14017	Hutchinson (MS/MSD)	4/29/2021	< 0.50	51.2	< 2.0	215	--	--	--	--
	Reporting Limit		< 0.50	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50

Reported By:

Date: 5/17/2021

INDUSTRIAL SAMPLES
May 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
14018	CCI	5/4/2021	--	--	< 2.0	84.7	**121.9**	--	--	--	--
14019	AAR RT	5/4/2021	< 1.0	35.1	< 2.0	63.2	--	< 20.0	15.9	--	--
14020	AKWEL	5/5/2021	< 1.0	21.8	< 2.0	175.7	--	< 20.0	--	--	--
14021	AAR FAC	5/5/2021	< 1.0	46.6	4.71	392	--	--	--	--	--
14022	AKWEL 5th Street	5/6/2021	< 1.0	< 20.0	< 2.0	30.4	--	< 20.0	--	--	--
14023	AAR RT	5/6/2021	--	--	--	--	--	< 20.0	15.8	--	--
14024	AAR 6th Ave	5/7/2021	1.56	51.1	< 2.0	40.7	--	< 20.0	21.7	--	--
14025	Arvco (8 hr)	5/11/2021	< 1.0	71.0	< 2.0	105.9	--	--	< 3.0	--	--
14026	Arvco (8 hr)	5/13/2021	< 1.0	56.9	< 2.0	57.4	--	--	< 3.0	--	--
14027	Piranha	5/18/2021	--	22.1	--	--	--	--	--	--	--
14028	Rec Boat Holdings (8hr)	5/18/2021	--	**756**	--	--	< 10.0	--	--	--	CN
14029	CCI	5/19/2021	--	--	< 2.0	41.5	**287**	--	--	--	--
14030	AKWEL	5/20/2021	< 1.0	< 20.0	< 2.0	84.2	--	< 20.0	--	--	--
14031	AKWEL 5th Street	5/21/2021	< 1.0	< 20.0	< 2.0	< 20.0	--	< 20.0	--	--	--
	Reporting Limit		< 1.0	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 1.0	

Reported By: **Date:** 6/9/2021

INDUSTRIAL SAMPLES
June 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
14042	CCI	6/2/2021	--	--	< 2.0	54.0	**158**	--	--	--	--
14043	AAR RT	6/2/2021	< 0.20	< 20.0	< 2.0	44.7	--	< 20.0	6.8	--	--
14044	AKWEL	6/3/2021	< 0.20	22.0	7.05	99.7	--	< 20.0	--	--	--
14045	AAR FAC	6/3/2021	0.21	45.9	< 2.0	325	--	--	--	--	--
14046	AKWEL 5th Street	6/4/2021	< 0.20	< 20.0	< 2.0	26.8	--	< 20.0	--	--	--
14047	AAR RT	6/4/2021	--	--	--	--	--	< 20.0	13.7	--	--
14048	Arvco (8 hr)	6/7/2021	0.86	**239**	< 2.0	195	--	--	< 3.0	--	--
14049	Piranha	6/8/2021	--	26.3	--	--	--	--	--	--	--
14050	Rec Boat Holdings (8hr)	6/8/2021	0.47	**748**	13.5	**19200**	28.2	--	--	--	--
14051	AAR 6th Ave	6/9/2021	1.2	46.9	< 2.0	42.4	--	< 20.0	24.9	--	--
14052	Arvco (8 hr)	6/9/2021	0.89	**313**	8.7	381	--	--	4.1	--	--
14053	AKWEL 5th Street	6/10/2021	< 0.20	< 20.0	< 2.0	22.4	--	< 20.0	--	--	--
14054	CCI	6/11/2021	--	--	< 2.0	79.5	**160**	--	--	--	--
14055	AKWEL	6/15/2021	< 0.20	< 20.0	< 2.0	88.8	--	< 20.0	--	--	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50	

Reported By: **Date:** 7/15/2021

INDUSTRIAL SAMPLES
July 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
14086	Munson	7/8/2021	--	--	--	--	--	--	--	0.75	Hg
14087	AAR RT	7/8/2021	< 0.20	< 20.0	< 2.0	< 20.0	--	< 20.0	13.6	--	--
14088	Piranha	7/9/2021	--	< 20.0	--	--	--	--	--	--	--
14089	AAR FAC	7/9/2021	1.14	**233**	6.8	594	--	--	--	--	--
14090	CCI	7/13/2021	--	--	67.6	**2170**	56.5	--	--	--	--
14091	AAR RT	7/13/2021	--	--	--	--	--	< 20.0	11.5	--	--
14092	AKWEL	7/14/2021	< 0.20	38.7	< 2.0	90.9	--	< 20.0	--	--	--
14093	Arvco (8 hr)	7/14/2021	1.00	**201**	< 2.0	146	--	--	< 3.0	--	--
14094	AKWEL 5th Street	7/15/2021	< 0.20	21.9	< 2.0	< 20.0	--	< 20.0	--	--	--
14095	AAR 6th Ave	7/16/2021	0.70	38.0	< 2.0	61.0	--	< 20.0	40.8	--	--
14096	AKWEL 5th Street	7/20/2021	< 0.20	20.4	< 2.0	< 20.0	--	< 20.0	--	--	--
14097	Arvco (8 hr)	7/20/2021	**2.47**	**326**	< 2.0	183	--	--	< 3.0	--	--
14098	CCI	7/21/2021	--	--	15.0	602	**229**	--	--	--	--
14099	Rec Boat Holdings (8hr)	7/21/2021	0.32	**1200**	10.4	**11400**	22.2	--	--	--	CN
14100	AKWEL	7/22/2021	< 0.20	< 20.0	< 2.0	53.7	--	< 20.0	--	--	--
14101	Borg Warner	7/23/2021	0.45	31.0	< 2.0	156	--	--	--	--	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50	

Reported By: **Date:** 8/9/2021

in-house ICP/MS batch:	210805	210805	210805	210805	210805	210805	210805	210805	(PACE)
permit limit:	2	99	100	835	118	185	369	3	

INDUSTRIAL SAMPLES
August 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
14117	Hutchinson	7/27/2021	0.21	62.2	< 2.0	307	--	--	--	--	--
14118	MI Rubber	7/28/2021	0.23	**99.8**	< 2.0	79.9	--	--	--	--	--
14119	FIAMM	7/29/2021	< 0.20	**190**	5.9	374	--	--	--	< 0.50	--
14120	CRE	7/30/2021	< 0.20	40.8	< 2.0	20.9	19.8	< 20.0	6.5	--	--
14121	CCI	8/3/2021	**2.10**	40.3	66.4	**1940**	79.9	< 20.0	--	**3.09**	--
14122	AAR RT	8/3/2021	< 0.20	23.5	3.2	43.8	--	< 20.0	10.5	--	VOC, CN
14123	AAR 6th Ave	8/4/2021	1.30	42.4	< 2.0	62.3	--	< 20.0	31.9	--	VOC, CN
14124	AAR FAC	8/4/2021	< 0.20	84.1	4.3	412	--	--	19.9	--	VOC
14125	AKWEL	8/5/2021	< 0.20	< 20.0	< 2.0	91.0	--	< 20.0	--	--	--
14126	AAR RT	8/5/2021	--	--	--	--	--	< 20.0	8.9	--	--
14127	AKWEL 5th Street	8/6/2021	< 0.20	< 20.0	12.3	22.2	--	< 20.0	--	--	--
14128	Spencer Plastics	8/10/2021	< 0.20	43.3	3.0	139	--	--	--	--	--
14129	Arvco (8 hr)	8/9/2021	0.96	**509**	5.1	578	--	--	8.5	--	--
14130	Rec Boat Holdings (8hr)	8/10/2021	--	**881**	--	--	15.7	32.0	30.6	--	--
14131	Piranha	8/11/2021	--	27.4	--	--	--	--	--	--	--
14132	Arvco (8 hr)	8/11/2021	0.42	**370**	2.5	356	--	--	< 3.0	--	--
14133	CCI	8/12/2021	--	--	**1930**	**59200**	**149**	--	--	--	--
14134	AKWEL	8/13/2021	< 0.20	< 20.0	< 2.0	79.7	--	< 20.0	--	--	--
14135	AKWEL 5th Street	8/17/2021	< 0.20	27.5	< 2.0	28.0	--	< 20.0	--	--	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50	

Reported By: **Date:** 9/15/2021

batch:	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS	ICP/MS

14136	Padnos (grab)	8/10/2021	**14.9**	**2370**	**609**	**1820**	**148**	**387**	31.1	**13.1**	Al=6040 Se<100
14137	CCI (grab)	8/25/2021	1.19	< 20.0	5.5	215.0	90.9	< 20.0	< 3.0	< 0.50	Al=536 Se=2.0
14135	CCI (comp)	8/26/2021	1.63	25.5	6.5	185.8	**241.9**	< 20.0	< 3.0	0.59	Al=423 Se=4.5

INDUSTRIAL SAMPLES
September 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
14146	Piranha	9/2/2021	--	< 50.0	--	--	--	--	--	--	--
14147	AAR 6th Ave	9/3/2021	0.82	< 50.0	< 2.0	46.0	--	< 20.0	19.4	--	--
14148	CCI	9/8/2021	--	--	5.7	156	**246**	--	--	--	--
14149	AAR RT (MS/MSD)	9/8/2021	< 0.20	< 50.0	< 2.0	< 20.0	--	< 20.0	6.6	--	--
14150	AKWEL 5th Street	9/9/2021	< 0.20	< 50.0	< 2.0	22.4	--	< 20.0	--	--	--
14151	AAR FAC	9/9/2021	0.22	63.0	< 2.0	222	--	--	--	--	--
14152	AKWEL 5th Street	9/10/2021	< 0.20	< 50.0	< 2.0	< 20.0	--	< 20.0	--	--	--
14153	AAR RT	9/10/2021	--	--	--	--	--	< 20.0	14.0	--	--
14154	AKWEL	9/14/2021	< 0.20	< 50.0	< 2.0	48.3	--	< 20.0	--	--	--
14155	Arvco (8 hr)	9/14/2021	0.26	**230**	< 2.0	85.6	--	--	< 3.0	--	--
14156	AKWEL	9/15/2021	< 0.20	< 50.0	< 2.0	110	--	< 20.0	--	--	--
14157	Rec Boat Holdings (8hr)	9/15/2021	0.39	**718**	9.9	**12200**	23.4	--	--	--	--
14158	CCI	9/16/2021	--	--	< 2.0	46.9	**245**	--	--	--	--
14159	Arvco (8 hr)	9/16/2021	< 0.20	**209**	5.1	104	--	--	< 3.0	--	--
	Reporting Limit		< 0.20	< 50.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 50	

Reported By:

Date: 10/14/2021

INDUSTRIAL SAMPLES
October 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)
14182	CCI	10/5/2021	1.43	43.3	19.5	329	**206**	< 20.0	--	--
14183	AAR RT	10/5/2021	< 0.20	< 20.0	2.4	< 20.0	--	< 20.0	12.4	--
14184	AKWEL	10/6/2021	< 0.20	< 20.0	< 2.0	82.0	--	< 20.0	--	--
14185	AAR FAC	10/6/2021	< 0.20	38.2	2.5	194	--	--	--	--
14186	AKWEL 5th Street	10/7/2021	< 0.20	< 20.0	2.1	82.8	--	< 20.0	--	--
14187	AAR RT	10/7/2021	--	--	--	--	--	< 20.0	21.0	--
14188	AAR 6th Ave	10/8/2021	0.82	53.8	< 2.0	53.5	--	< 20.0	18.0	--
14189	Arvco (8 hr)	10/11/2021	< 0.20	**187**	6.5	55.4	--	--	< 3.0	--
14190	Borg Warner (MS/MSD)	10/12/2021	< 0.20	< 20.0	< 2.0	59.0	--	--	--	--
14191	Rec Boat Holdings (8hr)	10/12/2021	--	**578**	--	--	32.0	29.1	26.6	--
14192	Spencer Plastics (MS/MSD)	10/14/2021	< 0.20	**120**	4.4	115	--	--	--	--
14193	Piranha	10/15/2021	--	23.3	--	--	--	--	--	--
14194	Arvco (8 hr)	10/15/2021	0.75	**574**	5.8	529	--	--	10.2	--
14195	CRE	10/19/2021	< 0.20	25.5	3.0	26.4	10.1	< 20.0	< 3.0	--
14196	AKWEL 5th Street	10/20/2021	< 0.20	21.5	< 2.0	52.0	--	< 20.0	--	--
14197	AKWEL	10/21/2021	< 0.20	21.4	< 2.0	104	--	< 20.0	--	--
14198	CCI	10/22/2021	--	--	7.1	164	**167**	--	--	--
14199	FIAMM (MS/MSD)	10/26/2021	< 0.20	**140**	4.1	366	--	--	--	**5.7**
14200	MI Rubber (MS/MSD)	10/27/2021	< 0.20	82.2	< 2.0	45.8	--	--	--	--
14201	Hutchinson	10/28/2021	0.38	55.3	2.4	291	--	--	--	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 2.0

Reported By:

Date: 11/18/2021

INDUSTRIAL SAMPLES
November 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
14204	AKWEL 5th Street (MS/MSD)	11/2/2021	< 0.50	24.4	< 2.0	< 20.0	--	< 20.0	--	--	--
14205	AAR RT	11/2/2021	< 0.50	21.9	2.2	27.8	--	< 20.0	9.7	--	--
14206	AKWEL	11/3/2021	< 0.50	< 20.0	< 2.0	86.4	--	< 20.0	--	--	--
14207	AAR FAC	11/3/2021	< 0.50	54.9	2.7	328	--	--	--	--	--
14208	AAR 6th Ave	11/4/2021	0.54	41.9	< 2.0	44.5	--	< 20.0	17.4	--	--
14209	AAR RT	11/4/2021	--	--	--	--	--	< 20.0	11.0	--	--
14210	Piranha	11/5/2021	--	26.4	--	--	--	--	--	--	--
14211	Arvco (8 hr)	11/5/2021	0.92	**278**	7.3	387	--	--	3.2	--	--
14212	Arvco (8 hr)	11/8/2021	< 0.50	89.5	< 2.0	55.8	--	--	< 3.0	--	--
14213	AKWEL (MS/MSD)	11/9/2021	< 0.50	< 20.0	< 2.0	133	--	< 20.0	--	--	--
14214	Rec Boat Holdings (8hr)	11/9/2021	< 0.50	**370**	4.1	**2520**	30.6	--	--	--	--
14215	AKWEL 5th Street	11/10/2021	< 0.50	< 20.0	< 2.0	< 20.0	--	< 20.0	--	--	--
14216	CCI	11/11/2021	--	--	11.1	294	**220**	--	--	--	--
14217	CCI	11/12/2021	--	--	9.3	259	**171**	--	--	**12.6**	--
	Reporting Limit		< 0.50	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	**< 1.0**	

Reported By:

Date: 12/16/2021

INDUSTRIAL SAMPLES
December 2021

Lab ID#	Location and Date		Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
14218	Arvco (8 hr)	12/1/2021	0.62	201	< 2.0	114	--	--	< 3.0	--	--
14219	Piranha	12/2/2021	--	20.8	--	--	--	--	--	--	--
14220	Rec Boat Holdings (8hr)	12/2/2021	--	332	--	--	< 10.0	--	--	--	Zn=16000
14221	AAR 6th Ave	12/3/2021	0.42	52.3	< 2.0	52.4	--	< 20.0	5.0	--	--
14222	Arvco (8 hr)	12/3/2021	0.85	353	2.5	218	--	--	3.3	--	--
14223	AKWEL 5th Street	12/7/2021	< 0.20	21.3	2.4	20.8	--	< 20.0	--	--	--
14224	AAR RT	12/7/2021	< 0.20	< 20.0	< 2.0	32.4	--	< 20.0	6.6	--	--
14225	AKWEL 5th Street	12/8/2021	< 0.20	29.8	< 2.0	33.7	--	< 20.0	--	--	--
14226	AAR FAC	12/8/2021	0.55	27.2	5.6	76.8	--	--	--	--	--
14227	AKWEL	12/9/2021	< 0.20	< 20.0	< 2.0	90.6	--	< 20.0	--	--	--
14228	AAR RT	12/9/2021	--	--	--	--	--	< 20.0	7.6	--	--
14229	AKWEL	12/10/2021	< 0.20	< 20.0	< 2.0	121	--	< 20.0	--	--	--
14230	CCI	12/14/2021	--	--	4.4	96.9	68.9	--	--	--	--
14231	CCI (MS/MSD)	12/15/2021	--	--	3.6	76.8	81.5	--	--	--	--
	Reporting Limit		< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0		

Reported By:

Date: 1/13/2022

INDUSTRIAL SAMPLES
CCI summary 2021

Lab ID#	Location and Date	Cd (µg/L)	Cu (µg/L)	Pb (µg/L)	Zn (µg/L)	Mo (µg/L)	Ni (µg/L)	Cr (µg/L)	Ag (µg/L)	OTHER
13949	CCI 1/5/2021	< 0.50	23	4.60	148	11.2	< 20.0	--	1.3	--
13963	CCI 1/20/2021	--	--	2.5	**5410**	39.3	--	--	--	--
13970	CCI 2/2/2021	--	--	< 2.0	52.3	44.0	--	--	--	--
13977	CCI 2/18/2021	--	--	< 2.0	81.6	86.6	--		--	--
13984	CCI 3/2/2021	--	--	< 2.0	92.9	**245**	--	--	--	--
13994	CCI 3/10/2021	--	--	2.3	52.7	**246**	< 20.0	--	--	--
13998	CCI (MS/MSD) 4/6/2021	1.6	< 20.0	< 2.0	43.2	**172**	< 20.0	--	< 0.50	--
14012	CCI 4/20/2021	--	--	3.5	122	80.0	--	--	--	--
14018	CCI 5/4/2021	--	--	< 2.0	84.7	**121.9**	--	--	--	--
14029	CCI 5/19/2021	--	--	< 2.0	41.5	**287**	--	--	--	--
14042	CCI 6/2/2021	--	--	< 2.0	54.0	**158**	--	--	--	--
14054	CCI 6/11/2021	--	--	< 2.0	79.5	**160**	--	--	--	--
14090	CCI 7/13/2021	--	--	67.6	**2170**	56.5	--	--	--	--
14098	CCI 7/21/2021	--	--	15.0	602	**229**	--	--	--	--
14121	CCI 8/3/2021	**2.10**	40.3	66.4	**1940**	79.9	< 20.0	--	**3.09**	--
14133	CCI 8/12/2021	--	--	**1930**	**59200**	**149**	--	--	--	--
14148	CCI 9/8/2021	--	--	5.7	156	**246**	--	--	--	--
14158	CCI 9/16/2021	--	--	< 2.0	46.9	**245**	--	--	--	--
14182	CCI 10/5/2021	1.43	43.3	19.5	329	**206**	< 20.0	--	--	--
14198	CCI 10/22/2021	--	--	7.13	164	**167**	--	--	--	--
14216	CCI 11/11/2021	--	--	11.1	294	**220**	--	--	--	--
14217	CCI 11/12/2021	--	--	9.33	259	**171**	--	--	12.639	--
14230	CCI 12/14/2021	--	--	4.39	96.9	68.9	--	--	--	--
14231	CCI (MS/MSD) 12/15/2021	--	--	3.56	76.8	81.5	--	--	--	--
	Reporting Limit	< 0.20	< 20.0	< 2.0	< 20.0	< 10.0	< 20.0	< 3.0	< 0.50	
	permit limit:	2	99	100	835	118	185	369	3	

APPENDIX B

PERMIT NO. MI0020257

STATE OF MICHIGAN
DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

AUTHORIZATION TO DISCHARGE UNDER THE NATIONAL POLLUTANT DISCHARGE ELIMINATION SYSTEM

In compliance with the provisions of the Federal Water Pollution Control Act, 33 U.S.C., Section 1251 *et seq*., as amended; Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended (NREPA); Part 41, Sewerage Systems, of the NREPA; and Michigan Executive Order 2019-06,

City of Cadillac
200 North Lake Street
Cadillac, MI 49601

is authorized to discharge from the **Cadillac Wastewater Treatment Plant** located at

1121 Plett Road
Cadillac, MI 49601

designated as **Cadillac WWTP**

to the receiving water named the Clam River in accordance with effluent limitations, monitoring requirements, and other conditions set forth in this permit.

This permit is based on a complete application submitted on April 3, 2017, as amended through May 11, 2017.

This permit takes effect on September 1, 2019. The provisions of this permit are severable. After notice and opportunity for a hearing, this permit may be modified, suspended, or revoked in whole or in part during its term in accordance with applicable laws and rules. On its effective date, this permit shall supersede National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257 (expiring October 1, 2017).

This permit and the authorization to discharge shall expire at midnight on **October 1, 2022**. In order to receive authorization to discharge beyond the date of expiration, the permittee shall submit an application that contains such information, forms, and fees as are required by the Michigan Department of Environment, Great Lakes, and Energy (Department) by **April 4, 2022**.

Issued: August 30, 2019.

Original signed by Christine Alexander
Christine Alexander, Manager
Permits Section
Water Resources Division

PERMIT FEE REQUIREMENTS

In accordance with Section 324.3120 of the NREPA, the permittee shall make payment of an annual permit fee to the Department for each October 1 the permit is in effect regardless of occurrence of discharge. The permittee shall submit the fee in response to the Department's annual notice. The fee shall be postmarked by January 15 for notices mailed by December 1. The fee is due no later than 45 days after receiving the notice for notices mailed after December 1.

Annual Permit Fee Classification: Municipal Major, less than 10 MGD (Individual Permit)

In accordance with Section 324.3132 of the NREPA, the permittee shall make payment of an annual biosolids land application fee to the Department if the permittee land applies biosolids. In response to the Department's annual notice, the permittee shall submit the fee, which shall be postmarked no later than January 31 of each year.

CONTACT INFORMATION

Unless specified otherwise, all contact with the Department required by this permit shall be made to the Cadillac District Office of the Water Resources Division. The Cadillac District Office is located at 120 West Chapin Street, Cadillac, MI 49601-2158, Telephone: 231-775-3960, Fax: 231-775-1511.

CONTESTED CASE INFORMATION

Any person who is aggrieved by this permit may file a sworn petition with the Michigan Administrative Hearing System within the Michigan Department of Licensing and Regulatory Affairs, c/o the Michigan Department of Environment, Great Lakes, and Energy, setting forth the conditions of the permit which are being challenged and specifying the grounds for the challenge. The Department of Licensing and Regulatory Affairs may reject any petition filed more than 60 days after issuance as being untimely.

PART I

Section A. Limitations and Monitoring Requirements

1. Final Effluent Limitations, Monitoring Point 001A

During the period beginning on the effective date of this permit and lasting until the expiration date of this permit, the permittee is authorized to discharge treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River at Latitude 44.2657, Longitude -85.3985. Such discharge shall be limited and monitored by the permittee as specified below.

	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration					
Parameter	**Monthly**	**7-Day**	**Daily**	**Units**	**Monthly**	**7-Day**	**Daily**	**Units**	**Monitoring Frequency**	**Sample Type**
Flow	(report)	---	(report)	MGD	---	---	---	---	Daily	Report Total Daily Flow
Carbonaceous Biochemical Oxygen Demand ($CBOD_5$)										
June – September	190	270	(report)	lbs/day	7	---	10	mg/l	5x Weekly	24-Hr Composite
October – November	290	480	(report)	lbs/day	11	---	17	mg/l	5x Weekly	24-Hr Composite
December – March	670	910	(report)	lbs/day	25	---	34	mg/l	5x Weekly	24-Hr Composite
April – May	450	670	(report)	lbs/day	17	---	25	mg/l	5x Weekly	24-Hr Composite
Total Suspended Solids (TSS)										
June – September	530	800	(report)	lbs/day	20	30	(report)	mg/l	5x Weekly	24-Hr Composite
October – May	800	1200	(report)	lbs/day	30	45	(report)	mg/l	5x Weekly	24-Hr Composite
Ammonia Nitrogen (as N)										
June – September	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite
October – November	88	---	(report)	lbs/day	1.6	---	2.8	mg/l	5x Weekly	24-Hr Composite
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite
April – May	130	---	(report)	lbs/day	4.8	---	(report)	mg/l	5x Weekly	24-Hr Composite
Total Phosphorus (as P)										
May – March	13	---	(report)	lbs/day	0.5	---	(report)	mg/l	5x Weekly	24-Hr Composite
April	17	---	(report)	lbs/day	0.65	---	(report)	mg/l	5x Weekly	24-Hr Composite
Fecal Coliform Bacteria	---	---	---	---	200	400	(report)	cts/100 ml	5x Weekly	Grab
Perfluorooctane sulfonate (PFOS)	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	2x Annually	Grab
Perfluorooctanoic acid (PFOA)	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	2x Annually	Grab
Total Copper	0.3	---	0.82	lbs/day	11	---	31	ug/l	Weekly	24-Hr Composite
Total Nickel	(report)	---	(report)	lbs/day	(report)	---	(report)	ug/l	Quarterly	24-Hr Composite
Total Selenium	0.15	---	(report)	lbs/day	5.7	---	(report)	ug/l	Monthly	24-Hr Composite
Bis(2-ethylhexyl)phthalate	0.82	---	(report)	lbs/day	31	---	(report)	ug/l	Monthly	24-Hr Composite
Total Mercury										
Corrected	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	Quarterly	Calculation
Uncorrected	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab
Field Duplicate	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab
Field Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation

PART I

Section A. Limitations and Monitoring Requirements

Parameter							Daily	Units	Monitoring Frequency	Sample Type
Laboratory Method Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation
	12-Month Rolling Avg				**12-Month Rolling Avg**					
Total Mercury	0.000053	---	---	lbs/day	2.0	---	---	ng/l	Quarterly	Calculation
					Minimum % Monthly		**Minimum % Daily**			
TSS Minimum % Removal										
October – May	---	---	---	---	85	---	(report)	%	Monthly	Calculation
					Minimum Daily		**Maximum Daily**			
pH	---	---	---	---	6.5	---	9.0	S.U.	5x Weekly	Grab
Dissolved Oxygen										
May – November	---	---	---	---	6.0	---	---	mg/l	5x Weekly	Grab
December – April	---	---	---	---	5.0	---	---	mg/l	5x Weekly	Grab

The following design flow was used in determining the above limitations, but is not to be considered a limitation or actual capacity: 3.2 MGD.

a. Narrative Standard
The receiving water shall contain no turbidity, color, oil films, floating solids, foams, settleable solids, or deposits as a result of this discharge in unnatural quantities which are or may become injurious to any designated use.

b. Sampling Locations
Samples for Carbonaceous Biochemical Oxygen Demand ($CBOD_5$), Total Suspended Solids (TSS), Ammonia Nitrogen (as N), Total Phosphorus (as P), Total Copper, Total Nickel, Total Selenium, and Bis(2-ethylhexyl)phthalate shall be taken prior to disinfection. Samples for Fecal Coliform Bacteria, Total Mercury, pH and Dissolved Oxygen shall be taken after disinfection. The Department may approve alternate sampling locations that are demonstrated by the permittee to be representative of the effluent.

c. Quarterly and 2x Annual Monitoring
Quarterly samples shall be taken during the months of January, April, July, and October. 2x Annual samples shall be taken during the months of April and October. If the facility does not discharge during these months, the permittee shall sample the next discharge occurring during the period in question. If the facility does not discharge during the period in question, a sample is not required for that period. For any month in which a sample is not taken, the permittee shall enter "*G" on the Discharge Monitoring Report (DMR). (For purposes of reporting on the Daily tab of the DMR, the permittee shall enter “*G” on the first day of the month only).

d. Ultraviolet Disinfection
It is understood that ultraviolet light will be used to achieve compliance with the fecal coliform limitations. If disinfection other than ultraviolet light will be used, the permittee shall notify the Department in accordance with Part II.C.12. of this permit.

e. Percent Removal Requirements
These requirements shall be calculated based on the monthly (30-day) effluent TSS concentrations and the monthly influent concentrations for approximately the same period.

Section A. Limitations and Monitoring Requirements

f. Analytical Method(s) and Quantification Level for Total Copper
The sampling procedures, preservation and handling, and analytical protocol for compliance monitoring for total copper shall be in accordance with an EPA Method listed in Title 40 of the Code of Federal Regulations (CFR), Part 136. The quantification level for total copper shall be 10 ug/l unless a higher level is appropriate because of sample matrix interference. Justification for higher quantification levels shall be submitted to the Department within 30 days of such determination. Upon approval from the Department, the permittee may use alternate analytical methods (for parameters with methods specified in Title 40 of the CFR, Part 136, the alternate methods are restricted to those listed in 40 CFR, Part 136).

g. Analytical Method(s) and Quantification Level for Total Selenium
The sampling procedures, preservation and handling, and analytical protocol for compliance monitoring for total selenium shall be in accordance with an EPA Method listed in Title 40 of the Code of Federal Regulations (CFR), Part 136. The quantification level for total selenium shall be 2 ug/l unless a higher level is appropriate because of sample matrix interference. Justification for higher quantification levels shall be submitted to the Department within 30 days of such determination. Upon approval from the Department, the permittee may use alternate analytical methods (for parameters with methods specified in Title 40 of the CFR, Part 136, the alternate methods are restricted to those listed in 40 CFR, Part 136).

h. Monitoring Frequency Reduction for Total Selenium, and Bis(2-ethylhexyl)phthalate
After the submittal of 12 months of data, the permittee may request, in writing, Department approval for a reduction in monitoring frequency for total selenium, and bis(2-ethylhexyl)phthalate. This request shall contain an explanation as to why the reduced monitoring is appropriate. Upon receipt of written approval and consistent with such approval, the permittee may reduce the monitoring frequency indicated in Part I.A.1. of this permit. The monitoring frequency for total selenium, and bis(2-ethylhexyl)phthalate shall not be reduced to less than quarterly. The Department may revoke the approval for reduced monitoring at any time upon notification to the permittee.

i. Monitoring Frequency Reduction for Perfluorooctane Sulfonate (PFOS) and/or Perfluorooctanoic Acid (PFOA)
After the submittal of at least 10 equally spaced data points over a minimum of 3 months, the permittee may request, in writing, Department approval of a reduction in monitoring frequency for PFOS and/or PFOA. This request shall contain an explanation as to why the reduced monitoring is appropriate. Upon receipt of written approval and consistent with such approval, the permittee may reduce the monitoring frequency indicated in Part I.A.1. of this permit. The monitoring frequency for PFOS and/or PFOA shall not be reduced to less than annually. The Department may revoke the approval for reduced monitoring at any time upon notification to the permittee.

j. Final Effluent Limitation for Total Mercury
The final limit for total mercury is the Discharge Specific Level Currently Achievable (LCA) based on a multiple discharger variance from the WQBEL of 1.3 ng/l, pursuant to Rule 1103(9) of the Water Quality Standards. Compliance with the LCA shall be determined as a 12-month rolling average, the calculation of which may be done using blank-corrected sample results. The 12-month rolling average shall be determined by adding the present monthly average result to the preceding 11 monthly average results then dividing the sum by 12. For facilities with quarterly monitoring requirements for total mercury, quarterly monitoring shall be equivalent to three (3) months of monitoring in calculating the 12-month rolling average. Facilities that monitor more frequently than monthly for total mercury must determine the monthly average result, which is the sum of the results of all data obtained in a given month divided by the total number of samples taken, in order to calculate the 12-month rolling average. If the 12-month rolling average for any quarter is less than or equal to the LCA, the permittee will be considered to be in compliance for total mercury for that quarter, provided the permittee is also in full compliance with the Pollutant Minimization Program for Total Mercury, set forth in Part I.A.4. of this permit.

PART I

Section A. Limitations and Monitoring Requirements

k. Total Mercury Testing and Additional Reporting Requirements
The analytical protocol for total mercury shall be in accordance with EPA Method 1631, Revision E, "Mercury in Water by Oxidation, Purge and Trap, and Cold Vapor Atomic Fluorescence Spectrometry." The quantification level for total mercury shall be 0.5 ng/l, unless a higher level is appropriate because of sample matrix interference. Justification for higher quantification levels shall be submitted to the Department within 30 days of such determination.

The use of clean technique sampling procedures is required unless the permittee can demonstrate to the Department that an alternate sampling procedure is representative of the discharge. Guidance for clean technique sampling is contained in EPA Method 1669, Sampling Ambient Water for Trace Metals at EPA Water Quality Criteria Levels (Sampling Guidance), EPA-821-R96-001, July 1996. Information and data documenting the permittee's sampling and analytical protocols and data acceptability shall be submitted to the Department upon request.

In order to demonstrate compliance with EPA Method 1631E and EPA Method 1669, the permittee shall report, on the daily sheet, the analytical results of all field blanks and field duplicates collected in conjunction with each sampling event, as well as laboratory method blanks when used for blank correction. The permittee shall collect at least one (1) field blank and at least one (1) field duplicate per sampling event. If more than ten (10) samples are collected during a sampling event, the permittee shall collect at least one (1) additional field blank AND field duplicate for every ten (10) samples collected. Only field blanks or laboratory method blanks may be used to calculate a concentration lower than the actual sample analytical results (i.e., a blank correction). Only one (1) blank (field OR laboratory method) may be used for blank correction of a given sample result, and only if the blank meets the quality control acceptance criteria. If blank correction is not performed on a given sample analytical result, the permittee shall report under "Total Mercury – Corrected" the same value reported under "Total Mercury – Uncorrected." The field duplicate is for quality control purposes only; its analytical result shall not be averaged with the sample result.

PART I

Section A. Limitations and Monitoring Requirements

2. Quantification Levels and Analytical Methods for Selected Parameters

Quantification levels (QLs) are specified for selected parameters in the table below. These QLs shall be considered the maximum acceptable unless a higher QL is appropriate because of sample matrix interference. Justification for higher QLs shall be submitted to the Department within 30 days of such determination. Where necessary to help ensure that the QLs specified can be achieved, analytical methods may also be specified in the table below. The sampling procedures, preservation and handling, and analytical protocol for all monitoring conducted in compliance with this permit, including monitoring conducted to meet the requirements of the application for permit reissuance, shall be in accordance with the methods specified in the table below, or in accordance with Part II.B.2. of this permit if no method is specified in the table below, unless an alternate method is approved by the Department. With the exception of total mercury, all units are in ug/l. The table is continued on the following page:

Parameter	QL	Units	Analytical Method
1,2-Diphenylhydrazine (as Azobenzene)	3.0	ug/l	
2,4,6-Trichlorophenol	5.0	ug/l	
2,4-Dinitrophenol	19	ug/l	
3,3'-Dichlorobenzidine	1.5	ug/l	EPA Method 605
4-Chloro-3-methylphenol	5.0	ug/l	
4,4'-DDD	0.05	ug/l	EPA Method 608
4,4'-DDE	0.01	ug/l	EPA Method 608
4,4'-DDT	0.01	ug/l	EPA Method 608
Acrylonitrile	1.0	ug/l	
Aldrin	0.01	ug/l	EPA Method 608
Alpha-Hexachlorocyclohexane	0.01	ug/l	EPA Method 608
Antimony, Total	1	ug/l	
Arsenic, Total	1	ug/l	
Barium, Total	5	ug/l	
Benzidine	0.1	ug/l	EPA Method 605
Beryllium, Total	1	ug/l	
Beta-Hexachlorocyclohexane	0.01	ug/l	EPA Method 608
Bis (2-Chloroethyl) Ether	1.0	ug/l	
Boron, Total	20	ug/l	
Cadmium, Total	0.2	ug/l	
Chlordane	0.01	ug/l	EPA Method 608
Chromium, Hexavalent	5	ug/l	
Chromium, Total	10	ug/l	
Cyanide, Available	2	ug/l	EPA Method OIA 1677
Cyanide, Total	5	ug/l	
Delta-Hexachlorocyclohexane	0.01	ug/l	EPA Method 608
Dieldrin	0.01	ug/l	EPA Method 608
Di-N-Butyl Phthalate	9.0	ug/l	
Endosulfan I	0.01	ug/l	EPA Method 608
Endosulfan II	0.01	ug/l	EPA Method 608
Endosulfan Sulfate	0.01	ug/l	EPA Method 608
Endrin	0.01	ug/l	EPA Method 608
Endrin Aldehyde	0.01	ug/l	EPA Method 608
Fluoranthene	1.0	ug/l	
Heptachlor	0.01	ug/l	EPA Method 608
Heptachlor Epoxide	0.01	ug/l	EPA Method 608

PART I

Section A. Limitations and Monitoring Requirements

Parameter	QL	Units	Analytical Method
Hexachlorobenzene	0.01	ug/l	EPA Method 612
Hexachlorobutadiene	0.01	ug/l	EPA Method 612
Hexachlorocyclopentadiene	0.01	ug/l	EPA Method 612
Hexachloroethane	5.0	ug/l	
Lead, Total	1	ug/l	
Lindane	0.01	ug/l	EPA Method 608
Lithium, Total	10	ug/l	
Mercury, Total	0.5	**ng/l**	EPA Method 1631E
Nickel, Total	5	ug/l	
PCB-1016	0.1	ug/l	EPA Method 608
PCB-1221	0.1	ug/l	EPA Method 608
PCB-1232	0.1	ug/l	EPA Method 608
PCB-1242	0.1	ug/l	EPA Method 608
PCB-1248	0.1	ug/l	EPA Method 608
PCB-1254	0.1	ug/l	EPA Method 608
PCB-1260	0.1	ug/l	EPA Method 608
Pentachlorophenol	1.8	ug/l	
Perfluorooctane sulfonate (PFOS)	2.0	ng/l	ASTM D7979 or an isotope dilution method (sometimes referred to as Method 537 modified)
Perfluorooctanoic acid (PFOA)	2.0	ng/l	ASTM D7979 or an isotope dilution method (sometimes referred to as Method 537 modified)
Phenanthrene	1.0	ug/l	
Silver, Total	0.5	ug/l	
Strontium, Total	1000	ug/l	
Sulfides, Dissolved	20	ug/l	
Thallium, Total	1	ug/l	
Toxaphene	0.1	ug/l	EPA Method 608
Vinyl Chloride	0.25	ug/l	
Zinc, Total	10	ug/l	

3. Additional Monitoring Requirements

As a condition of this permit, the permittee shall monitor the discharge from monitoring point 001A for the constituents listed below. This monitoring is an application requirement of 40 CFR 122.21(j), effective December 2, 1999. Testing shall be conducted in October 2019, May 2020, March 2021, and August 2021. Grab samples shall be collected for available cyanide, total phenols, and the Volatile Organic Compounds identified below. For all other parameters, 24-hour composite samples shall be collected.

Test species for whole effluent toxicity monitoring shall include fathead minnow **and** *Ceriodaphnia dubia,* for a total of four (4) tests on each species. Testing and reporting procedures shall follow procedures contained in EPA-821-R-02-013, “Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms” (Fourth Edition). When the effluent ammonia nitrogen (as N) concentration is greater than 3 mg/l, the pH of the toxicity test shall be maintained at a pH of 8 Standard Units. Acute and chronic toxicity data shall be included in the reporting for the toxicity test results. Toxicity test data acceptability is contingent upon the validation of the test method by the testing laboratory. Such validation shall be submitted to the Department upon request.

Section A. Limitations and Monitoring Requirements

The results of such additional monitoring shall be submitted with the application for reissuance (see the cover page of this permit for the application due date). The permittee shall notify the Department within 14 days of completing the monitoring for each month specified above in accordance with Part II.C.5. Additional reporting requirements are specified in Part II.C.11. The permittee shall report to the Department any whole effluent toxicity test results greater than 1.0 TU_A or 1.0 TU_C within five (5) days of becoming aware of the result. If, upon review of the analysis, it is determined that additional requirements are needed to protect the receiving waters in accordance with applicable water quality standards, the permit may then be modified by the Department in accordance with applicable laws and rules.

Whole Effluent Toxicity

acute toxicity	chronic toxicity		

Hardness

calcium carbonate

Metals (Total Recoverable), Cyanide and Total Phenols

antimony	arsenic	available cyanide	barium
beryllium	boron	cadmium	chromium
lead	silver	thallium	zinc
total phenolic compounds			

Volatile Organic Compounds

acrolein	acrylonitrile	benzene	bromoform
carbon tetrachloride	chlorobenzene	chlorodibromomethane	chloroethane
2-chloroethylvinyl ether	chloroform	dichlorobromomethane	1,1-dichloroethane
1,2-dichloroethane	trans-1,2-dichloroethylene	1,1-dichloroethylene	1,2-dichloropropane
1,3-dichloropropylene	ethylbenzene	methyl bromide	methyl chloride
methylene chloride	1,1,2,2,-tetrachloroethane	tetrachloroethylene	toluene
1,1,1-trichloroethane	1,1,2-trichloroethane	trichloroethylene	vinyl chloride

Acid-Extractable Compounds

4-chloro-3-methylphenol	2-chlorophenol	2,4-dichlorophenol	2,4-dimethylphenol
4,6-dinitro-o-cresol	2,4-dinitrophenol	2-nitrophenol	4-nitrophenol
Pentachlorophenol	phenol	2,4,6-trichlorophenol	

Base/Neutral Compounds

acenaphthene	acenaphthylene	anthracene	benzidine
benzo(a)anthracene	benzo(a)pyrene	3,4-benzofluoranthene	benzo(ghi)perylene
benzo(k)fluoranthene	bis(2-chloroethoxy)methane	bis(2-chloroethyl)ether	bis(2-chloroisopropyl)ether
4-bromophenyl phenyl ether	butyl benzyl phthalate	2-chloronaphthalene	4-chlorophenyl phenyl ether
chrysene	di-n-butyl phthalate	di-n-octyl phthalate	dibenzo(a,h)anthracene
1,2-dichlorobenzene	1,3-dichlorobenzene	1,4-dichlorobenzene	3,3'-dichlorobenzidine
diethyl phthalate	dimethyl phthalate	2,4-dinitrotoluene	2,6-dinitrotoluene
1,2-diphenylhydrazine	fluoranthene	fluorene	Hexachlorobenzene
hexachlorobutadiene	hexachlorocyclo-pentadiene	hexachloroethane	indeno(1,2,3-cd)pyrene
isophorone	naphthalene	nitrobenzene	n-nitrosodi-n-propylamine
n-nitrosodimethylamine	n-nitrosodiphenylamine	phenanthrene	pyrene
1,2,4-trichlorobenzene			

PART I

Section A. Limitations and Monitoring Requirements

4. Pollutant Minimization Program for Total Mercury

The goal of the Pollutant Minimization Program is to maintain the effluent concentration of total mercury at or below 1.3 ng/l. The permittee shall continue to implement the Pollutant Minimization Program approved on May 17, 2005, and modifications thereto, to proceed toward the goal. The Pollutant Minimization Program includes the following:

a. an annual review and semi-annual monitoring of potential sources of mercury entering the wastewater collection system;

b. a program for quarterly monitoring of influent and periodic monitoring of sludge for mercury; and

c. implementation of reasonable cost-effective control measures when sources of mercury are discovered. Factors to be considered include significance of sources, economic considerations, and technical and treatability considerations.

On or before March 31 of each year, the permittee shall submit a status report for the previous calendar year to the Department that includes 1) the monitoring results for the previous year, 2) an updated list of potential mercury sources, and 3) a summary of all actions taken to reduce or eliminate identified sources of mercury.

Any information generated as a result of the Pollutant Minimization Program set forth in this permit may be used to support a request to modify the approved program or to demonstrate that the Pollutant Minimization Program requirement has been completed satisfactorily.

A request for modification of the approved program and supporting documentation shall be submitted in writing to the Department for review and approval. The Department may approve modifications to the approved program (approval of a program modification does not require a permit modification), including a reduction in the frequency of the requirements under items a. and b.

This permit may be modified in accordance with applicable laws and rules to include additional mercury conditions and/or limitations as necessary.

5. Pollutant Minimization and Source Evaluation Program for Perfluorooctane Sulfonate (PFOS) and/or Perfluorooctanoic Acid (PFOA)

The goal of the Pollutant Minimization and Source Evaluation Program is to identify and address sources of PFOS and/or PFOA and to reduce and maintain the effluent concentrations of PFOS and/or PFOA at or below the water quality-based effluent limitations (WQBELs). The WQBELs are 12 ng/L for PFOS and 20 ug/L for PFOA.

Within 90 days of written notification by the Department or after the permittee notifies the Department that the final effluent concentration of PFOS and/or PFOA has exceeded the WQBELs, the permittee shall submit to the Department an approvable Pollutant Minimization and Source Evaluation Program for PFOS and/or PFOA to proceed toward the goal. The Pollutant Minimization and Source Evaluation Program shall continue work under the IPP Interim Initiative and shall include the following at a minimum:

a. identification of and strategies to identify any additional potential and probable PFOS and/or PFOA sources;

b. monitoring plan for the permitted facility’s influent and effluent and effluent from potential sources;

c. implemented measures thus far to eliminate, reduce, and/or control sources, and an assessment of the degree of success and the strategies used to measure success; and

d. proposed measures and implementation schedules for elimination, control, and/or reduction of the identified sources (prioritizing highest loadings and concentrations), and the strategies that will be used to measure success.

The Pollutant Minimization and Source Evaluation Program shall be implemented upon approval by the Department.

Section A. Limitations and Monitoring Requirements

On or before May 1 of each year following Pollutant Minimization and Source Evaluation Program implementation, the permittee shall submit to the Department a status report for the previous calendar year. Upon written notification by the Department, the permittee may be required to submit more frequent status reports. Status reports at a minimum shall include:

a. complete listing of PFOS and/or PFOA sources;

b. summary of influent and effluent monitoring data;

c. summary of monitoring data from known or potential sources;

d. history and compliance status for sources;

e. implemented measures to eliminate, reduce, or control sources, (prioritizing highest loadings and concentrations), and an assessment of the degree of success and the strategies used to measure success;

f. proposed measures and schedules for elimination, control, or reduction of any newly identified PFOS and/or PFOA sources (prioritizing highest loadings and concentrations), and the strategies that will be used to measure success;

g. barriers to implementation and revisions to the implementation schedule; and

h. laboratory reports, if not previously supplied.

Any information generated as a result of the Pollutant Minimization and Source Evaluation Program set forth in this permit may be used to support a request to modify the Pollutant Minimization and Source Evaluation Program or to demonstrate that the requirement has been completed satisfactorily.

A request for modification of the approved Pollutant Minimization and Source Evaluation Program shall be submitted in writing to the Department along with supporting documentation for review and approval. The Department may approve modifications to the approved Pollutant Minimization and Source Evaluation Program, including a reduction in the frequency of the influent and known or potential source monitoring requirements. Approval of a Pollutant Minimization and Source Evaluation Program modification does not require a permit modification.

This permit may be modified in accordance with applicable laws and rules to include additional PFOS and/or PFOA conditions and/or limitations as necessary.

6. Untreated or Partially Treated Sewage Discharge Reporting and Testing Requirements

In accordance with Section 324.3112a of the NREPA, if untreated or partially treated sewage is directly or indirectly discharged from a sewer system onto land or into the waters of the state, the permittee shall immediately, but not more than 24 hours after the discharge begins, notify local health departments, a daily newspaper of general circulation in the county in which the permittee is located, and a daily newspaper of general circulation in the county or counties in which the municipalities whose waters may be affected by the discharge are located, that the discharge is occurring. The permittee shall also notify the Department via its MiWaters system on the form entitled “Report of Discharge (CSO\SSO\RTB).” The MiWaters website is located at https://miwaters.deq.state.mi.us. At the conclusion of the discharge, the permittee shall make all such notifications specified in, and in accordance with, Section 324.3112a of the NREPA, and shall notify the Department via its MiWaters system on the form entitled “Report of Discharge (CSO\SSO\RTB).”

The permittee shall also annually contact municipalities, including the superintendent of a public drinking water supply with potentially affected intakes, whose waters may be affected by the permittee's discharge of untreated or partially treated sewage, and if those municipalities wish to be notified in the same manner as specified above, the permittee shall provide such notification.

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Section A. Limitations and Monitoring Requirements

Additionally, in accordance with Section 324.3112a of the NREPA, each time a discharge of untreated or partially treated sewage occurs, the permittee shall test the affected waters for *Escherichia coli* to assess the risk to the public health as a result of the discharge and shall provide the test results to the affected local county health departments and to the Department. The results of this testing shall be submitted to the Department via MiWaters as part of the notification specified above, or, if the results are not yet available, submitted as soon as they become available. This testing is not required if it has been waived by the local health department, or if the discharge(s) did not affect surface waters. The testing shall be done at locations specified by each affected local county health department but shall not exceed 10 tests for each separate discharge event. The affected local county health department may waive this testing requirement if it determines that such testing is not needed to assess the risk to the public health as a result of the discharge event.

Permittees accepting sanitary or municipal sewage from other sewage collection systems are encouraged to notify the owners of those systems of the above reporting and testing requirements.

7. Facility Contact

The "Facility Contact" was specified in the application. The permittee may replace the facility contact at any time, and shall notify the Department in writing within 10 days after replacement (including the name, address and telephone number of the new facility contact).

a. The facility contact shall be (or a duly authorized representative of this person):
   - for a corporation, a principal executive officer of at least the level of vice president; or a designated representative if the representative is responsible for the overall operation of the facility from which the discharge originates, as described in the permit application or other NPDES form,
   - for a partnership, a general partner,
   - for a sole proprietorship, the proprietor, or
   - for a municipal, state, or other public facility, either a principal executive officer, the mayor, village president, city or village manager or other duly authorized employee.

b. A person is a duly authorized representative only if:
   - the authorization is made in writing to the Department by a person described in paragraph a. of this section; and
   - the authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the facility (a duly authorized representative may thus be either a named individual or any individual occupying a named position).

Nothing in this section releases the permittee from properly submitting reports and forms as required by law.

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Section A. Limitations and Monitoring Requirements

8. Monthly Operating Reports

Part 41 of Act 451 of 1994 as amended, specifically Section 324.4106 and associated R 299.2953, requires that the permittee file with the Department, on forms prescribed by the Department, operating reports showing the effectiveness of the treatment facility operation and the quantity and quality of liquid wastes discharged into waters of the state.

Within 30 days of the effective date of this permit, the permittee shall submit to the Department a revised treatment facility monitoring program to address monitoring requirement changes reflected in this permit, or submit justification explaining why monitoring requirement changes reflected in this permit do not necessitate revisions to the treatment facility monitoring program. The permittee shall implement the revised treatment facility monitoring program upon approval from the Department. Applicable forms and guidance are available on the Department's web site at http://www.michigan.gov/deq/0,1607,7-135-3313_44117---,00.html. The permittee may use alternate forms if they are consistent with the approved treatment facility monitoring program. Unless the Department provides written notification to the permittee that monthly submittal of operating reports is required, operating reports that result from implementation of the approved treatment facility monitoring program shall be maintained on site for a minimum of three (3) years and shall be made available to the Department for review upon request.

9. Asset Management

The permittee shall at all times properly operate and maintain all facilities (i.e., the sewer system and treatment works as defined in Part 41 of the NREPA), and control systems installed or used by the permittee to operate the sewer system and treatment works and achieve and maintain compliance with the conditions of this permit (also see Part II.D.3 of this permit). The requirements of an Asset Management Program function to achieve the goals of effective performance, adequate funding, and adequate operator staffing and training. Asset management is a planning process for ensuring that optimum value is gained for each asset and that financial resources are available to rehabilitate and replace those assets when necessary. Asset management is centered on a framework of five (5) core elements: the current state of the assets; the required sustainable level of service; the assets critical to sustained performance; the minimum life-cycle costs; and the best long-term funding strategy.

a. Asset Management Program Requirements
On or before December 1, 2019, the permittee shall submit to the Department an Asset Management Plan for review and approval. An approvable Asset Management Plan shall contain a schedule for the development and implementation of an Asset Management Program that meets the requirements outlined below in 1) – 4). A copy of any Asset Management Program requirements already completed by the permittee should be submitted as part of the Asset Management Plan. Upon approval by the Department the permittee shall implement the Asset Management Plan. (The permittee may choose to include the Operation and Maintenance Manual required under Part II.C.14. of this permit as part of their Asset Management Program).

1) *Maintenance Staff.* The permittee shall provide an adequate staff to carry out the operation, maintenance, repair, and testing functions required to ensure compliance with the terms and conditions of this permit. The level of staffing needed shall be determined by taking into account the work involved in operating the sewer system and treatment works, planning for and conducting maintenance, and complying with this permit.

2) *Collection System Map.* The permittee shall complete a map of the sewer collection system it owns and operates. The map shall be of sufficient detail and at a scale to allow easy interpretation. The collection system information shown on the map shall be based on current conditions and shall be kept up-to-date and available for review by the Department. **Note: Items below referencing combined sewer systems are not applicable to separate sewer systems.** Such map(s) shall include but not be limited to the following:

Section A. Limitations and Monitoring Requirements

a) all sanitary sewer lines and related manholes;

b) all combined sewer lines, related manholes, catch basins and CSO regulators;

c) all known or suspected connections between the sanitary sewer or combined sewer and storm drain systems;

d) all outfalls, including the treatment plant outfall(s), combined sewer treatment facility outfalls, untreated CSOs, and any known SSOs;

e) all pump stations and force mains;

f) the wastewater treatment facility(ies), including all treatment processes;

g) all surface waters (labeled);

h) other major appurtenances such as inverted siphons and air release valves;

i) a numbering system which uniquely identifies manholes, catch basins, overflow points, regulators and outfalls;

j) the scale and a north arrow;

k) the pipe diameter, date of installation, type of material, distance between manholes, and the direction of flow; and

l) the manhole interior material, rim elevation (optional), and invert elevations.

3) *Inventory and assessment of fixed assets.* The permittee shall complete an inventory and assessment of operations-related fixed assets. Fixed assets are assets that are normally stationary (e.g., pumps, blowers, and buildings). The inventory and assessment shall be based on current conditions and shall be kept up-to-date and available for review by the Department.

a) The fixed asset inventory shall include the following:

(1) a brief description of the fixed asset, its design capacity (e.g., pump: 120 gallons per minute), its level of redundancy, and its tag number if applicable;

(2) the location of the fixed asset;

(3) the year the fixed asset was installed;

(4) the present condition of the fixed asset (e.g., excellent, good, fair, poor); and

(5) the current fixed asset (replacement) cost in dollars for year specified in accordance with approved schedules;

b) The fixed asset assessment shall include a “Business Risk Evaluation” that combines the probability of failure of the fixed asset and the criticality of the fixed asset, as follows:

(1) Rate the probability of failure of the fixed asset on a scale of 1-5 (low to high) using criteria such as maintenance history, failure history, and remaining percentage of useful life (or years remaining);

(2) Rate the criticality of the fixed asset on a scale of 1-5 (low to high) based on the consequence of failure versus the desired level of service for the facility; and

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Section A. Limitations and Monitoring Requirements

(3) Compute the Business Risk Factor of the fixed asset by multiplying the failure rating from (1) by the criticality rating from (2).

4) *Operation, Maintenance & Replacement (OM&R) Budget and Rate Sufficiency for the Sewer System and Treatment Works.* The permittee shall complete an assessment of its user rates and replacement fund, including the following:

a) beginning and end dates of fiscal year;

b) name of the department, committee, board, or other organization that sets rates for the operation of the sewer system and treatment works;

c) amount in the permittee's replacement fund in dollars for year specified in accordance with approved schedules;

d) replacement fund strategy of all assets with a useful life of 20 years or less;

e) expenditures for maintenance, corrective action and capital improvement taken during the fiscal year;

f) OM&R budget for the fiscal year; and

g) rate calculation demonstrating sufficient revenues to cover OM&R expenses. If the rate calculation shows there are insufficient revenues to cover OM&R expenses, the permittee shall document, within three (3) fiscal years after submittal of the Asset Management Plan, that there is at least one rate adjustment that reduces the revenue gap by at least 10 percent. The permittee may prepare and submit an alternate plan, subject to Department approval, for addressing the revenue gap. The ultimate goal of the Asset Management Program is to ensure sufficient revenues to cover OM&R expenses.

b. Reporting
Following Department approval of the permittee's Asset Management Plan, the permittee shall develop a written report that summarizes asset management activities completed during the previous year and planned for the upcoming year. The written report shall be submitted to the Department on or before <u>July 31 of each year</u>. The written report shall include:

1) a description of the staffing levels maintained during the year;

2) a description of inspections and maintenance activities conducted and corrective actions taken during the previous year;

3) expenditures for collection system maintenance activities, treatment works maintenance activities, corrective actions, and capital improvement during the previous year;

4) a summary of assets/areas identified for inspection/action (including capital improvement) in the upcoming year based on the five (5) core elements and the Business Risk Factors;

5) a maintenance budget and capital improvement budget for the upcoming year that take into account implementation of an effective Asset Management Program that meets the five (5) core elements;

6) an updated asset inventory based on the original submission; and

7) an updated OM&R budget with an updated rate schedule that includes the amount of insufficient revenues, if any.

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Section A. Limitations and Monitoring Requirements

10. Discharge Monitoring Report – Quality Assurance Study Program

The permittee shall participate in the Discharge Monitoring Report – Quality Assurance (DMR-QA) Study Program. The purpose of the DMR-QA Study Program is to annually evaluate the proficiency of all in-house and/or contract laboratory(ies) that perform, on behalf of the facility authorized to discharge under this permit, the analytical testing required under this permit. In accordance with Section 308 of the Clean Water Act (33 U.S.C. § 1318); and R 323.2138 and R 323.2154 of Part 21, Wastewater Discharge Permits, promulgated under Part 31 of the NREPA, participation in the DMR-QA Study Program is required for all major facilities, and for minor facilities selected for participation by the Department.

Annually and in accordance with DMR-QA Study Program requirements and submittal due dates, the permittee shall submit to the Michigan DMR-QA Study Program state coordinator all documentation required by the DMR-QA Study. DMR-QA Study Program participation is required only for the analytes required under this permit and only when those analytes are also identified in the DMR-QA Study.

If the permitted facility's status as a major facility should change, participation in the DMR-QA Study Program may be reevaluated. Questions concerning participation in the DMR-QA Study Program should be directed to the Michigan DMR-QA Study Program state coordinator.

All forms and instructions required for participation in the DMR-QA Study Program, including submittal due dates and state coordinator contact information, can be found at http://www.epa.gov/compliance/discharge-monitoring-report-quality-assurance-study-program.

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Section B. Storm Water Pollution Prevention

Section B. Storm Water Pollution Prevention is not required for this permit.

PART I

Section C. Industrial Waste Pretreatment Program

1. Michigan Industrial Pretreatment Program

a. The permittee shall implement the Michigan Industrial Pretreatment Program (MIPP) approved on July 23, 1985, and any subsequent modifications approved up to the issuance of this permit.

b. The permittee shall comply with R 323.2301 through R 323.2317 of the Michigan Administrative Code (Part 23 Rules) and the approved MIPP.

c. The permittee shall have the legal authority and necessary interjurisdictional agreements that provide the basis for the implementation and enforcement of the approved MIPP throughout the service area. The legal authority and necessary interjurisdictional agreements shall include, at a minimum, the authority to carry out the activities specified in R 323.2306(a).

d. The permittee shall develop procedures which describe, in sufficient detail, program commitments which enable implementation of the approved MIPP and the Part 23 Rules in accordance with R 323.2306(c).

e. The permittee shall establish an interjurisdictional agreement (or comparable document) with all tributary governmental jurisdictions. Each interjurisdictional agreement shall contain, at a minimum, the following:

1) identification of the agency responsible for the implementation and enforcement of the approved MIPP within the tributary governmental jurisdiction's boundaries; and

2) the provision of the legal authority which provides the basis for the implementation and enforcement of the approved MIPP within the tributary governmental jurisdiction's boundaries.

f. The permittee shall prohibit discharges that:

1) cause, in whole or in part, the permittee's failure to comply with any condition of this permit or the NREPA;

2) restrict, in whole or in part, the permittee's management of biosolids;

3) cause, in whole or in part, operational problems at the treatment facility or in its collection system;

4) violate any of the general or specific prohibitions identified in R 323.2303(1) and (2);

5) violate categorical standards identified in R 323.2311; and

6) violate local limits established in accordance with R 323.2303(4).

g. The permittee shall maintain a list of its nondomestic users that meet the criteria of a significant industrial user as identified in R 323.2302(cc).

h. The permittee shall develop an enforcement response plan which describes, in sufficient detail, program commitments which will enable the enforcement of the approved MIPP and the Part 23 Rules in accordance with R 323.2306(g).

i. The Department may require modifications to the approved MIPP which are necessary to ensure compliance with the Part 23 Rules in accordance with R 323.2309.

j. The permittee shall not implement changes or modifications to the approved MIPP without notification to the Department.

Section C. Industrial Waste Pretreatment Program

k. The permittee shall maintain an adequate revenue structure and staffing level for effective implementation of the approved MIPP.

l. The permittee shall develop and maintain, for a minimum of three (3) years, all records and information necessary to determine nondomestic user compliance with the Part 23 Rules and the approved MIPP. This period of retention shall be extended during the course of any unresolved enforcement action or litigation regarding a nondomestic user or when requested by the Department or the United States Environmental Protection Agency. All of the aforementioned records and information shall be made available upon request for inspection and copying by the Department and the United States Environmental Protection Agency.

m. The permittee shall evaluate the approved MIPP for compliance with the Part 23 Rules and the prohibitions set forth in item f. above. Based upon this evaluation, the permittee shall propose to the Department all necessary changes or modifications to the approved MIPP no later than the next Industrial Pretreatment Program Annual Report due date (see item o. below).

n. The permittee shall develop and enforce local limits to implement the prohibitions set forth in item f. above. Local limits shall be based upon data representative of actual conditions demonstrated in a maximum allowable headworks loading analysis.

o. On or before April 1 of each year, the permittee shall submit to the Department, as required by R 323.2310(8), an Industrial Pretreatment Program Annual Report on the status of program implementation and enforcement activities. The reporting period shall begin on January 1 and end on December 31. At a minimum, the Industrial Pretreatment Program Annual Report shall include:

1) the Pretreatment Program Reports data identified in Appendix A to 40 CFR Part 127 – NPDES Electronic Reporting;

2) a summary of changes to the approved MIPP that have not been previously reported to the Department;

3) a summary of results of all the sampling and analyses performed of the wastewater treatment plant's influent, effluent, and biosolids conducted in accordance with approved methods during the reporting period. The summary shall include the monthly average, daily maximum, quantification level, and number of samples analyzed for each pollutant. At a minimum, the results of analyses for all locally limited parameters for at least one monitoring event that tests influent, effluent and biosolids during the reporting period shall be submitted with each report, unless otherwise required by the Department. Sample collection shall be at intervals sufficient to provide pollutant removal rates, unless the pollutant is not measurable; and

4) any other relevant information requested by the Department.

p. The permittee is required under this permit and R 323.2303(4) of the Michigan Administrative Code to review and update their local limits when:

1) New pollutants are introduced

2) New pollutants that were previously unevaluated are identified

3) New water quality or biosolids standards are established or additional information becomes available about the nature of pollutants, such as removal rates and accumulation in biosolids.

4) Substantial increases of pollutants are proposed as required in the notification of new or increased uses in accordance with the provisions of 40 CFR 122.42.

Section D. Residuals Management Program

1. Residuals Management Program for Land Application of Biosolids

The permittee is authorized to land-apply bulk biosolids or prepare bulk biosolids for land application in accordance with the permittee's approved Residuals Management Program (RMP) approved on May 6, 2002, and approved modifications thereto, in accordance with the requirements established in R 323.2401 through R 323.2418 of the Michigan Administrative Code (Part 24 Rules). The approved RMP, and any approved modifications thereto, are enforceable requirements of this permit. Incineration, landfilling and other residual disposal activities shall be conducted in accordance with Part II.D.7. of this permit. The Part 24 Rules can be obtained via the internet (http://www.michigan.gov/deq/ and on the left side of the screen click on Water, Biosolids & Industrial Pretreatment, Biosolids then click on Biosolids Laws and Rules Information which is under the Laws & Rules banner in the center of the screen).

a. Annual Report
On or before October 30 of each year, the permittee shall submit an annual report to the Department for the previous fiscal year of October 1 through September 30. The report shall be submitted electronically via the Department's MiWaters system at https://miwaters.deq.state.mi.us. At a minimum, the report shall contain:

1) a certification that current residuals management practices are in accordance with the approved RMP, or a proposal for modification to the approved RMP; and

2) a completed Biosolids Annual Report Form, available at https://miwaters.deq.state.mi.us.

b. Modifications to the Approved RMP
Prior to implementation of modifications to the RMP, the permittee shall submit proposed modifications to the Department for approval. The approved modification shall become effective upon the date of approval. Upon written notification, the Department may impose additional requirements and/or limitations to the approved RMP as necessary to protect public health and the environment from any adverse effect of a pollutant in the biosolids.

c. Record Keeping
Records required by the Part 24 Rules shall be kept for a minimum of five years. However, the records documenting cumulative loading for sites subject to cumulative pollutant loading rates shall be kept as long as the site receives biosolids.

d. Contact Information
RMP-related submittals shall be made to the Department.

PART II

Part II may include terms and /or conditions not applicable to discharges covered under this permit.

Section A. Definitions

Acute toxic unit (TU_A) means $100/LC_{50}$ where the LC_{50} is determined from a whole effluent toxicity (WET) test which produces a result that is statistically or graphically estimated to be lethal to 50% of the test organisms.

Annual monitoring frequency refers to a calendar year beginning on January 1 and ending on December 31. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Authorized public agency means a state, local, or county agency that is designated pursuant to the provisions of section 9110 of Part 91 of the NREPA to implement soil erosion and sedimentation control requirements with regard to construction activities undertaken by that agency.

Best management practices (BMPs) means structural devices or nonstructural practices that are designed to prevent pollutants from entering into storm water, to direct the flow of storm water, or to treat polluted storm water.

Bioaccumulative chemical of concern (BCC) means a chemical which, upon entering the surface waters, by itself or as its toxic transformation product, accumulates in aquatic organisms by a human health bioaccumulation factor of more than 1000 after considering metabolism and other physiochemical properties that might enhance or inhibit bioaccumulation. The human health bioaccumulation factor shall be derived according to R 323.1057(5). Chemicals with half-lives of less than 8 weeks in the water column, sediment, and biota are not BCCs. The minimum bioaccumulation concentration factor (BAF) information needed to define an organic chemical as a BCC is either a field-measured BAF or a BAF derived using the biota-sediment accumulation factor (BSAF) methodology. The minimum BAF information needed to define an inorganic chemical as a BCC, including an organometal, is either a field-measured BAF or a laboratory-measured bioconcentration factor (BCF). The BCCs to which these rules apply are identified in Table 5 of R 323.1057 of the Water Quality Standards.

Biosolids are the solid, semisolid, or liquid residues generated during the treatment of sanitary sewage or domestic sewage in a treatment works. This includes, but is not limited to, scum or solids removed in primary, secondary, or advanced wastewater treatment processes and a derivative of the removed scum or solids.

Bulk biosolids means biosolids that are not sold or given away in a bag or other container for application to a lawn or home garden.

Certificate of Coverage (COC) is a document, issued by the Department, which authorizes a discharge under a general permit.

Chronic toxic unit (TU_C) means 100/MATC or $100/IC_{25}$, where the maximum acceptable toxicant concentration (MATC) and IC_{25} are expressed as a percent effluent in the test medium.

Class B biosolids refers to material that has met the Class B pathogen reduction requirements or equivalent treatment by a Process to Significantly Reduce Pathogens (PSRP) in accordance with the Part 24 Rules. Processes include aerobic digestion, composting, anaerobic digestion, lime stabilization and air drying.

Combined sewer system is a sewer system in which storm water runoff is combined with sanitary wastes.

PART II

Section A. Definitions

Daily concentration is the sum of the concentrations of the individual samples of a parameter divided by the number of samples taken during any calendar day. The daily concentration will be used to determine compliance with any maximum and minimum daily concentration limitations (except for pH and dissolved oxygen). When required by the permit, report the maximum calculated daily concentration for the month in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the Discharge Monitoring Reports (DMRs).

For pH, report the maximum value of any *individual* sample taken during the month in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs and the minimum value of any *individual* sample taken during the month in the "MINIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs. For dissolved oxygen, report the minimum concentration of any *individual* sample in the "MINIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs.

Daily loading is the total discharge by weight of a parameter discharged during any calendar day. This value is calculated by multiplying the daily concentration by the total daily flow and by the appropriate conversion factor. The daily loading will be used to determine compliance with any maximum daily loading limitations. When required by the permit, report the maximum calculated daily loading for the month in the "MAXIMUM" column under "QUANTITY OR LOADING" on the DMRs.

Daily monitoring frequency refers to a 24-hour day. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Department means the Michigan Department of Environment, Great Lakes, and Energy.

Detection level means the lowest concentration or amount of the target analyte that can be determined to be different from zero by a single measurement at a stated level of probability.

Discharge means the addition of any waste, waste effluent, wastewater, pollutant, or any combination thereof to any surface water of the state.

EC_{50} means a statistically or graphically estimated concentration that is expected to cause 1 or more specified effects in 50% of a group of organisms under specified conditions.

Fecal coliform bacteria monthly
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – Fecal coliform bacteria monthly is the geometric mean of all daily concentrations determined during a discharge event. Days on which no daily concentration is determined shall not be used to determine the calculated monthly value. The calculated monthly value will be used to determine compliance with the maximum monthly fecal coliform bacteria limitations. When required by the permit, report the calculated monthly value in the "AVERAGE" column under "QUALITY OR CONCENTRATION" on the DMR. If the period in which the discharge event occurred was partially in each of two months, the calculated monthly value shall be reported on the DMR of the month in which the last day of discharge occurred.

FOR ALL OTHER DISCHARGES – Fecal coliform bacteria monthly is the geometric mean of all daily concentrations determined during a reporting month. Days on which no daily concentration is determined shall not be used to determine the calculated monthly value. The calculated monthly value will be used to determine compliance with the maximum monthly fecal coliform bacteria limitations. When required by the permit, report the calculated monthly value in the "AVERAGE" column under "QUALITY OR CONCENTRATION" on the DMR.

PART II

Section A. Definitions

Fecal coliform bacteria 7-day
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – Fecal coliform bacteria 7-day is the geometric mean of the daily concentrations determined during any 7 consecutive days of discharge during a discharge event. If the number of daily concentrations determined during the discharge event is less than 7 days, the number of actual daily concentrations determined shall be used for the calculation. Days on which no daily concentration is determined shall not be used to determine the value. The calculated 7-day value will be used to determine compliance with the maximum 7-day fecal coliform bacteria limitations. When required by the permit, report the maximum calculated 7-day geometric mean value for the month in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs. If the 7-day period was partially in each of two months, the value shall be reported on the DMR of the month in which the last day of discharge occurred.

FOR ALL OTHER DISCHARGES – Fecal coliform bacteria 7-day is the geometric mean of the daily concentrations determined during any 7 consecutive days in a reporting month. If the number of daily concentrations determined is less than 7, the actual number of daily concentrations determined shall be used for the calculation. Days on which no daily concentration is determined shall not be used to determine the value. The calculated 7-day value will be used to determine compliance with the maximum 7-day fecal coliform bacteria limitations. When required by the permit, report the maximum calculated 7-day geometric mean for the month in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs. The first calculation shall be made on day 7 of the reporting month, and the last calculation shall be made on the last day of the reporting month.

Flow-proportioned sample is a composite sample with the sample volume proportional to the effluent flow.

General permit means a National Pollutant Discharge Elimination System permit issued authorizing a category of similar discharges.

Geometric mean is the average of the logarithmic values of a base 10 data set, converted back to a base 10 number.

Grab sample is a single sample taken at neither a set time nor flow.

IC_{25} means the toxicant concentration that would cause a 25% reduction in a nonquantal biological measurement for the test population.

Illicit connection means a physical connection to a municipal separate storm sewer system that primarily conveys non-storm water discharges other than uncontaminated groundwater into the storm sewer; or a physical connection not authorized or permitted by the local authority, where a local authority requires authorization or a permit for physical connections.

Illicit discharge means any discharge to, or seepage into, a municipal separate storm sewer system that is not composed entirely of storm water or uncontaminated groundwater. Illicit discharges include non-storm water discharges through pipes or other physical connections; dumping of motor vehicle fluids, household hazardous wastes, domestic animal wastes, or litter; collection and intentional dumping of grass clippings or leaf litter; or unauthorized discharges of sewage, industrial waste, restaurant wastes, or any other non-storm water waste directly into a separate storm sewer.

Individual permit means a site-specific NPDES permit.

Inlet means a catch basin, roof drain, conduit, drain tile, retention pond riser pipe, sump pump, or other point where storm water or wastewater enters into a closed conveyance system prior to discharge off site or into waters of the state.

PART II

Section A. Definitions

Interference is a discharge which, alone or in conjunction with a discharge or discharges from other sources, both: 1) inhibits or disrupts the POTW, its treatment processes or operations, or its sludge processes, use or disposal; and 2) therefore, is a cause of a violation of any requirement of the POTW's NPDES permit (including an increase in the magnitude or duration of a violation) or, of the prevention of sewage sludge use or disposal in compliance with the following statutory provisions and regulations or permits issued thereunder (or more stringent state or local regulations): Section 405 of the Clean Water Act, the Solid Waste Disposal Act (SWDA) (including Title II, more commonly referred to as the Resource Conservation and Recovery Act (RCRA), and including state regulations contained in any state sludge management plan prepared pursuant to Subtitle D of the SWDA), the Clean Air Act, the Toxic Substances Control Act, and the Marine Protection, Research and Sanctuaries Act. [This definition does not apply to sample matrix interference].

Land application means spraying or spreading biosolids or a biosolids derivative onto the land surface, injecting below the land surface, or incorporating into the soil so that the biosolids or biosolids derivative can either condition the soil or fertilize crops or vegetation grown in the soil.

LC_{50} means a statistically or graphically estimated concentration that is expected to be lethal to 50% of a group of organisms under specified conditions.

Maximum acceptable toxicant concentration (MATC) means the concentration obtained by calculating the geometric mean of the lower and upper chronic limits from a chronic test. A lower chronic limit is the highest tested concentration that did not cause the occurrence of a specific adverse effect. An upper chronic limit is the lowest tested concentration which did cause the occurrence of a specific adverse effect and above which all tested concentrations caused such an occurrence.

Maximum extent practicable means implementation of best management practices by a public body to comply with an approved storm water management program as required by a national permit for a municipal separate storm sewer system, in a manner that is environmentally beneficial, technically feasible, and within the public body’s legal authority.

MGD means million gallons per day.

Monthly concentration is the sum of the daily concentrations determined during a reporting period divided by the number of daily concentrations determined. The calculated monthly concentration will be used to determine compliance with any maximum monthly concentration limitations. Days with no discharge shall not be used to determine the value. When required by the permit, report the calculated monthly concentration in the “AVERAGE” column under “QUALITY OR CONCENTRATION” on the DMR.

For minimum percent removal requirements, the monthly influent concentration and the monthly effluent concentration shall be determined. The calculated monthly percent removal, which is equal to 100 times the quantity [1 minus the quantity (monthly effluent concentration divided by the monthly influent concentration)], shall be reported in the "MINIMUM" column under "QUALITY OR CONCENTRATION" on the DMRs.

Monthly loading is the sum of the daily loadings of a parameter divided by the number of daily loadings determined during a reporting period. The calculated monthly loading will be used to determine compliance with any maximum monthly loading limitations. Days with no discharge shall not be used to determine the value. When required by the permit, report the calculated monthly loading in the “AVERAGE” column under “QUANTITY OR LOADING” on the DMR.

Monthly monitoring frequency refers to a calendar month. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Municipal separate storm sewer means a conveyance or system of conveyances designed or used for collecting or conveying storm water which is not a combined sewer and which is not part of a publicly-owned treatment works as defined in the Code of Federal Regulations at 40 CFR 122.2.

PART II

Section A. Definitions

Municipal separate storm sewer system (MS4) means all separate storm sewers that are owned or operated by the United States, a state, city, village, township, county, district, association, or other public body created by or pursuant to state law, having jurisdiction over disposal of sewage, industrial wastes, storm water, or other wastes, including special districts under state law, such as a sewer district, flood control district, or drainage district, or similar entity, or a designated or approved management agency under Section 208 of the Federal Act that discharges to the waters of the state. This term includes systems similar to separate storm sewer systems in municipalities, such as systems at military bases, large hospital or prison complexes, and highways and other thoroughfares. The term does not include separate storm sewers in very discrete areas, such as individual buildings.

National Pretreatment Standards are the regulations promulgated by or to be promulgated by the Federal Environmental Protection Agency pursuant to Section 307(b) and (c) of the Federal Act. The standards establish nationwide limits for specific industrial categories for discharge to a POTW.

No observed adverse effect level (NOAEL) means the highest tested dose or concentration of a substance which results in no observed adverse effect in exposed test organisms where higher doses or concentrations result in an adverse effect.

Noncontact cooling water is water used for cooling which does not come into direct contact with any raw material, intermediate product, by-product, waste product or finished product.

Nondomestic user is any discharger to a POTW that discharges wastes other than or in addition to water-carried wastes from toilet, kitchen, laundry, bathing or other facilities used for household purposes.

Outfall is the location at which a point source discharge enters the surface waters of the state.

Part 91 agency means an agency that is designated by a county board of commissioners pursuant to the provisions of section 9105 of Part 91 of the NREPA; an agency that is designated by a city, village, or township in accordance with the provisions of section 9106 of Part 91 of the NREPA; or the Department for soil erosion and sedimentation activities under Part 615, Part 631, or Part 632 pursuant to the provisions of section 9115 of Part 91 of the NREPA.

Part 91 permit means a soil erosion and sedimentation control permit issued by a Part 91 agency pursuant to the provisions of Part 91 of the NREPA.

Partially treated sewage is any sewage, sewage and storm water, or sewage and wastewater, from domestic or industrial sources that is treated to a level less than that required by the permittee's National Pollutant Discharge Elimination System permit, or that is not treated to national secondary treatment standards for wastewater, including discharges to surface waters from retention treatment facilities.

Point of discharge is the location of a point source discharge where storm water is discharged directly into a separate storm sewer system.

Point source discharge means a discharge from any discernible, confined, discrete conveyance, including but not limited to any pipe, ditch, channel, tunnel, conduit, well, discrete fissure, container, or rolling stock. Changing the surface of land or establishing grading patterns on land will result in a point source discharge where the runoff from the site is ultimately discharged to waters of the state.

Polluting material means any material, in solid or liquid form, identified as a polluting material under the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code).

POTW is a publicly owned treatment work.

PART II

Section A. Definitions

Pretreatment is reducing the amount of pollutants, eliminating pollutants, or altering the nature of pollutant properties to a less harmful state prior to discharge into a public sewer. The reduction or alteration can be by physical, chemical, or biological processes, process changes, or by other means. Dilution is not considered pretreatment unless expressly authorized by an applicable National Pretreatment Standard for a particular industrial category.

Public (as used in the MS4 individual permit) means all persons who potentially could affect the authorized storm water discharges, including, but not limited to, residents, visitors to the area, public employees, businesses, industries, and construction contractors and developers.

Public body means the United States; the state of Michigan; a city, village, township, county, school district, public college or university, or single-purpose governmental agency; or any other body which is created by federal or state statute or law.

Qualified Personnel means an individual who meets qualifications acceptable to the Department and who is authorized by an Industrial Storm Water Certified Operator to collect the storm water sample.

Qualifying storm event means a storm event causing greater than 0.1 inch of rainfall and occurring at least 72 hours after the previous measurable storm event that also caused greater than 0.1 inch of rainfall. Upon request, the Department may approve an alternate definition meeting the condition of a qualifying storm event.

Quantification level means the measurement of the concentration of a contaminant obtained by using a specified laboratory procedure calculated at a specified concentration above the detection level. It is considered the lowest concentration at which a particular contaminant can be quantitatively measured using a specified laboratory procedure for monitoring of the contaminant.

Quarterly monitoring frequency refers to a three month period, defined as January through March, April through June, July through September, and October through December. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

Regional Administrator is the Region 5 Administrator, U.S. EPA, located at R-19J, 77 W. Jackson Blvd., Chicago, Illinois 60604.

Regulated area means the permittee's urbanized area, where urbanized area is defined as a place and its adjacent densely-populated territory that together have a minimum population of 50,000 people as defined by the United States Bureau of the Census and as determined by the latest available decennial census.

Secondary containment structure means a unit, other than the primary container, in which significant materials are packaged or held, which is required by State or Federal law to prevent the escape of significant materials by gravity into sewers, drains, or otherwise directly or indirectly into any sewer system or to the surface or ground waters of this state.

Separate storm sewer system means a system of drainage, including, but not limited to, roads, catch basins, curbs, gutters, parking lots, ditches, conduits, pumping devices, or man-made channels, which is not a combined sewer where storm water mixes with sanitary wastes, and is not part of a POTW.

Significant industrial user is a nondomestic user that: 1) is subject to Categorical Pretreatment Standards under 40 CFR 403.6 and 40 CFR Chapter I, Subchapter N; or 2) discharges an average of 25,000 gallons per day or more of process wastewater to a POTW (excluding sanitary, noncontact cooling and boiler blowdown wastewater); contributes a process waste stream which makes up five (5) percent or more of the average dry weather hydraulic or organic capacity of the POTW treatment plant; or is designated as such by the permittee as defined in 40 CFR 403.12(a) on the basis that the industrial user has a reasonable potential for adversely affecting the POTW's treatment plant operation or violating any pretreatment standard or requirement (in accordance with 40 CFR 403.8(f)(6)).

PART II

Section A. Definitions

Significant materials Significant Materials means any material which could degrade or impair water quality, including but not limited to: raw materials; fuels; solvents, detergents, and plastic pellets; finished materials such as metallic products; hazardous substances designated under Section 101(14) of Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) (see 40 CFR 372.65); any chemical the facility is required to report pursuant to Section 313 of Emergency Planning and Community Right-to-Know Act (EPCRA); polluting materials as identified under the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code); Hazardous Wastes as defined in Part 111 of the NREPA; fertilizers; pesticides; and waste products such as ashes, slag, and sludge that have the potential to be released with storm water discharges.

Significant spills and significant leaks means any release of a polluting material reportable under the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code).

Special-use area means secondary containment structures required by state or federal law; lands on Michigan's List of Sites of Environmental Contamination pursuant to Part 201, Environmental Remediation, of the NREPA; and/or areas with other activities that may contribute pollutants to the storm water for which the Department determines monitoring is needed.

Stoichiometric means the quantity of a reagent calculated to be necessary and sufficient for a given chemical reaction.

Storm water means storm water runoff, snow melt runoff, surface runoff and drainage, and non-storm water included under the conditions of this permit.

Storm water discharge point is the location where the point source discharge of storm water is directed to surface waters of the state or to a separate storm sewer. It includes the location of all point source discharges where storm water exits the facility, including *outfalls* which discharge directly to surface waters of the state, and *points of discharge* which discharge directly into separate storm sewer systems.

SWPPP means the Storm Water Pollution Prevention Plan prepared in accordance with this permit.

Tier I value means a value for aquatic life, human health or wildlife calculated under R 323.1057 of the Water Quality Standards using a tier I toxicity database.

Tier II value means a value for aquatic life, human health or wildlife calculated under R 323.1057 of the Water Quality Standards using a tier II toxicity database.

Total maximum daily loads (TMDLs) are required by the Federal Act for waterbodies that do not meet water quality standards. TMDLs represent the maximum daily load of a pollutant that a waterbody can assimilate and meet water quality standards, and an allocation of that load among point sources, nonpoint sources, and a margin of safety.

Toxicity reduction evaluation (TRE) means a site-specific study conducted in a stepwise process designed to identify the causative agents of effluent toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in effluent toxicity.

Water Quality Standards means the Part 4 Water Quality Standards promulgated pursuant to Part 31 of the NREPA, being R 323.1041 through R 323.1117 of the Michigan Administrative Code.

Weekly monitoring frequency refers to a calendar week which begins on Sunday and ends on Saturday. When required by this permit, an analytical result, reading, value or observation shall be reported for that period if a discharge occurs during that period.

WWSL is a wastewater stabilization lagoon.

WWSL discharge event is a discrete occurrence during which effluent is discharged to the surface water up to 10 days of a consecutive 14 day period.

PART II

Section A. Definitions

3-portion composite sample is a sample consisting of three equal-volume grab samples collected at equal intervals over an 8-hour period.

7-day concentration
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – The 7-day concentration is the sum of the daily concentrations determined during any 7 consecutive days of discharge during a WWSL discharge event divided by the number of daily concentrations determined. If the number of daily concentrations determined during the WWSL discharge event is less than 7 days, the number of actual daily concentrations determined shall be used for the calculation. The calculated 7-day concentration will be used to determine compliance with any maximum 7-day concentration limitations. When required by the permit, report the maximum calculated 7-day concentration for the WWSL discharge event in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the DMR. If the WWSL discharge event was partially in each of two months, the value shall be reported on the DMR of the month in which the last day of discharge occurred.

FOR ALL OTHER DISCHARGES – The 7-day concentration is the sum of the daily concentrations determined during any 7 consecutive days in a reporting month divided by the number of daily concentrations determined. If the number of daily concentrations determined is less than 7, the actual number of daily concentrations determined shall be used for the calculation. The calculated 7-day concentration will be used to determine compliance with any maximum 7-day concentration limitations in the reporting month. When required by the permit, report the maximum calculated 7-day concentration for the month in the "MAXIMUM" column under "QUALITY OR CONCENTRATION" on the DMR. The first 7-day calculation shall be made on day 7 of the reporting month, and the last calculation shall be made on the last day of the reporting month.

7-day loading
FOR WWSLs THAT COLLECT AND STORE WASTEWATER AND ARE AUTHORIZED TO DISCHARGE ONLY IN THE SPRING AND/OR FALL ON AN INTERMITTENT BASIS – The 7-day loading is the sum of the daily loadings determined during any 7 consecutive days of discharge during a WWSL discharge event divided by the number of daily loadings determined. If the number of daily loadings determined during the WWSL discharge event is less than 7 days, the number of actual daily loadings determined shall be used for the calculation. The calculated 7-day loading will be used to determine compliance with any maximum 7-day loading limitations. When required by the permit, report the maximum calculated 7-day loading for the WWSL discharge event in the "MAXIMUM" column under "QUANTITY OR LOADING" on the DMR. If the WWSL discharge event was partially in each of two months, the value shall be reported on the DMR of the month in which the last day of discharge occurred

FOR ALL OTHER DISCHARGES – The 7-day loading is the sum of the daily loadings determined during any 7 consecutive days in a reporting month divided by the number of daily loadings determined. If the number of daily loadings determined is less than 7, the actual number of daily loadings determined shall be used for the calculation. The calculated 7-day loading will be used to determine compliance with any maximum 7-day loading limitations in the reporting month. When required by the permit, report the maximum calculated 7-day loading for the month in the "MAXIMUM" column under "QUANTITY OR LOADING" on the DMR. The first 7-day calculation shall be made on day 7 of the reporting month, and the last calculation shall be made on the last day of the reporting month.

24-hour composite sample is a flow-proportioned composite sample consisting of hourly or more frequent portions that are taken over a 24-hour period. A time-proportioned composite sample may be used upon approval of the Department if the permittee demonstrates it is representative of the discharge.

PART II

Section B. Monitoring Procedures

1. Representative Samples

Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge.

2. Test Procedures

Test procedures for the analysis of pollutants shall conform to regulations promulgated pursuant to Section 304(h) of the Federal Act (40 CFR Part 136 – Guidelines Establishing Test Procedures for the Analysis of Pollutants), unless specified otherwise in this permit. **Test procedures used shall be sufficiently sensitive to determine compliance with applicable effluent limitations**. Requests to use test procedures not promulgated under 40 CFR Part 136 for pollutant monitoring required by this permit shall be made in accordance with the Alternate Test Procedures regulations specified in 40 CFR 136.4. These requests shall be submitted to the Manager of the Permits Section, Water Resources Division, Michigan Department of Environment, Great Lakes, and Energy, P.O. Box 30458, Lansing, Michigan, 48909-7958. The permittee may use such procedures upon approval.

The permittee shall periodically calibrate and perform maintenance procedures on all analytical instrumentation at intervals to ensure accuracy of measurements. The calibration and maintenance shall be performed as part of the permittee's laboratory Quality Control/Quality Assurance program.

3. Instrumentation

The permittee shall periodically calibrate and perform maintenance procedures on all monitoring instrumentation at intervals to ensure accuracy of measurements.

4. Recording Results

For each measurement or sample taken pursuant to the requirements of this permit, the permittee shall record the following information: 1) the exact place, date, and time of measurement or sampling; 2) the person(s) who performed the measurement or sample collection; 3) the dates the analyses were performed; 4) the person(s) who performed the analyses; 5) the analytical techniques or methods used; 6) the date of and person responsible for equipment calibration; and 7) the results of all required analyses.

5. Records Retention

All records and information resulting from the monitoring activities required by this permit including all records of analyses performed and calibration and maintenance of instrumentation and recordings from continuous monitoring instrumentation shall be retained for a minimum of three (3) years, or longer if requested by the Regional Administrator or the Department.

Section C. Reporting Requirements

1. Start-up Notification

If the permittee will not discharge during the first 60 days following the effective date of this permit, the permittee shall notify the Department within 14 days following the effective date of this permit, and then 60 days prior to the commencement of the discharge.

2. Submittal Requirements for Self-Monitoring Data

Part 31 of the NREPA (specifically Section 324.3110(7)); and R 323.2155(2) of Part 21, Wastewater Discharge Permits, promulgated under Part 31 of the NREPA, allow the Department to specify the forms to be utilized for reporting the required self-monitoring data. Unless instructed on the effluent limitations page to conduct "Retained Self-Monitoring," the permittee shall submit self-monitoring data via the Department's MiWaters system.

The permittee shall utilize the information provided on the MiWaters website, located at https://miwaters.deq.state.mi.us, to access and submit the electronic forms. Both monthly summary and daily data shall be submitted to the Department no later than the 20th day of the month following each month of the authorized discharge period(s). The permittee may be allowed to submit the electronic forms after this date if the Department has granted an extension to the submittal date.

3. Retained Self-Monitoring Requirements

If instructed on the effluent limits page (or otherwise authorized by the Department in accordance with the provisions of this permit) to conduct retained self-monitoring, the permittee shall maintain a year-to-date log of retained self-monitoring results and, upon request, provide such log for inspection to the staff of the Department. Retained self-monitoring results are public information and shall be promptly provided to the public upon request.

The permittee shall certify, in writing, to the Department, on or before January 10th (April 1st for animal feeding operation facilities) of each year, that: 1) all retained self-monitoring requirements have been complied with and a year-to-date log has been maintained; and 2) the application on which this permit is based still accurately describes the discharge. With this annual certification, the permittee shall submit a summary of the previous year's monitoring data. The summary shall include maximum values for samples to be reported as daily maximums and/or monthly maximums and minimum values for any daily minimum samples.

Retained self-monitoring may be denied to a permittee by notification in writing from the Department. In such cases, the permittee shall submit self-monitoring data in accordance with Part II.C.2., above. Such a denial may be rescinded by the Department upon written notification to the permittee. Reissuance or modification of this permit or reissuance or modification of an individual permittee's authorization to discharge shall not affect previous approval or denial for retained self-monitoring unless the Department provides notification in writing to the permittee.

4. Additional Monitoring by Permittee

If the permittee monitors any pollutant at the location(s) designated herein more frequently than required by this permit, using approved analytical methods as specified above, the results of such monitoring shall be included in the calculation and reporting of the values required in the Discharge Monitoring Report. Such increased frequency shall also be indicated.

Monitoring required pursuant to Part 41 of the NREPA or Rule 35 of the Mobile Home Park Commission Act (Act 96 of the Public Acts of 1987) for assurance of proper facility operation shall be submitted as required by the Department.

Section C. Reporting Requirements

5. Compliance Dates Notification

Within 14 days of every compliance date specified in this permit, the permittee shall submit a *written* notification to the Department indicating whether or not the particular requirement was accomplished. If the requirement was not accomplished, the notification shall include an explanation of the failure to accomplish the requirement, actions taken or planned by the permittee to correct the situation, and an estimate of when the requirement will be accomplished. If a written report is required to be submitted by a specified date and the permittee accomplishes this, a separate written notification is not required.

6. Noncompliance Notification

Compliance with all applicable requirements set forth in the Federal Act, Parts 31 and 41 of the NREPA, and related regulations and rules is required. All instances of noncompliance shall be reported as follows:

a. 24-Hour Reporting
Any noncompliance which may endanger health or the environment (including maximum and/or minimum daily concentration discharge limitation exceedances) shall be reported, verbally, within 24 hours from the time the permittee becomes aware of the noncompliance. A written submission shall also be provided within five (5) days.

b. Other Reporting
The permittee shall report, in writing, all other instances of noncompliance not described in a. above at the time monitoring reports are submitted; or, in the case of retained self-monitoring, within five (5) days from the time the permittee becomes aware of the noncompliance.

Written reporting shall include: 1) a description of the discharge and cause of noncompliance; and 2) the period of noncompliance, including exact dates and times, or, if not yet corrected, the anticipated time the noncompliance is expected to continue, and the steps taken to reduce, eliminate and prevent recurrence of the noncomplying discharge.

7. Spill Notification

The permittee shall immediately report any release of any polluting material which occurs to the surface waters or groundwaters of the state, unless the permittee has determined that the release is not in excess of the threshold reporting quantities specified in the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code), by calling the Department at the number indicated on the second page of this permit (or, if this is a general permit, on the COC); or, if the notice is provided after regular working hours, call the Department's 24-hour Pollution Emergency Alerting System telephone number, 1-800-292-4706 (calls from **out-of-state** dial 1-517-373-7660).

Within ten (10) days of the release, the permittee shall submit to the Department a full written explanation as to the cause of the release, the discovery of the release, response (clean-up and/or recovery) measures taken, and preventive measures taken or a schedule for completion of measures to be taken to prevent reoccurrence of similar releases.

PART II

Section C. Reporting Requirements

8. Upset Noncompliance Notification

If a process "upset" (defined as an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the permittee) has occurred, the permittee who wishes to establish the affirmative defense of upset, shall notify the Department by telephone within 24 hours of becoming aware of such conditions; and within five (5) days, provide in writing, the following information:

a. that an upset occurred and that the permittee can identify the specific cause(s) of the upset;

b. that the permitted wastewater treatment facility was, at the time, being properly operated and maintained (note that an upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation); and

c. that the permittee has specified and taken action on all responsible steps to minimize or correct any adverse impact in the environment resulting from noncompliance with this permit.

No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review.

In any enforcement proceedings, the permittee, seeking to establish the occurrence of an upset, has the burden of proof.

9. Bypass Prohibition and Notification

a. Bypass Prohibition
Bypass is prohibited, and the Department may take an enforcement action, unless:

1) bypass was unavoidable to prevent loss of life, personal injury, or severe property damage;

2) there were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate backup equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass; and

3) the permittee submitted notices as required under 9.b. or 9.c. below.

b. Notice of Anticipated Bypass
If the permittee knows in advance of the need for a bypass, it shall submit prior notice to the Department, if possible at least ten (10) days before the date of the bypass, and provide information about the anticipated bypass as required by the Department. The Department may approve an anticipated bypass, after considering its adverse effects, if it will meet the three (3) conditions listed in 9.a. above.

c. Notice of Unanticipated Bypass
The permittee shall submit notice to the Department of an unanticipated bypass by calling the Department at the number indicated on the second page of this permit (if the notice is provided after regular working hours, use the following number: 1-800-292-4706) as soon as possible, but no later than 24 hours from the time the permittee becomes aware of the circumstances.

Section C. Reporting Requirements

d. Written Report of Bypass
A written submission shall be provided within five (5) working days of commencing any bypass to the Department, and at additional times as directed by the Department. The written submission shall contain a description of the bypass and its cause; the period of bypass, including exact dates and times, and if the bypass has not been corrected, the anticipated time it is expected to continue; steps taken or planned to reduce, eliminate, and prevent reoccurrence of the bypass; and other information as required by the Department.

e. Bypass Not Exceeding Limitations
The permittee may allow any bypass to occur which does not cause effluent limitations to be exceeded, but only if it also is for essential maintenance to ensure efficient operation. These bypasses are not subject to the provisions of 9.a., 9.b., 9.c., and 9.d., above. This provision does not relieve the permittee of any notification responsibilities under Part II.C.11. of this permit.

f. Definitions

1) Bypass means the intentional diversion of waste streams from any portion of a treatment facility.

2) Severe property damage means substantial physical damage to property, damage to the treatment facilities which causes them to become inoperable, or substantial and permanent loss of natural resources which can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production.

10. Bioaccumulative Chemicals of Concern (BCC)

Consistent with the requirements of R 323.1098 and R 323.1215 of the Michigan Administrative Code, the permittee is prohibited from undertaking any action that would result in a lowering of water quality from an increased loading of a BCC unless an increased use request and antidegradation demonstration have been submitted and approved by the Department.

11. Notification of Changes in Discharge

The permittee shall notify the Department, in writing, as soon as possible but no later than 10 days of knowing, or having reason to believe, that any activity or change has occurred or will occur which would result in the discharge of: 1) detectable levels of chemicals on the current Michigan Critical Materials Register, priority pollutants or hazardous substances set forth in 40 CFR 122.21, Appendix D, or the Pollutants of Initial Focus in the Great Lakes Water Quality Initiative specified in 40 CFR 132.6, Table 6, which were not acknowledged in the application or listed in the application at less than detectable levels; 2) detectable levels of any other chemical not listed in the application or listed at less than detection, for which the application specifically requested information; or 3) any chemical at levels greater than five times the average level reported in the complete application (see the first page of this permit, for the date(s) the complete application was submitted). Any other monitoring results obtained as a requirement of this permit shall be reported in accordance with the compliance schedules.

Section C. Reporting Requirements

12. Changes in Facility Operations

Any anticipated action or activity, including but not limited to facility expansion, production increases, or process modification, which will result in new or increased loadings of pollutants to the receiving waters must be reported to the Department by a) submission of an increased use request (application) and all information required under R 323.1098 (Antidegradation) of the Water Quality Standards or b) by notice if the following conditions are met: 1) the action or activity will not result in a change in the types of wastewater discharged or result in a greater quantity of wastewater than currently authorized by this permit; 2) the action or activity will not result in violations of the effluent limitations specified in this permit; 3) the action or activity is not prohibited by the requirements of Part II.C.10.; and 4) the action or activity will not require notification pursuant to Part II.C.11. Following such notice, the permit or, if applicable, the facility's COC may be modified according to applicable laws and rules to specify and limit any pollutant not previously limited.

13. Transfer of Ownership or Control

In the event of any change in control or ownership of facilities from which the authorized discharge emanates, the permittee shall submit to the Department 30 days prior to the actual transfer of ownership or control a written agreement between the current permittee and the new permittee containing: 1) the legal name and address of the new owner; 2) a specific date for the effective transfer of permit responsibility, coverage and liability; and 3) a certification of the continuity of or any changes in operations, wastewater discharge, or wastewater treatment.

If the new permittee is proposing changes in operations, wastewater discharge, or wastewater treatment, the Department may propose modification of this permit in accordance with applicable laws and rules.

14. Operations and Maintenance Manual

For wastewater treatment facilities that serve the public (and are thus subject to Part 41 of the NREPA), Section 4104 of Part 41 and associated Rule 2957 of the Michigan Administrative Code allow the Department to require an Operations and Maintenance (O&M) Manual from the facility. An up-to-date copy of the O&M Manual shall be kept at the facility and shall be provided to the Department upon request. The Department may review the O&M Manual in whole or in part at its discretion and require modifications to it if portions are determined to be inadequate.

At a minimum, the O&M Manual shall include the following information: permit standards; descriptions and operation information for all equipment; staffing information; laboratory requirements; record keeping requirements; a maintenance plan for equipment; an emergency operating plan; safety program information; and copies of all pertinent forms, as-built plans, and manufacturer's manuals.

Certification of the existence and accuracy of the O&M Manual shall be submitted to the Department at least sixty days prior to start-up of a new wastewater treatment facility. Recertification shall be submitted sixty days prior to start-up of any substantial improvements or modifications made to an existing wastewater treatment facility.

Section C. Reporting Requirements

15. Signatory Requirements

All applications, reports, or information submitted to the Department in accordance with the conditions of this permit and that require a signature shall be signed and certified as described in the Federal Act and the NREPA.

The Federal Act provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this permit, including monitoring reports or reports of compliance or noncompliance, shall, upon conviction, be punished by a fine of not more than $10,000 per violation, or by imprisonment for not more than 6 months per violation, or by both.

The NREPA (Section 3115(2)) provides that a person who at the time of the violation knew or should have known that he or she discharged a substance contrary to this part, or contrary to a permit, COC, or order issued or rule promulgated under this part, or who intentionally makes a false statement, representation, or certification in an application for or form pertaining to a permit or COC or in a notice or report required by the terms and conditions of an issued permit or COC, or who intentionally renders inaccurate a monitoring device or record required to be maintained by the Department, is guilty of a felony and shall be fined not less than $2,500.00 or more than $25,000.00 for each violation. The court may impose an additional fine of not more than $25,000.00 for each day during which the unlawful discharge occurred. If the conviction is for a violation committed after a first conviction of the person under this subsection, the court shall impose a fine of not less than $25,000.00 per day and not more than $50,000.00 per day of violation. Upon conviction, in addition to a fine, the court in its discretion may sentence the defendant to imprisonment for not more than 2 years or impose probation upon a person for a violation of this part. With the exception of the issuance of criminal complaints, issuance of warrants, and the holding of an arraignment, the circuit court for the county in which the violation occurred has exclusive jurisdiction. However, the person shall not be subject to the penalties of this subsection if the discharge of the effluent is in conformance with and obedient to a rule, order, permit, or COC of the Department. In addition to a fine, the attorney general may file a civil suit in a court of competent jurisdiction to recover the full value of the injuries done to the natural resources of the state and the costs of surveillance and enforcement by the state resulting from the violation.

16. Electronic Reporting

Upon notice by the Department that electronic reporting tools are available for specific reports or notifications, the permittee shall submit electronically all such reports or notifications as required by this permit, on forms provided by the Department.

PART II

Section D. Management Responsibilities

1. Duty to Comply

All discharges authorized herein shall be consistent with the terms and conditions of this permit. The discharge of any pollutant identified in this permit, more frequently than, or at a level in excess of, that authorized, shall constitute a violation of the permit.

It is the duty of the permittee to comply with all the terms and conditions of this permit. Any noncompliance with the Effluent Limitations, Special Conditions, or terms of this permit constitutes a violation of the NREPA and/or the Federal Act and constitutes grounds for enforcement action; for permit or Certificate of Coverage (COC) termination, revocation and reissuance, or modification; or denial of an application for permit or COC renewal.

It shall not be a defense for a permittee in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this permit.

2. Operator Certification

The permittee shall have the waste treatment facilities under direct supervision of an operator certified at the appropriate level for the facility certification by the Department, as required by Sections 3110 and 4104 of the NREPA. Permittees authorized to discharge storm water shall have the storm water treatment and/or control measures under direct supervision of a storm water operator certified by the Department, as required by Section 3110 of the NREPA.

3. Facilities Operation

The permittee shall, at all times, properly operate and maintain all treatment or control facilities or systems installed or used by the permittee to achieve compliance with the terms and conditions of this permit. Proper operation and maintenance includes adequate laboratory controls and appropriate quality assurance procedures.

4. Power Failures

In order to maintain compliance with the effluent limitations of this permit and prevent unauthorized discharges, the permittee shall either:

a. provide an alternative power source sufficient to operate facilities utilized by the permittee to maintain compliance with the effluent limitations and conditions of this permit; or

b. upon the reduction, loss, or failure of one or more of the primary sources of power to facilities utilized by the permittee to maintain compliance with the effluent limitations and conditions of this permit, the permittee shall halt, reduce or otherwise control production and/or all discharge in order to maintain compliance with the effluent limitations and conditions of this permit.

5. Adverse Impact

The permittee shall take all reasonable steps to minimize or prevent any adverse impact to the surface waters or groundwaters of the state resulting from noncompliance with any effluent limitation specified in this permit including, but not limited to, such accelerated or additional monitoring as necessary to determine the nature and impact of the discharge in noncompliance.

Section D. Management Responsibilities

6. Containment Facilities

The permittee shall provide facilities for containment of any accidental losses of polluting materials in accordance with the requirements of the Part 5 Rules (R 324.2001 through R 324.2009 of the Michigan Administrative Code). For a Publicly Owned Treatment Work (POTW), these facilities shall be approved under Part 41 of the NREPA.

7. Waste Treatment Residues

Residuals (i.e. solids, sludges, biosolids, filter backwash, scrubber water, ash, grit, or other pollutants or wastes) removed from or resulting from treatment or control of wastewaters, including those that are generated during treatment or left over after treatment or control has ceased, shall be disposed of in an environmentally compatible manner and according to applicable laws and rules. These laws may include, but are not limited to, the NREPA, Part 31 for protection of water resources, Part 55 for air pollution control, Part 111 for hazardous waste management, Part 115 for solid waste management, Part 121 for liquid industrial wastes, Part 301 for protection of inland lakes and streams, and Part 303 for wetlands protection. Such disposal shall not result in any unlawful pollution of the air, surface waters or groundwaters of the state.

8. Right of Entry

The permittee shall allow the Department, any agent appointed by the Department, or the Regional Administrator, upon the presentation of credentials and, for animal feeding operation facilities, following appropriate biosecurity protocols:

a. to enter upon the permittee’s premises where an effluent source is located or any place in which records are required to be kept under the terms and conditions of this permit; and

b. at reasonable times to have access to and copy any records required to be kept under the terms and conditions of this permit; to inspect process facilities, treatment works, monitoring methods and equipment regulated or required under this permit; and to sample any discharge of pollutants.

9. Availability of Reports

Except for data determined to be confidential under Section 308 of the Federal Act and Rule 2128 (R 323.2128 of the Michigan Administrative Code), all reports prepared in accordance with the terms of this permit, shall be available for public inspection at the offices of the Department and the Regional Administrator. As required by the Federal Act, effluent data shall not be considered confidential. Knowingly making any false statement on any such report may result in the imposition of criminal penalties as provided for in Section 309 of the Federal Act and Sections 3112, 3115, 4106 and 4110 of the NREPA.

10. Duty to Provide Information

The permittee shall furnish to the Department, within a reasonable time, any information which the Department may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this permit or the facility’s COC, or to determine compliance with this permit. The permittee shall also furnish to the Department, upon request, copies of records required to be kept by this permit.

Where the permittee becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Department, it shall promptly submit such facts or information.

PART II

Section E. Activities Not Authorized by This Permit

1. Discharge to the Groundwaters

This permit does not authorize any discharge to the groundwaters. Such discharge may be authorized by a groundwater discharge permit issued pursuant to the NREPA.

2. POTW Construction

This permit does not authorize or approve the construction or modification of any physical structures or facilities at a POTW. Approval for the construction or modification of any physical structures or facilities at a POTW shall be by permit issued under Part 41 of the NREPA.

3. Civil and Criminal Liability

Except as provided in permit conditions on "Bypass" (Part II.C.9. pursuant to 40 CFR 122.41(m)), nothing in this permit shall be construed to relieve the permittee from civil or criminal penalties for noncompliance, whether or not such noncompliance is due to factors beyond the permittee’s control, such as accidents, equipment breakdowns, or labor disputes.

4. Oil and Hazardous Substance Liability

Nothing in this permit shall be construed to preclude the institution of any legal action or relieve the permittee from any responsibilities, liabilities, or penalties to which the permittee may be subject under Section 311 of the Federal Act except as are exempted by federal regulations.

5. State Laws

Nothing in this permit shall be construed to preclude the institution of any legal action or relieve the permittee from any responsibilities, liabilities, or penalties established pursuant to any applicable state law or regulation under authority preserved by Section 510 of the Federal Act.

6. Property Rights

The issuance of this permit does not convey any property rights in either real or personal property, or any exclusive privileges, nor does it authorize violation of any federal, state or local laws or regulations, nor does it obviate the necessity of obtaining such permits, including any other Department of Environment, Great Lakes, and Energy permits, or approvals from other units of government as may be required by law.

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

TO: Claire Lowe, Cadillac District Office, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: April 19, 2022

SUBJECT: Local Limits Review
MiWaters Submission Number: HPG-XCC7-NCG09
Cadillac Wastewater Treatment Plant (WWTP)
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

Pursuant to your request, we have reviewed the Industrial Pretreatment Program Local Limits request dated April 13, 2022 for the Cadillac WWTP. The facility is currently authorized to continuously discharge treated municipal wastewater from Monitoring Point 001A through Outfall 001 to the Clam River at Latitude 44.2657, Longitude -85.3985. A design flow of 3.2 million gallons per day (MGD) was used in determining the final effluent limitations and monitoring requirements in the current NPDES permit, but this flow is not considered a limitation or actual capacity.

The Clam River has a lowest 95 percent exceedance low flow, harmonic mean low flow, and 90dQ10 low flow of 2.6 cubic feet per second (cfs), 13 cfs, and 3.4 cfs (Low Flow File Number 10194). An ambient total hardness value for the Clam River of 70 milligrams per liter (mg/l) was used for the parameters that are total hardness dependent and was collected by Permits Section staff on August 19, 2021 from the Clam River at Latitude 44.263376, Longitude -85.4011.

The theoretical water quality-based effluent limits (WQBELs) are presented in micrograms per liter (ug/l) and pounds per day (lbs./day) for the parameters requested.

Parameter	CAS Number	Daily Maximum WQBEL (µg/L)	Daily Maximum Load (lbs./day)	Monthly Average WQBEL (µg/l)	Monthly Average Load (lbs./day)
Arsenic #	7440382	680	18	16	0.42
Cadmium	7440439	12	0.32	4.1	0.11
Chromium	7440473	1,300	34	94	2.5
Copper	7440508	29	0.77	11	0.29
Cyanide	57125	44	1.2	5.9	0.16
Lead	7439921	820	22	94	2.5
Mercury @	7439976	---	---	0.0013	3.5×10^{-5}
Molybdenum	7439987	58,000	1,500	3,600	97
Nickel	7440020	760	20	47	1.3
Selenium	7782492	120	3.2	5.7	0.15
Silver	7440224	22	0.59	1.4	0.036
Zinc	7440666	360	9.7	210	5.5
Phenol	108952	6,800	180	510	14

Parameter	CAS Number	Daily Maximum WQBEL (µg/L)	Daily Maximum Load (lbs./day)	Monthly Average WQBEL (µg/l)	Monthly Average Load (lbs./day)
1,2-Dichlorobenzene	95501	240	6.4	15	0.39
Chloroform #	67663	11,000	290	710	19
Ethylbenzene #	100414	370	9.9	24	0.63
Lindane # @	58899	---	---	0.026	0.00069
Polychlorinated Biphenyls (PCBs) # @	1336363	---	---	0.000026	6.9×10^{-7}
Styrene #	100425	2,900	77	130	3.5
Toluene	108883	2,600	69	290	7.8
Xylene	1330207	890	24	55	1.5

\# - Carcinogen
@ - bioaccumulative chemical of concern (BCC)

1) The current NPDES permit includes a monthly average total phosphorus limit of 0.5 milligrams per liter (mg/l) and a corresponding load limitation of 13 pounds per day (lbs./day) from May through March and a monthly average total phosphorus limit of 0.65 mg/l and a corresponding load limit of 17 lbs./day during the month of April. These limits are to protect the Clam River and downstream surface waters from excessive nutrients.

2) The recommendations contained herein are based on the Part 4 Water Quality Standards and Part 8 Water Quality-Based Effluent Limit Development for Toxic Substances. Treatment methods or economic concerns have not been addressed. We understand that other considerations may be used in deciding permit limits and requirements.

Please contact me at (517) 290-6424 or schmittg1@michigan.gov if you have any questions regarding the recommendations above.

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

TO: Claire Lowe, Cadillac District Office, Water Resources Division

FROM: Phillip DePetro, Permits Section, Water Resources Division

DATE: April 19, 2022

SUBJECT: Local Limits Review (MiWaters Submission HPG-XCC7-NCG09)
Cadillac WWTP
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

Permits Section has evaluated your request for a local limits review for the Cadillac WWTP. This memo summarizes the calculated effluent limits for five-day carbonaceous biochemical oxygen demand ($CBOD_5$), ammonia (NH3-N), minimum effluent dissolved oxygen (DO), and total suspended solids (TSS). The Cadillac WWTP discharges treated municipal wastewater to the Clam River at 44.2657°N, -85.3985°E in Wexford County. The facility has an annual average design flow of 3.2 million gallons per day (MGD)

Note that the calculated limits provided in this memo are based on water quality and not the current permit effluent limits. In most cases the effluent limits set in the current permit are based on anti-degredation. In the 1980s, WQBELs were developed based on a trout stream designation (coldwater protection) for the Clam River between Lake Cadillac and the Wexford-Missaukee County line in the non-summer seasons. The Michigan DNR Fisheries order in 1997 and the most current fisheries order list the Clam River as being protected for coldwater fish downstream of the Wexford-Missaukee County line, and as warmwater protected between the lake and the county line. The difference in stream designation between the most recent modeling and the modeling done in the 1980's is the primary driver for the difference in limits.

The Clam River from Lake Cadillac to the Wexford-Missaukee county line in protected for warmwater fish species, other indigenous aquatic life and wildlife, agriculture, navigation, industrial water supply, public water supply at the point of intake, partial body contact recreation, total body contact recreation from May 1 to October 31, and fish consumption. The Clam River downstream of the county line is protected for the same designated uses in addition to being protected for coldwater fish.

The monthly exceedance flows (in cfs) for the Clam River at the Cadillac WWTP outfall are as follows based on Low Flow Discharge Request 10314:

	JAN.	FEB.	MAR.	APR.	MAY	JUNE
50%	17	17	36	78	35	13
95%	5.6	5.1	9.0	14	6.9	4.1

	JULY	AUG.	SEP.	OCT.	NOV.	DEC.
50%	7.4	6.3	6.3	10	18	19
95%	3.0	3.0	3.1	3.7	5.5	5.5

The lowest monthly 95% exceedance flows in up to four seasons are used as the background flow of the receiving stream for the purposes of developing WQBELs. The following seasons are used, consistent with the current permit structure: June-September, October-November, December-March, and April-May.

A multiple reach Streeter-Phelps model was used to model the impact of the Cadillac WWTP on the Clam River at critical low-flow conditions. The Haring Twp WWTP (NPDES Permit No. MI0059076) was included in this model given its proximity downstream. Effluent limit recommendations are determined that are always protective of the applicable DO standards. A diurnal variation of 0.75 mg/l per day between the daily average and daily minimum DO concentration was assumed based on a 2006 field study for the warmwater-protected stream reaches. A diurnal variation of 0.51 mg/l was assumed for the coldwater-protected reaches based on the same study. Kinetic rates were also set based on department procedure defaults for $CBOD_5$. A 1983 field study determined that nitrification was not occurring in the Clam River due to a lack of substrate for nitrifying bacteria. Site-specific time of passage and reaeration studies completed in 1983 were used to determine stream velocity and reaeration. Stream slopes were updated using LiDAR data. The recommended local limits based on existing stream designations and water quality for the Cadillac WWTP are presented in Table 1.

As previously stated, the effluent limit recommendations of the Cadillac WWTP in the summer season are less restrictive than current permit effluent limits due to the change in stream designation. The current permit effluent limits are based on anti-degradation in the basis of decision memo associated with the last permit reissuance. There is no daily maximum effluent limit recommendation for NH3-N based on the assumption that nitrification is not occurring in the Clam River.

Effluent limit recommendations based on federal secondary treatment standards (STS) are sufficient to protect the DO standard in the Clam River in the remaining three seasons. A minimum effluent DO concentration of 3.0 mg/l is recommended to protect against acute toxicity in the effluent-receiving water mixing zone. The STS-based effluent limits are less restrictive than current permit effluent limits for the Cadillac WWTP.

The Cadillac WWTP discharge was also evaluated to determine 30-day average NH3-N effluent limits that are protective of the chronic NH3-N toxicity criterion. A new chronic toxicity criterion was adopted by the department in 2019 and is being implemented for permits being issued in the current fiscal year. An effluent pH of 7.3 SU was assumed for the Cadillac WWTP based on two years of discharge monitoring data. The recommended effluent limits to protect against chronic toxicity are more restrictive than current permit effluent limits at the Cadillac WWTP in April. The limits for the remaining months at the Cadillac WWTP are equivalent to or less restrictive than current permit effluent limits.

There is no applicable numeric water quality standard for TSS in the Part 4 Water Quality Standards. The narrative standard set in Rule 50 [R323.1050] would apply for TSS.

These effluent limit recommendations are based on water quality standards. We have not addressed treatment practicality or cost effectiveness. Our recommendations do not imply that other considerations, such as the existing permit effluent limits, should not be taken into account when deciding on permit limits.

Table 2. Cadillac WWTP (MI0020257) Local Limits Review Recommendations

Parameter	Months	Concentration (mg/l)	Basis	Rationale
$CBOD_5$	June-September	17	Daily Maximum	DO Standard
	October-May	25	7-Day Average	Secondary Treatment
		40	30-Day Average	Secondary Treatment
NH3-N	June-September	0.9	30-Day Average	Chronic Toxicity
	October-November	3.1	30-Day Average	Chronic Toxicity
	December-March	5.4	30-Day Average	Chronic Toxicity
	April-May	3.0	30-Day Average	Chronic Toxicity
D.O.	June-September	6.0	Minimum	DO Standard
	October-May	3.0	Minimum	Acute DO Toxicity

Design Flow = 3.2 MGD

filter. The vacuum filter removes water from the sludge and procudes a sludge cake which may be disposed of at a landfill. Water removed from the sludge is returned to the wastewater at the quick-mix tank.

B. PLANT FLOW PATTERN

Figure I shows a simplified flow diagram of main process flow at the Cadillac Wastewater Treatment Plant.

C. PRINCIPLE DESIGN CRITERIA

1. Design population (year of 1990) - 15,000 capita

2. Influent characteristics

   a. BOD5(five day biological oxygen demand) @ 0.22 pounds/day - 3,300 lbs./day or 200 mg/l.

   b. SS (suspended solids) @ 0.28 pounds/day -4200 lbs/day or 252 mg/l.

   c. NH_3-N (ammonia-nitrogen found in the ammonia form) - 20 mg/l.

   d. PO_4-P (phosphate-phosphorous or phosphorous found in the phosphate form) - 40 mg/l.

4. Design Flows

Average day flow	2.0 MGD
Minimum day flow	0.8 MGD
Maximum day flow	4.5 MGD

D. EXPECTED EFFICIENCY OF TREATMENT

Overall plant efficiency is expected to be 97 percent for BOD, 96 percent for suspended solids, 92.5 percent for ammonia and 80 percent for phosphates. Individually, the primary, secondary and tertiary phases of treatment are expected to perform as follows:

PERCENT REMOVAL

	BOD	SS	NH_3-N	PO_4-P
Primary	45	65	---	60
Secondary	82	65	---	---
Tertiary	70	70	92.5	50
Total	97	96	92.5	80

E. OPERATION AND MANAGERIAL RESPONSIBILITY

Personnel for the Treatment Plant should consist of a superintendent, a laboratory technician, and six to eight assistant operators. The superintendent is responsible for the management and total operation of the Plant. In general, his duties include supervision and training of all subordinate personnel and responsibility for daily operations, laboratory procedures, equipment maintenance and the general care and upkeep of the facility. He is expected to acquire a thorough knowledge of the Plant, the wastewater it treats, the effects of the effluent on the receiving stream and applicable regulations or laws governing his operations.

A more detailed discussion of plant personnel and their responsibilities is included in Chapter X.